CH10　樑的應力

10-1.樑的彎曲應力

一、樑的彎曲

研究樑在彈性限度內之彎曲狀態時，必需先假設：

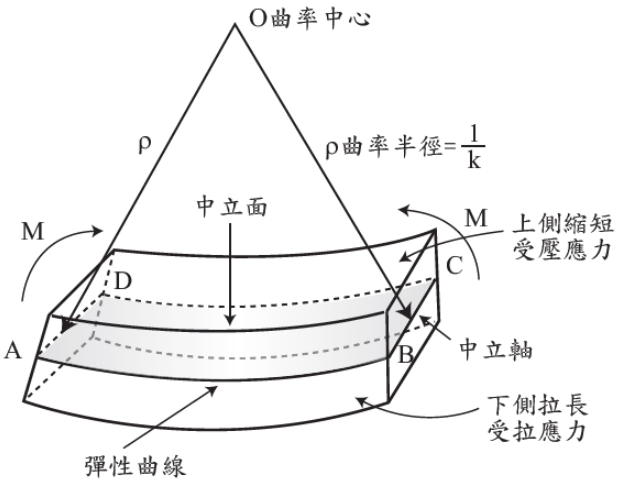
(1)所有的負載皆作用於樑之對稱面。

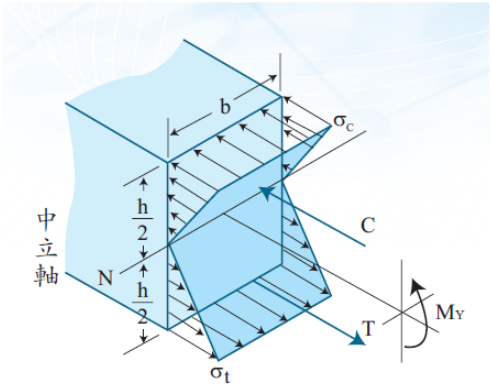
(2)材料為均質、斷面一致之材料，受負荷後，須完全符合虎克定律，即其應力與應變須成正比。

(3)材料拉伸與壓縮之剛性模數相同。

(4)樑受彎矩前後，其截面仍為平面，其截面不可發生扭曲現象。

如圖12.3(A)(B)所示，樑受彎矩M作用時，該樑會造成彎曲，其中該曲面向上撓曲，頂部被壓縮而縮短，底部被拉伸而伸長，由壓縮到拉伸之間，伸長量由負到正間必存在一伸長量為零的直線，此直線的位置稱為中性軸(neutral axis)；中性軸所在垂直於對稱軸的平面稱為中立面(neutral surface)，我們設作用於樑上面之最大壓應力為σ_c，作用於下面之最大張應力為σ_t，因符合虎克定律，故其應力會與上下表面距中立軸的距離成正比。





二、樑的彎曲應力

(一)梁的彎曲應力

樑的彎曲應力

(1)線應變

如圖12.4(a)樑受彎曲應力後，O點為樑受彎矩後，相鄰兩個截面延長線的交點，稱之為曲率中心，由曲率中心到中性軸的距離，稱之為曲率半徑，如圖示之符號ρ，圖中L爲中立面與樑縱截面之交線，稱之彈性曲線，亦即該樑受彎曲應力後，樑的伸長量為零之曲線，由圖示可知該橫截面間相對轉過的角度爲θ，$\overline{OA}$ 曲率半徑爲ρ，距中性面y處的圓弧曲線為 $\overline{nn'}$，其受力後伸長爲

$$\delta=\overline{nn'}-\overline{nB'}=\overline{nn'}-\overline{AB}=(\rho+y)\theta-L=(\rho+y)\theta-\rho\theta=y\theta$$

線應變：$\varepsilon=\frac{\triangle L}{AB}=\frac{y\theta}{\rho\theta}=\frac{y}{\rho}$

(2)彎曲應力

A.如圖12.4(b)所示樑的x方向只是發生簡單拉伸或壓縮，橫截面上的正應力不超過材料的比例極限時，可由虎克定律得到橫截面上座標爲y處各點的正應力爲 $\sigma=E\varepsilon=\frac{E}{\rho}y$，該式表明，橫截面上各點的正應力σ與點的座標y成正比。

B.中性軸上部橫截面的各點均爲壓應力，而下部各點則均爲拉應力。

C.彎曲應力推導

如圖12.4(b)所示，將此截面以平行中立軸之直線分割成許多微小面積A_1、A_2……，每一個微小面積與中立軸的距離為y_1、y_2、y_3……。我們以σ_1、σ_2、σ_3……表示微小截面上所受之正交應力，則每一微小截面所受之力為σ_1A_1、σ_2A_2……。我們用M來表示其中立軸所受之力矩代

數和為

$$M = \sigma_1 A_1 y_1 + \sigma_2 A_2 y_2 \quad \cdots\cdots (1)$$

因符合虎克定律，故其應力會與上下表面距中立軸的距離成正比。所以可知 $\frac{\sigma}{y} = \frac{\sigma_c}{y_c} = \frac{\sigma_t}{y_t}$

所以

$$\sigma_1 = \frac{\sigma_t}{y_t} Y_1 \ ， \ \sigma_2 = \frac{\sigma_t}{y_t} Y_2 \quad \cdots\cdots (2)$$

代入(1)式得

上式中括號內之值為該截面對中立軸之慣性矩I來表示，則

$$M = \sigma_t \frac{I}{y_t}$$

同理，則另一側受壓側為 $M = \sigma_c \frac{I}{y_c}$

$\Rightarrow$ 正應力計算公式 $\sigma = \frac{My}{I}$

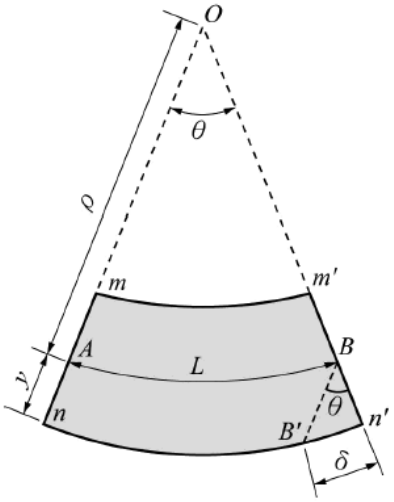


圖12.4(a)　樑的彎矩應力

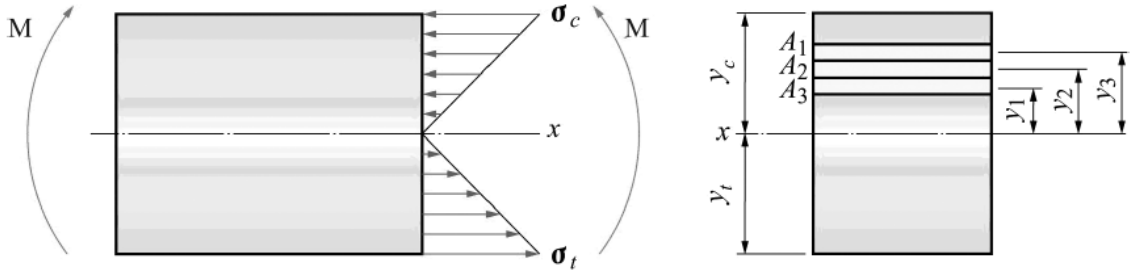


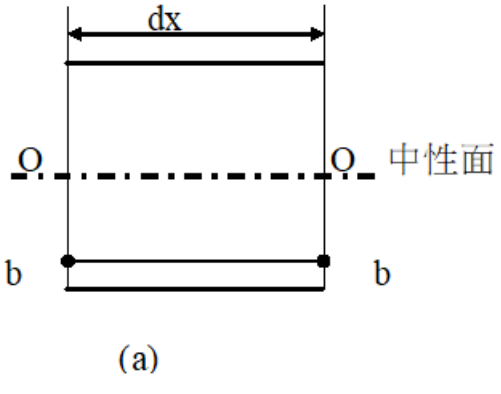
圖12.4(b)　樑的彎矩應力

(二)彎曲應力證明

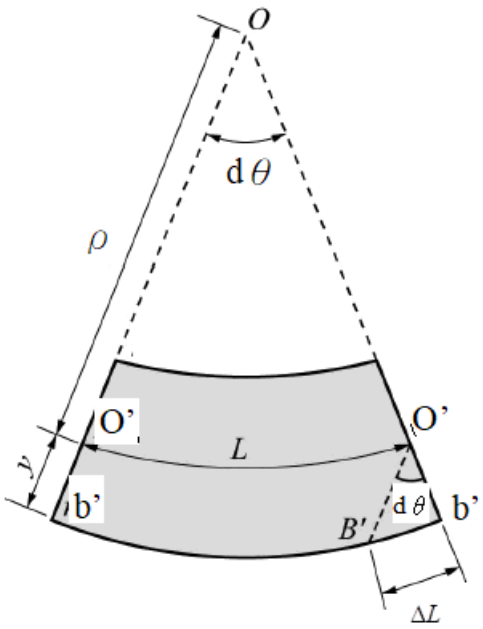
1. 基本假設:
   (1)純彎矩彎曲斷面上僅有彎矩作用
   (2)變形幾何關係(材料均質且具線彈性)。
   (3)受拉受壓之楊氏係數 E 相同
   (4)受彎矩負載下產生小變形效應。
   (5)受彎矩負載後橫斷面仍然保持平面(沒有翹曲(warping))現象。
   (6)對稱直樑斷面至少有一對稱軸。
2. 變形過程中，樑受正彎矩作用時產生凹面向上的彎曲，樑的頂部縮短，底部伸長，撓曲後斷面仍保持為平面，無翹曲。
3. 曲面向上撓曲的直樑，頂部的直線縮短，底部的直線伸長，伸長量由負到正間必存在一伸長量為零的直線。此直線的位置稱為中性軸(neutral axis)；中性軸所在垂直於對稱軸的平面稱為中性面(neutral surface)。在中性面上的直線撓曲時應變量為零。
4. 若軸受純彎矩作用，樑之原長為 L，曲一小段來看，若曲率半徑為 ρ，撓曲角為 $d\theta$，在距離中性軸上 y 的一長軸縮短為，橫截面選用如圖所示的 y-z 坐標系，圖中 y 軸爲橫截面的對稱軸，z 軸爲中性軸，如圖(a)(b)可以看到，從中截取出長爲 dx 的一個微段，橫截面間相對轉過的角度爲 $d\theta$，中性面 $\overline{O'O'}$ 曲率半徑爲 ρ，距中性面 y 處的任一縱線為 $\overline{bb}$，其受力後變為 $\overline{b'b'}$ 的圓弧曲線，因此縱線 $\overline{bb}$ 的伸長爲

   $$\Delta L = (\rho + y)d\theta - dx = (\rho + y)d\theta - \rho d\theta = y d\theta$$

   線應變：$$\boxed{\varepsilon = \frac{\Delta L}{\overline{bb}} = \frac{y d\theta}{\rho d\theta} = \frac{y}{\rho}}$$



(a)



(b)

5.彎曲應力推導

(1) 如圖橫截面上座標爲(y,z)的點的正應力爲σ，截面上各點的微內力σdA組成與橫截面垂直的空間平行力系。

(2) $\sum F_x = 0 \Rightarrow N = \int_A \sigma dA = 0$(平行$x$軸的軸力)

$\sum M_y = 0 \Rightarrow M_y = \int_A z\sigma dA = 0$(y 軸的力偶矩)

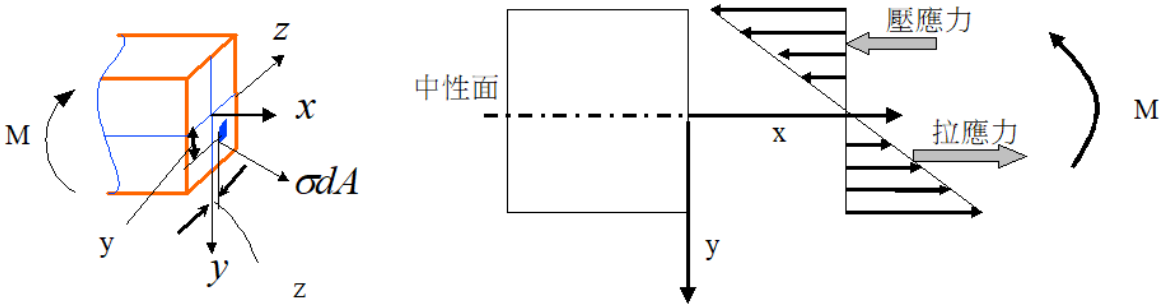
$M_z = \int_A y\sigma dA = \frac{E}{\rho}\int_A y^2 dA = M$ (其中慣性矩$I_z = \int_A y^2 dA$)

$\Rightarrow \boxed{\frac{1}{\rho} = \frac{M}{EI_z}}$代入$\sigma = E\varepsilon = \frac{E}{\rho}y$

$\Rightarrow$正應力計算公式$\boxed{\sigma = \frac{My}{I_z}}$

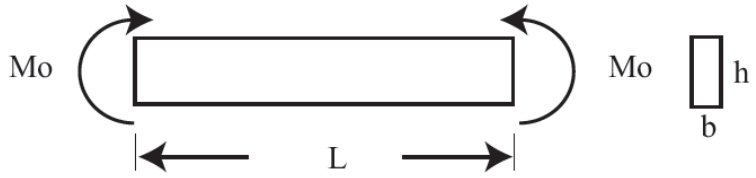
(3) EI_Z稱爲樑的抗彎剛度，EI_Z越大，則曲率$\frac{1}{\rho}$越小，變形越小。

(4) 截面圖形的斷面模數$\boxed{Z = \frac{I_z}{y}}$，只與截面圖形的幾何性質有關。


z
x
M
σdA
y
y
z
中性面
壓應力
拉應力
x
y
M

範例 1

.如圖所示，長度為L，斷面寬為b，斷面高為h，楊氏係數為E之均勻桿件，承受純彎矩M_o作用。求(1)桿件彎曲之曲率半徑為何？(2)桿件兩端點之相對轉角為何？(3)桿件最大應力為何？。(95年土木地特四等)

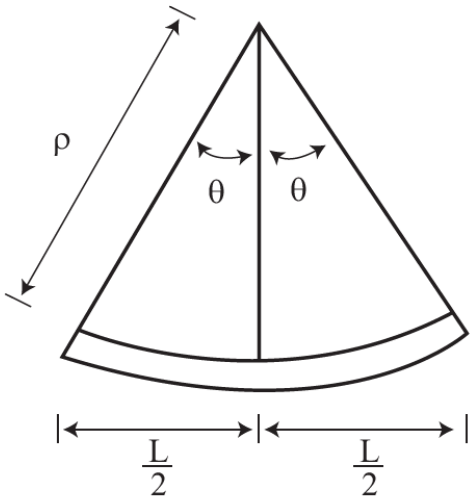


【解析】

答：(1) $K = \frac{1}{\rho} = \frac{M}{EI}$

$$\rho = \frac{EI}{M} = \frac{E \times \frac{1}{12} bh^3}{M_0} = \frac{Ebh^3}{12M_0}$$

(2) 如圖所示：



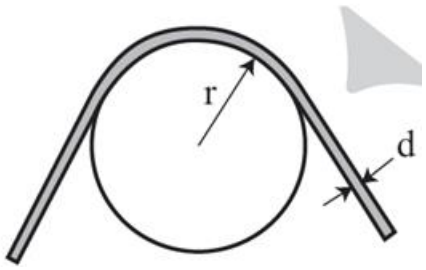
$$\sin\theta = \frac{L/2}{\rho} = \frac{6M_0L}{Ebh^3}$$

若θ很小則 $\theta \approx \sin\theta = \frac{6M_0L}{Ebh^3}$

(3) $\sigma = \frac{My}{I} = \frac{M_0 \times \frac{h}{2}}{\frac{1}{12}bh^3} = \frac{6M_0}{bh^2}$

範例 2

如圖所示，直徑為d的鋼索以彎曲方式沿一半徑為r的輪鼓圖繞，試求鋼索的最大彎曲應力以及彎曲力矩。假設楊氏係數 $E = 200\ \text{GPa}$，$d = 4\ \text{mm}$，以及 $r = 0.5\ \text{m}$。 (97年關務四等)



【解析】

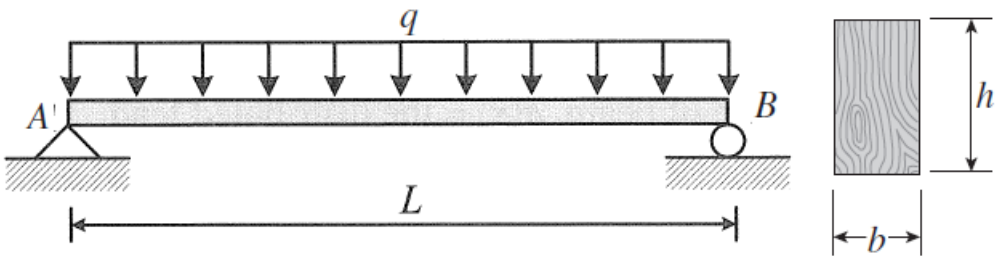
答：$$\sigma = \frac{yE}{\rho} = \frac{(\frac{d}{2})\times\ E}{\frac{d}{2}+r} = \frac{\frac{4}{2}\times\ 200\times\ 10^{9}}{\frac{4}{2}+500} = 796812749\,(\text{Pa}) = 796.81\,(\text{MPa})$$

$$\sigma = \frac{My}{I} = \frac{16M}{\pi d^{3}} = \frac{32\times\ M}{\pi\times\ (4)^{3}} \Rightarrow M = 5006.5\ (\text{N-mm})$$

彎

範例 3

A simply supported wood beam AB with span length $L = 3.5\text{m}$ carries a uniform load of intensity $q = 6.4\text{kN/m}$ (see figure). Calculate the maximum bending stress $\sigma_{\max}$ due to the load q if the beam has a rectangular cross section with width $b = 140mm$ and height h=240 mm.



【105 成大醫工】【中正機械】

【中文翻譯】

簡支木樑 AB 具跨距長 $L = 3.5\text{m}$，承載強度 $q = 6.4\text{kN/m}$ 均勻負載 (見圖)。求由於負載 q 之最大彎曲應力 $\sigma_{\max}$；樑為矩形橫截面，寬 b 高 h 分別為 140 mm 及 240 mm。

【解析】

$$M_{\max} = \frac{qL^2}{8} \qquad S = \frac{bh^2}{6}$$

$$\sigma_{\max} = \frac{M_{\max}}{S} = \frac{3qL^2}{4bh^2}$$

$$\sigma_{\max} = \frac{3(6.4\text{ kN/m})(3.5\text{ m})^2}{4(140\text{ mm})(240\text{ mm})^2} = 7.29\text{ MPa}$$

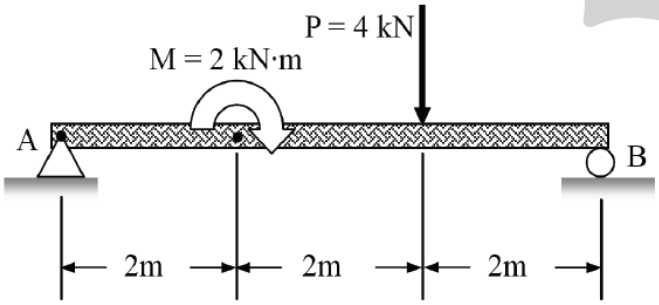
範例 4

有一簡支梁上受到一個順時針力矩M＝2KN．m以及一個集中負荷P＝4kN如右圖所示。

(1) **求支點A、B的反作用力。**

(2) **繪出該梁的剪力圖與彎矩圖。**

(3) **求簡支梁內的最大彎曲應力。梁橫截面為矩形，寬**0.1m，**高**0.2m。(100關四)



【100 關四】

【解析】

答：(1) 取梁之F、B、D

$$\sum M_A = 0 +\circlearrowleft$$

$$-2-4\times 4+R_B\times 6=0$$

$$R_B = 3(KN)(\uparrow)$$

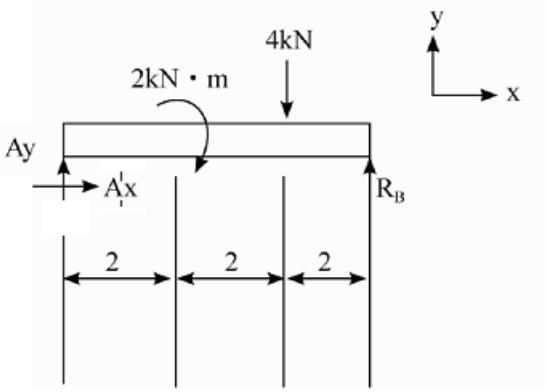
$$\sum F_X = 0 \rightarrow +$$

$$A_x = 0$$

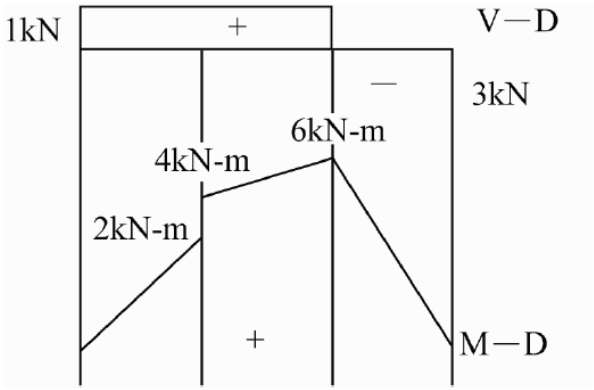
$$\sum F_y = 0 \uparrow +$$

$$A_y - 4+3=0$$

$$A_y = 1(KN)(\uparrow)$$



(2) V－D及M－D圖



(3) 已知

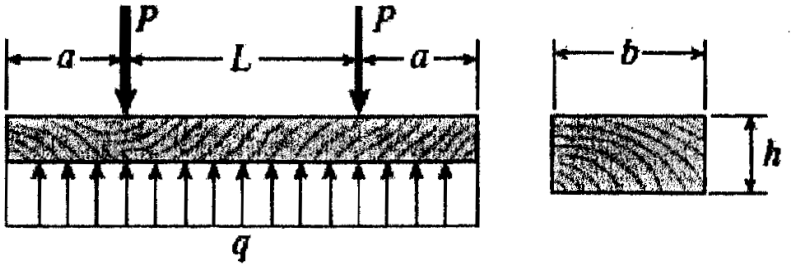
$M_{max}=6kN-m$

$$\sigma=\frac{My}{I}=\frac{6M}{6h^2}$$

$$\Rightarrow\frac{6\times6\times10^3}{0.1\times0.2^2}=9\ (MPa)$$

範例 5

A railroad tie (or sleeper) is subjected to two rail loads, each of magnitude $P = 175\text{kN}$, acting as shown in the figure. The reaction q of the ballast is assumed to be uniformly distributed over the length of the tie, which has cross-sectional dimensions $b = 300\text{mm}$ and $h = 250\text{mm}$. Calculate the maximum bending stress max in the tie due to the loads P, assuming the distance $L = 1.5m$ and the overhang length $a = 0.5m$.



【103 成大醫工】

【中文翻譯】

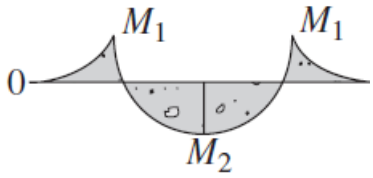
枕木受兩集中負載，每一負載大小為 $P = 175\text{kN}$，如圖所示作用著。枕石之反應均勻強度如圖在枕木整個長度。枕木寬 $b = 300\text{mm}$，高 $h = 250\text{mm}$。若 $L = 1.5m$ 及 $a = 0.5m$，求由於負載（們）P 之枕木的最大彎曲應力。

【解析】

(a)

$$P = 175\text{ kN} \quad b = 300\text{ mm} \quad h = 250\text{ mm}$$
$$L = 1500\text{ mm} \quad a = 500\text{ mm}$$

$$q = \frac{2P}{L + 2a} \qquad S = \frac{bh^2}{6} = 3.125 \times 10^{-3}\ \text{m}^3$$



$$M_1 = \frac{qa^2}{2} = \frac{Pa^2}{L + 2a}$$

$$M_2 = \frac{q}{2}\left(\frac{L}{2} + a\right)^2 - \frac{PL}{2}$$

$$= \frac{P}{L + 2a}\left(\frac{L}{2} + a\right)^2 - \frac{PL}{2}$$

$$= \frac{P}{4}(2a - L)$$

(b)

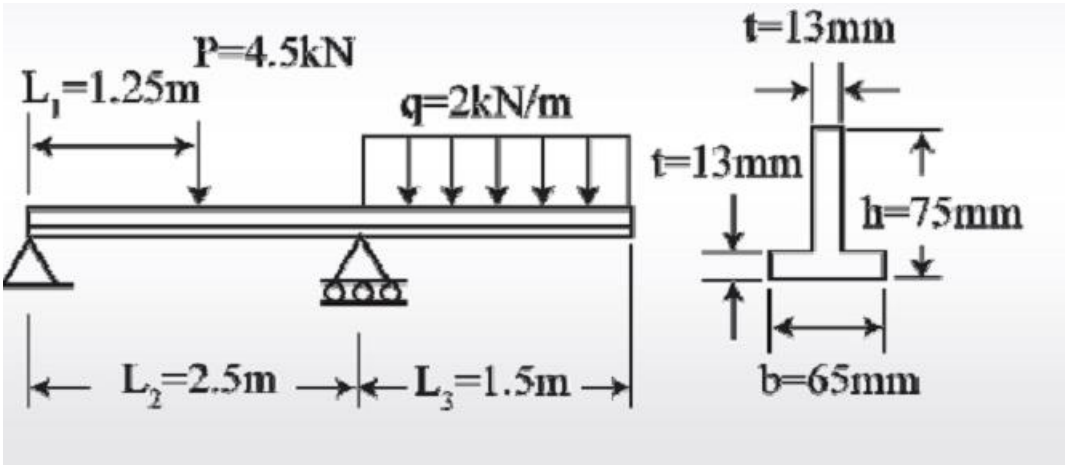
$$M_1 = 17{,}500 \text{ N} \cdot \text{m} \qquad M_2 = -21{,}875 \text{ N} \cdot \text{m}$$

$$M_{\max} = 21{,}875 \text{ N} \cdot \text{m}$$

$$\sigma_{\max} = \frac{M_{\max}}{5} = \frac{21{,}875 \text{ N} \cdot \text{m}}{3.125 \times 10^{-3} \text{ m}^3} = 7.0 \text{ MPa}$$

範例 6

beam of T-section is supported and loaded as shown in the figure. The cross section has width $b=65mm$., height $h=75\text{mm}$., and thickness $t=13\text{mm}$. Determine the maximum tensile and compressive stresses in the beam.



【台大工科】【104 台大土木】【105 北科機電】

【中文翻譯】

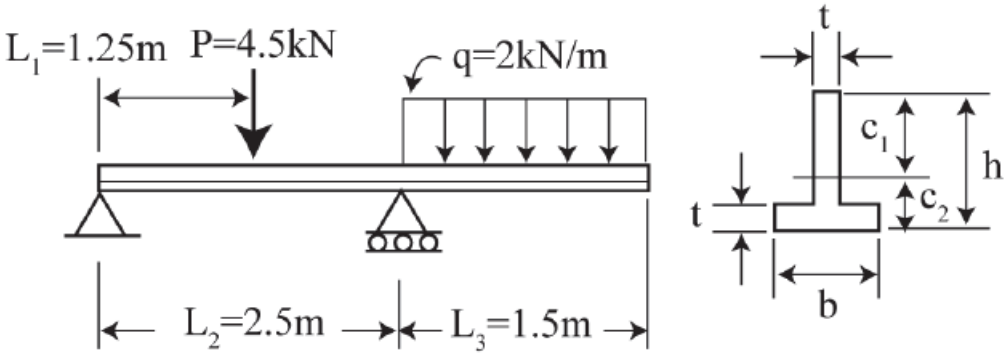
T 橫截面外伸樑如圖示；橫截面寬 $b=65mm$ ，高 $h=75\text{mm}$ ，且厚 $t=13\text{mm}$ 。求樑之最大拉伸及壓縮應力。

【解析】

解 1. **截面性質：**

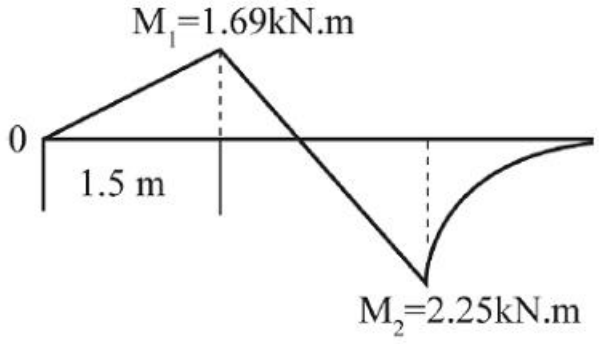
$t=13\text{mm}$， $b=65\text{mm}$， $h=75\text{mm}$， $a=1613\text{mm}^2$

$c_1x=50.2\text{mm}, c_2=24.8\text{mm}$ $I=850195\text{mm}^4$



2. **彎矩圖：**

M_1=1.69kN・m，M_2=−2.25kN・m



M_1最大彎曲應力：

$$\sigma_t = \frac{M_1 c_2}{I} = 49.3\text{MPa}$$

$$\sigma_c = \frac{M_1 c_1}{I} = 99.8\text{MPa}$$

M_2最大彎曲應力：

$$\sigma_t = \frac{M_2 c_1}{I} = 132.9\text{MPa}$$

$$\sigma_c = \frac{M_2 c_2}{I} = 65.6\text{MPa}$$

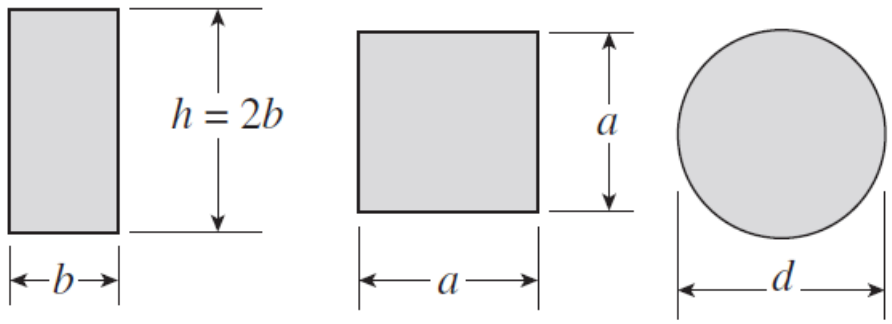
3. **最大應力：**

$\sigma_t = 132.9\text{MPa}$

$\sigma_c = 99.8\text{MPa}$

範例 7

Determine the ratios of the weights of three beams that have the same length, are made of the same material, are subjected to the same maximum bending moment, and have the same maximum bending stress if their cross sections are (1) a rectangle with height equal to twice the width, (2) a square, and (3) a circle (see figures).



【104 成大醫工】

【中文翻譯】

求三種樑的重量比，三種均為相同的材料，而且長相同，受有相同的最大彎（曲力）矩，且有相同的最大彎曲應力，如果：(a) 矩形之高為寬之兩倍、(b) 正方形、及 (c) 圓形（見圖）。

【解析】

$M = S\sigma_m$　M 相等、σ_m 相等 → S 相等

(一)矩形

$$S = \frac{bh^2}{6} = \frac{2b^3}{3} \qquad b = \left(\frac{3S}{2}\right)^{1/3}$$

$$A_1 = 2b^2 = 2\left(\frac{3S}{2}\right)^{2/3} = 2.6207S^{2/3}$$

(二)正方形

$$S = \frac{a^3}{6} \qquad a = (6S)^{1/3}$$

$$A_2 = a^2 = (6S)^{2/3} = 3.3019S^{2/3}$$

(三)圓形

$$S = \frac{\pi d^3}{32} \qquad d = \left(\frac{32S}{\pi}\right)^{1/3}$$

$$A_3 = \frac{\pi d^2}{4} = \frac{\pi}{4}\left(\frac{32S}{\pi}\right)^{2/3} = 3.6905\,S^{2/3}$$

重量比=面積比(長度相同)

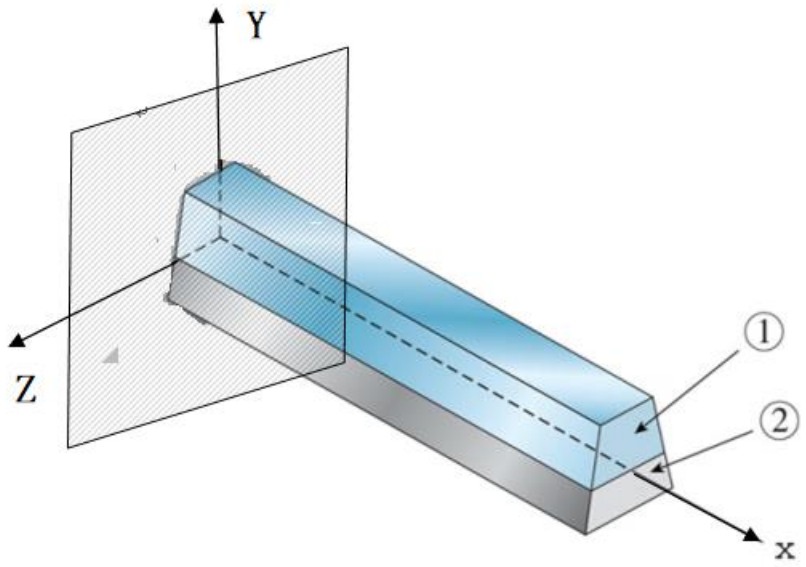
$$W_1 : W_2 : W_3 = A_1 : A_2 : A_3$$
$$A_1 : A_2 : A_3 = 2.6207 : 3.3019 : 3.6905$$
$$W_1 : W_2 : W_3 = 1 : 1.260 : 1.408$$

10-2.<u>複合梁(composite beam)之彎曲應力分析</u>

一、轉換斷面法之彎曲應力分析：

(一)複合樑

梁由二種或兩種以上材料所組成者，稱為複合梁，如鋼筋混凝土梁。複合梁承受彎矩時，因平面保持平面，因此應變仍隨距中性軸距離直線變化，惟因材料的彈性係數不同，因此同一位置不同材料的應力不同



(二) 中性軸(neutral axis)位置之決定

若樑以正曲率彎曲，則應變 ε_x 將如圖所示變化，其中 ε_A 為樑頂部壓應變、 ε_B 為底部之拉應變、 ε_C 為接觸面之應變，可由應變得到剖面上的正應力，利用虎克定律可將距離中性軸 y 處的正應力表示為:

$\sigma_{x1} = -E_1 ky$ 、 $\sigma_{x2} = -E_2 ky$

作用在橫剖面上的合軸向力為零

$$\int \sigma_{x1} dA + \int \sigma_{x2} dA = 0 \Rightarrow -\int E_1 ky dA - \int E_2 ky dA = 0$$

橫剖面處的曲率不變

$$E_1 \int y dA + E_2 \int y dA = 0$$

上式積分式代表兩部分剖面對中性軸的一次矩

(三)力矩-曲率關係式

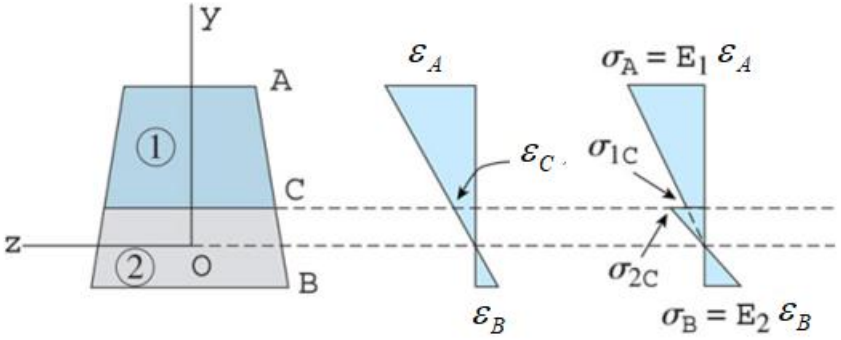
彎矩應力的合力矩等於作用在剖面上的彎矩 M

$$\mathrm{M} = -\int \sigma_{x1} y dA - \int \sigma_{x2} y dA = kE_1 \int y^2 dA + kE_2 \int y^2 dA = 0$$

$$\Rightarrow M = k(E_1 I_1 + E_2 I_2)\ kE_1 \int y^2 dA + kE_2 \int y^2 dA = 0$$

彎矩應力

$$\sigma_{x1} = \frac{-MyE_1}{E_1 I_1 + E_2 I_2} \text{ 、 } \sigma_{x2} = \frac{-MyE_2}{E_1 I_1 + E_2 I_2}$$



二、轉換斷面法之彎曲應力分析：

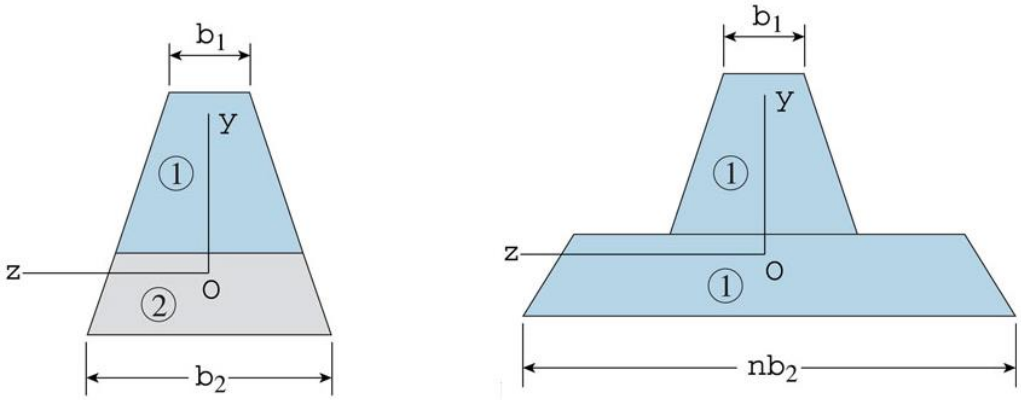
分析複合梁時通常將不同材料化成同一種材料，如圖所示之梁由兩種材料構成，第二種材料的彈性係數 E_2 較大，為第一種材料彈性係數 E_1 的 n 倍，將第二種材料視為第一種，但寬度變成原寬度之 n 倍，模數比(modular ratio)：$n = \frac{E_2}{E_1}$，然後視為第一種材料分析之，此稱為轉換斷面法。

材料1 的彎曲應力：

$$\sigma = -\frac{My}{I_z}$$

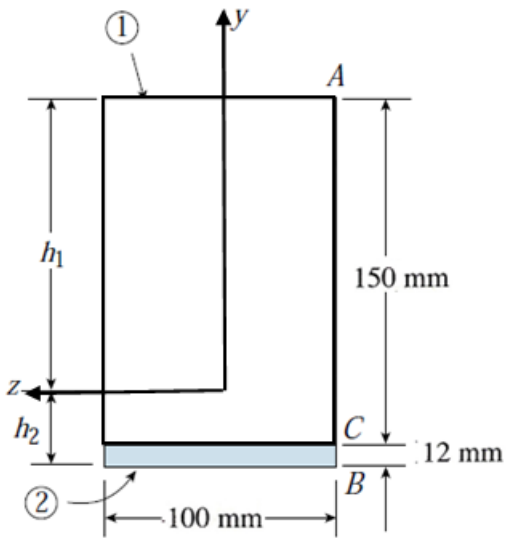
材料 2 的彎曲應力：

$$\sigma = -\frac{My}{I_z} \times n$$



範例 8

A composite beam (Fig.) is constructed from a wood beam (100 mm × 150 mm actual dimensions) and a steel reinforcing plate (100 mm wide and 12 mm thick). The wood and steel are securely fastened to act as a single beam. The beam is subjected to a positive bending moment M=6kN m. Calculate the maximum tesnile and compressive stresses in the wood (material 1) and the maximum and minimum tensile stresses in the steel (material 2) if E_1=10.5 GPa and E_2=210GPa.



【清大動機】

【中文翻譯】

一複合樑由木樑 (100×150 mm) 及一鋼補強板 (100×12 mm) 所製成。木及鋼牢固地接合而成一單一樑。樑受正彎矩 $M = 6$ kN·m 。求木 (材料 1) 之最大拉伸及壓縮應力，及鋼 (材料 2) 之 最大與最小應力；其中 $E_1 = 10.5$ GPa , $E_2 = 210$ GPa 。

【解析】

(a)先求中立軸位

$$E_1 \int_1 y dA + E_2 \int_2 y dA = 0$$

$$(10.5\text{ GPa})(h_1 - 75\text{ mm})(15{,}000\text{ mm}^2) + (210\text{ GPa})65.33\times10^6\text{ mm}^4 = 0$$

$$h_1 = 124.8\text{ mm}$$

$$h_2 = 162\text{ mm} - h_1 = 37.2\text{ mm}$$

(b)慣性矩

$$I_1 = \frac{1}{12}(100\text{ mm})(150\text{ mm})^3 + (100\text{ mm})(150\text{ mm})(h_1 - 75\text{ mm})^2$$
$$= 1.18\times10^6\text{ mm}^4$$
$$I_2 = \frac{1}{12}(100\text{ mm})(12\text{ mm})^3 + (100\text{ mm})(12\text{ mm})(h_2 - 6\text{ mm})^2$$
$$= 1.18\times10^6\text{ mm}^4$$

(c)

材料 1 之最大壓縮應力發生於樑頂 (A)

$$y = h_1 = 124.8\text{ mm}$$
$$\sigma_{1A} = -\frac{Mh_1E_1}{E_1I_1 + E_2I_2}$$
$$= -\frac{(6\text{ kN}\cdot\text{m})(124.8\text{ mm})(10.5\text{ GPa})}{(10.5\text{ GPa})(65.33\times10^6\text{ mm}^4) + (210\text{ GPa})(1.18\times10^6\text{ mm}^4)}$$
$$= -8.42\text{ MPa}$$

材料 1 之最大拉伸應力發生於兩材料的接觸面 (C)。

$$y = -(h_2 - 2\text{ mm}) = -25.2\text{ mm}$$
$$\sigma_{1C} = -\frac{(6\text{ kN}\cdot\text{m})(-25.2\text{ mm})(10.5\text{ GPa})}{(10.5\text{ GPa})(65.33\times10^6\text{ mm}^4) + (210\text{ GPa})(1.18\times10^6\text{ mm}^4)}$$
$$= 1.70\text{ MPa}$$

鋼板 (材料 2) 最大拉伸應力發生於樑之底 (B)

$$y = -h_2 = -37.2\text{ mm}$$
$$\sigma_{2B} = -\frac{M(-h_2)E_2}{E_1I_1 + E_2I_2}$$
$$= -\frac{(6\text{ kN}\cdot\text{m})(-37.2\text{ mm})(210\text{ GPa})}{(10.5\text{ GPa})(65.33\times10^6\text{ mm}^4) + (210\text{ GPa})(1.18\times10^6\text{ mm}^4)}$$
$$= 50.2\text{ MPa}$$

材料 2 之最小拉伸應力發生於接觸面 (C)

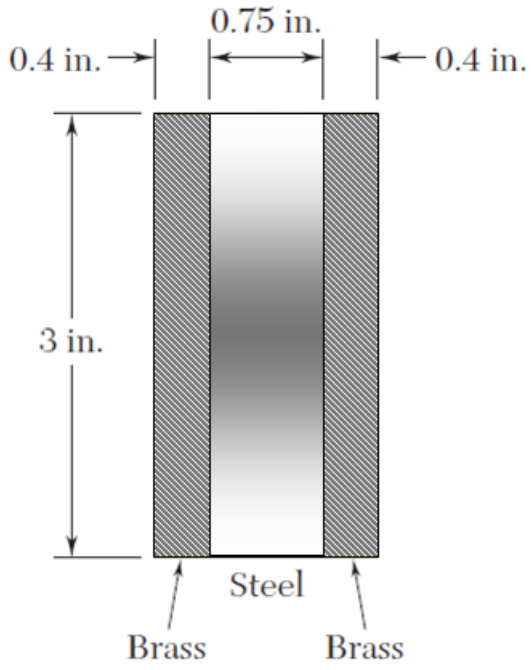
$y = -25.2$ mm

$$\sigma_{2C} = -\frac{(6\text{ kN}\cdot\text{m})(-25.2\text{ mm})(10.5\text{ GPa})}{(10.5\text{ GPa})(65.33\times10^{6}\text{ mm}^4)+(210\text{ GPa})(1.18\times10^{6}\text{ mm}^4)}$$

$$= 34\text{ MPa}$$

範例 9

A bar obtained by bonding together pieces of steel l (Es = 29x106 psi) and brass (E_b = 15x106 psi) has the cross section shown Fig. Determine the maximum stress in the steel and in the brass when the bar is in pure bending with a bending moment M= 40 kip-in.



【107 成大土木】

【中文翻譯】

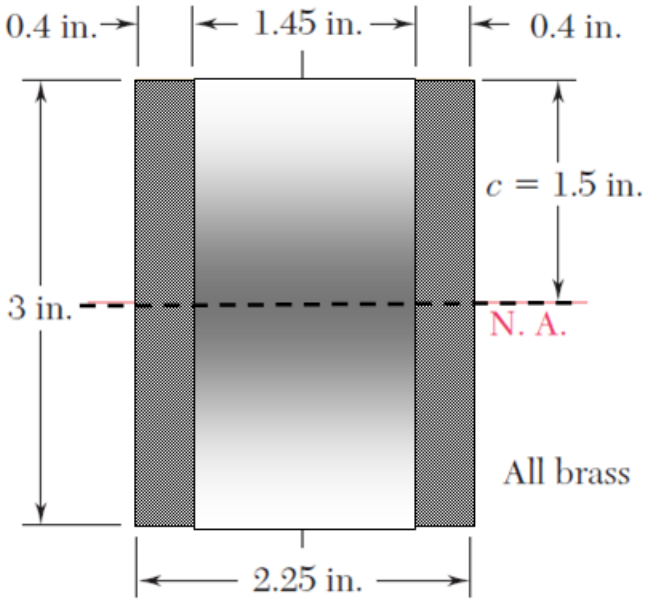
圖中二塊銅板與一塊鋼板構成之複合梁，黃銅及鋼之彈性模數分別為Es = 29x106 psi及Eb = 15x106 psi；當梁之斷面承受彎矩M= 40 kip-in.，試求銅板與鋼板內所生之最大應力？

【解析】

(一)轉換斷面

$$n = \frac{E_s}{E_b} = \frac{29 \times 10^6 \text{ psi}}{15 \times 10^6 \text{ psi}} = 1.933$$

$$(0.75 \text{ in.})(1.933) = 1.45 \text{ in.}$$



(二)

$$I = \tfrac{1}{12}bh^3 = \tfrac{1}{12}(2.25\text{ in.})(3\text{ in.})^3 = 5.063\text{ in}^4$$

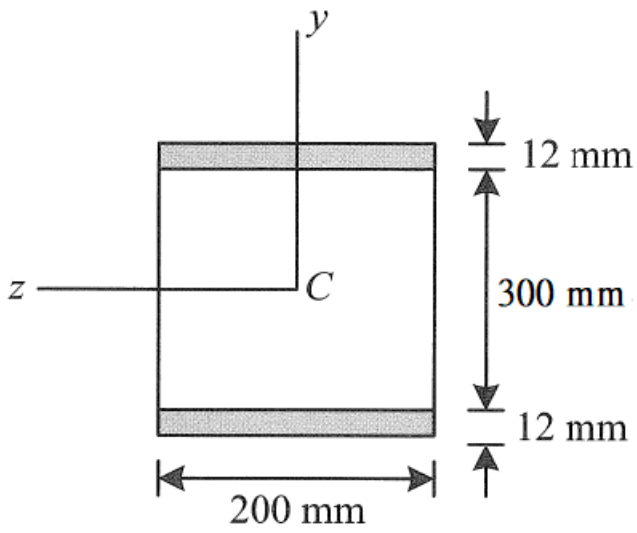
$$\sigma_m = \frac{Mc}{I} = \frac{(40\text{ kip}\cdot\text{in.})(1.5\text{ in.})}{5.063\text{ in}^4} = 11.85\text{ ksi}$$

(三)

$$(\sigma_{\text{brass}})_{\max} = 11.85\text{ ksi}$$
$$(\sigma_{\text{steel}})_{\max} = (1.933)(11.85\text{ ksi}) = 22.9\text{ ksi}$$

範例 10

A wood beam 200 mm wide and 300 mm deep is reinforced on top and bottom by 12 mm thick steel plates (see the figure). Find the allowable bending moment M_{max} about the z axis if the allowable stress in the wood is 17 MPa and in the steel is 120 MPa. (Assume that the ratio of the moduli of elasticity of steel and wood is 20.)



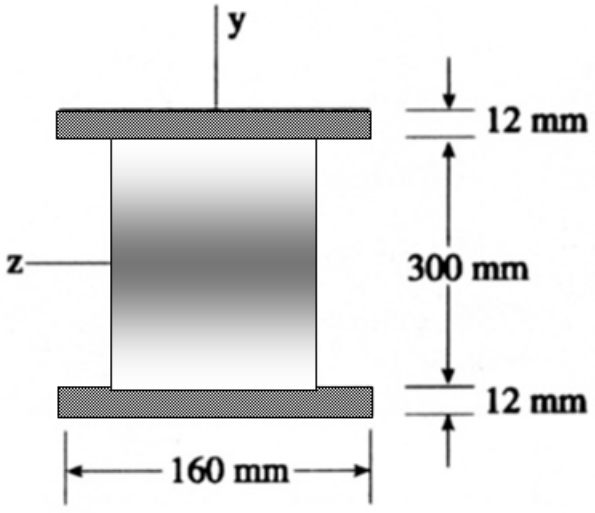
【台大應力所】

【中文翻譯】

木樑寬 200 mm 樑 300 mm，頂底以 12 mm 厚鋼板補強（ 見圖)。求對 z 軸之容許彎矩 M_{max} ；假定木之容許應力為 7 MPa，而鋼者則為 120 MPa。(假設鋼及木之彈性模數比率為 20)。

【解析】

(一)轉換斷面



$$n = E_S / E_W = 20$$

$$nb = (20)(200\,mm) = 4000\,mm$$

$$I_T = (1/12)(4\,m)(0.324\,m)^3 - (1/12)(4 - 0.2\,m)(0.3\,m)^3$$

$$= 0.00278\,m^4$$

(二)應力分析

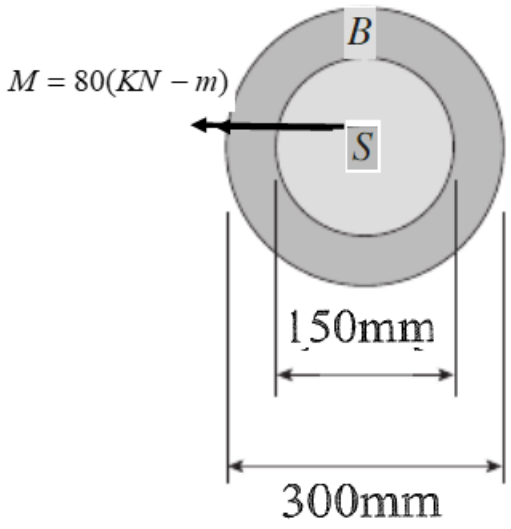
$$\sigma_1 = \frac{M(h_1/2)}{I_T}, M_1 = \frac{(\sigma_1)_{allow} I_T}{(h_1/2)} = 129\,kN \cdot m$$

$$\sigma_2 = \frac{M(h_2/2)n}{I_T}, M_2 = \frac{(\sigma_2)_{allow} I_T}{(h_2/2)n} = 103\,kN \cdot m$$

$$M_{max} = 103\,kN \cdot m$$

範例 11

A round brass tube of outside diameter $d_o = 300mm$ and a steel core of diameter $d = 150mm$ are bonded to form a composite beam, as shown in the figure($M = 80(KN - m)$).Determine the maximum stress in the steel and brass. (Assume that the moduli ofelasticity for the steel and brass are $E_s = 210GPa$ and $E_b = 70GPa$, respectively.)



【中文翻譯】

有一鋁圓空心管外直徑 $d_o = 300mm$ 包覆直徑 $d = 150mm$ 之鋼核心管接合而成複合樑，如圖所示 $M = 80(KN - m)$ 。求鋁及鋼所受之最大應力。(假設鋼與鋁之彈性模數分別為 $E_s = 210GPa$ 及 $E_b = 70GPa$)。

【解析】

(一)轉換斷面

模數比 $n = \dfrac{E_s}{E_a} = \dfrac{210}{70} = 3$

面積慣性矩 I_t

$$I_a = \frac{\pi}{64}[(0.3)^4 - (0.15)^4] = 3.73 \times 10^{-4}\,\mathrm{m}^4$$

$$I_s = \frac{\pi}{64}(0.15)^4 = 2.49 \times 10^{-5}\,\mathrm{m}^4$$

$$I_t = I_a + nI_s = 4.48 \times 10^{-4}\,\mathrm{m}^4$$

(二)應力分析

鋼：

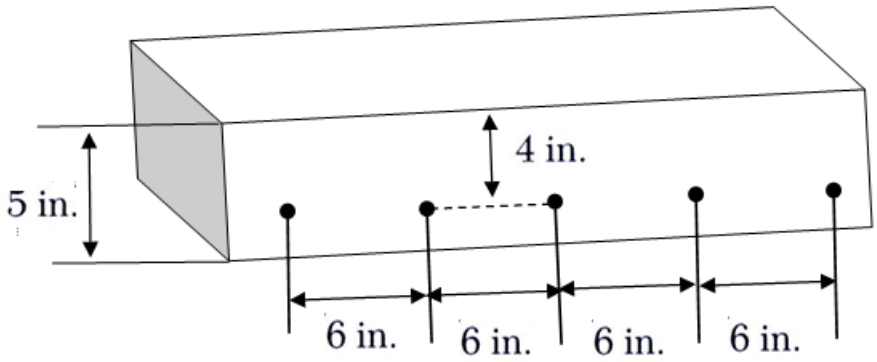
$$\sigma_{\max} = n \cdot \frac{Mc}{I_t} = 3 \times \frac{80 \times 10^3 \times (75 \times 10^{-3})}{4.48 \times 10^{-4}} = 40.2 \text{ MPa}$$

鋁：

$$\sigma_{\max} = \frac{Mc}{I_t} = \frac{80 \times 10^3 \times (150 \times 10^{-3})}{4.48 \times 10^{-4}} = 26.8 \text{ MPa}$$

範例 12

A concrete floor slab is reinforced by $\frac{5}{8}in$-diameter steel rods placed 1.5 in. above the lower face of the slab and spaced 6 in. on centers. The modulus of elasticity is $3.6\times10^6\ psi$ for the concrete used and $29\times10^6\ psi$ for the steel. Knowing that a bending moment of 40 (kip-in). is applied to each 1-ft width of the slab, determine (a) the maximum stress in the concrete, (b) the stress in the steel.



【台大土木】

【中文翻譯】

長方形斷面之鋼筋混凝土梁由$\frac{5}{8}in$直徑的鋼筋加固定，鋼筋位於鋼坯下表面深1.5英寸處，中心間距為6英寸，混凝土梁彈性模數$3.6\times10^6\ psi$、鋼筋彈性模數$29\times10^6\ psi$，已知鋼筋混凝土梁每1ft承受彎矩M = 40 (kip-in)，試求鋼筋及混凝土內之最大應力。

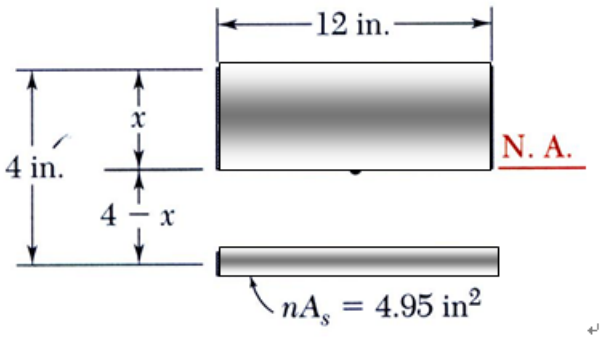
【解析】

水泥無法承受拉應力只能承受壓應力。拉應力的部分全部由鋼筋承受，故受拉側不考慮水泥的作用。

(一)轉換斷面

$$n=\frac{E_s}{E_c}=\frac{29\times10^6\ \text{psi}}{3.6\times10^6\ \text{psi}}=8.06$$

$$nA_s=8.06\times2\left[\frac{\pi}{4}\left(\frac{5}{8}\text{in}\right)^2\right]=4.95\text{in}^2$$

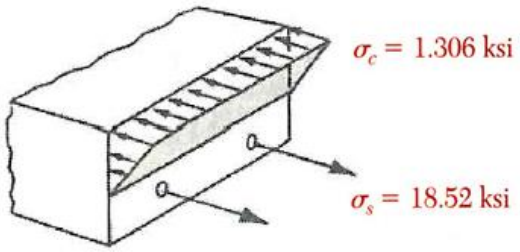


$$12x\left(\frac{x}{2}\right) - 4.95(4-x) = 0 \qquad x = 1.450\,\text{in}$$

$$I = \tfrac{1}{3}(12\,\text{in})(1.45\,\text{in})^3 + \left(4.95\,\text{in}^2\right)(2.55\,\text{in})^2 = 44.4\,\text{in}^4$$

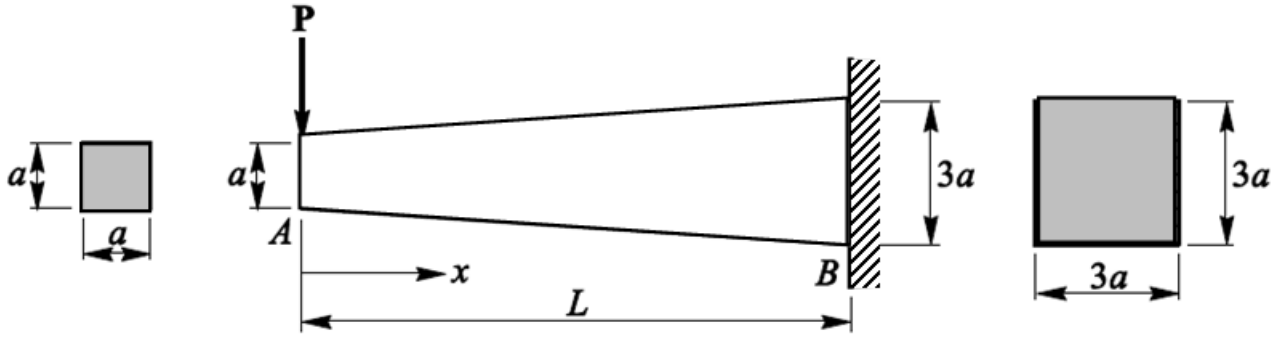
$$\sigma_c = \frac{Mc_1}{I} = \frac{40\,\text{kip}\cdot\text{in}\times 1.45\,\text{in}}{44.4\,\text{in}^4}$$

$$\sigma_s = n\frac{Mc_2}{I} = 8.06\frac{40\,\text{kip}\cdot\text{in}\times 2.55\,\text{in}}{44.4\,\text{in}^4}$$



範例 13

A tapered cantilever beam AB of length L has square cross sections and supports a concentrated load P at the free end (see figure on the next page). The width and height of the beam vary linearly from a at the free end to 3a at the fixed end. (a)Determine the distance x from the free end A to the cross section of maximum bending stress. (a)What is the magnitude σ_{max} of the maximum bending stress? (b)What is the ratio of the maximum stress to the largest stress σ_B at the support?

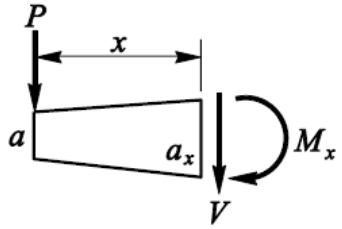


【中文翻譯】

圖中斷面為正方形之錐形懸臂梁，斷面之寬度由自由端之a直線變化至固定端之$3a$，今在自由端承受一集中負荷P，試求(a)最大彎曲應力σ_{max}及其位置(與之自由端之距離)，(b)若固定端斷面上之最大彎曲應為為σ_B，則σ_{max}/σ_B之比值為何？

【解析】

(一)內力分析



$$M_x = Px$$

(二)斷面分析

$$a_x = a\left(1+\frac{2x}{L}\right)$$

$$S_x = \frac{a^3}{6}\left(1+\frac{2x}{L}\right)^3$$

(三)應力分析

$$\sigma_x = \frac{M_x}{S_x} = \frac{Px}{\frac{a^3}{6}\left(1+\frac{2x}{L}\right)^3}$$

令 $\frac{d\sigma_x}{dx} = 0$ ，得 $x = \frac{L}{4}$ ，代入①式得 $\sigma_{\max} = \frac{4PL}{9a^3}$

(b)固定端之最大彎曲應力： $\sigma_B = \frac{M}{S} = \frac{PL}{\frac{(3a)^3}{6}} = \frac{2PL}{9a^3}$

故 $\frac{\sigma_{\max}}{\sigma_B} = 2$

10-3.樑的剪應力

一、樑的剪應力

樑的剪應力

樑受力彎曲時，樑內不僅有彎矩還有剪力，因而橫截面上既有彎曲正應力，又有彎曲剪應力，以下根據不同斷面分析其彎曲剪應力，首先考慮一橫斷面為長方形之樑，如圖12.5所示，此樑某一橫斷面上之剪力為V，將此橫斷面分成許多與Z軸平行之極細長條，在各細長條上由剪力V誘生與X軸平行之剪應力可視作為均佈於此細長條上，在之前章節已說明兩互成正交斷面上之餘剪應力，必等值反向，同理此元件其餘三面上必伴生等值之剪應力，因而樑在各橫斷間具有剪應力外，樑內與中立面平行之各層間亦具等值之剪應力，可以推導出矩形截面上距中性軸爲y處任意點的剪應力計算公式爲：

$$\tau = \frac{VQ}{I_z b}$$

式中　V：橫截面上的剪力、I_z：橫截面對中性軸的軸慣性矩

b：橫截面上所求剪應力點處截面的寬度（即矩形的寬度）

Q：分離面之面積對中性軸之一次矩

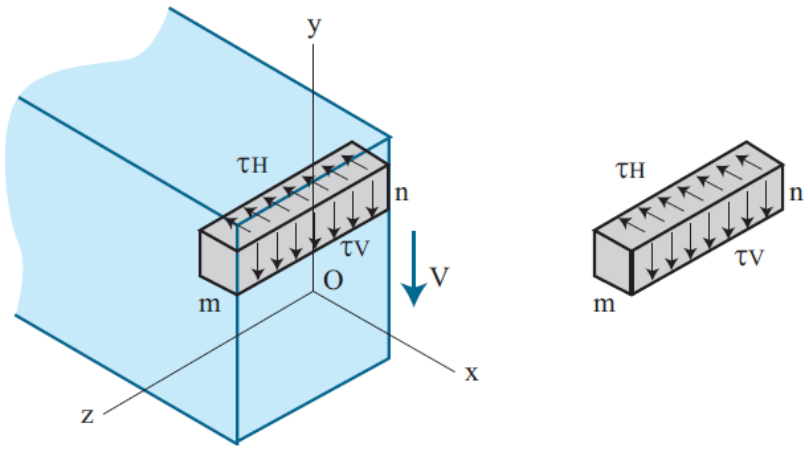


圖12.5　樑的剪應力

二、樑的剪應力證明

在梁上取微段dx，在微段上再取一塊如圖，列平衡方程：

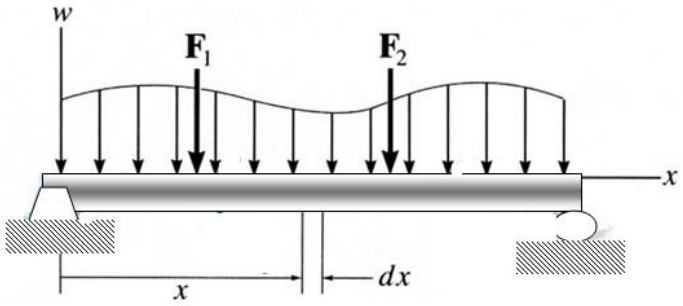
$$\sum F_x = N_2 - N_1 - \tau' b dx = 0 \quad (1)$$

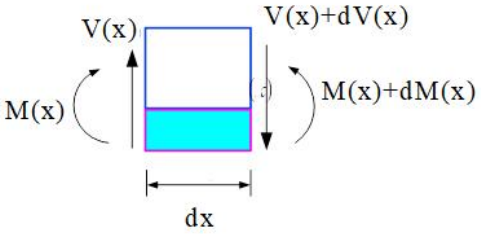
$$N_1 = \int_{A_1} \sigma_1 dA = \frac{M}{I_z}\int_{A_1} y\, dA = \frac{MQ}{I_z} \quad (2)$$

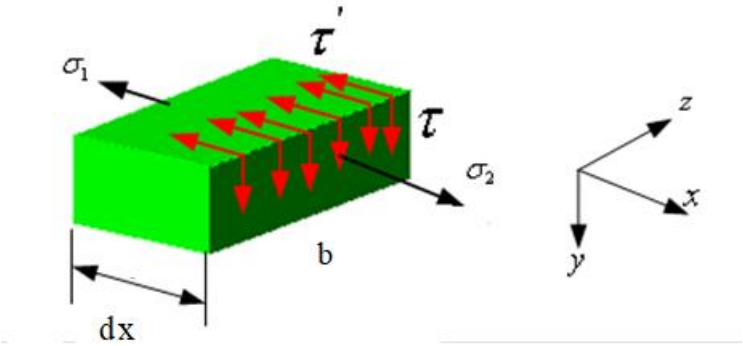
$$N_2=\int_{A_1}\sigma_2 dA=\frac{(M+dM)}{I_z}\int_{A_1} y\ dA=\frac{(M+dM)Q}{I_z}\quad (3)$$

（3）帶入（1）、（2）得 $\tau'=\frac{dM}{dx}\frac{\mathrm{Q}}{bI_z}$

$$\tau=\tau'=\frac{dM}{dx}\frac{\mathrm{Q}}{bI_z}=\frac{\mathrm{VQ}}{\mathrm{I}b}$$







三、各種斷面的剪應力

(一)矩形斷面

如圖所示矩形截面上取微面積 dA＝bdy，則距中性軸爲 y 的横線以下的面積 A^* 對中性軸 z 的一次矩爲

$$\mathrm{Q}=\int_{\mathrm{A}^*} y_1 dA=\int_y^{\frac{b}{2}} b y_1 d_{y_1}=\frac{b}{2}\left(\frac{h^2}{4}-y^2\right)$$

將此式代入剪應力公式，可得矩形截面剪應力計算公式的具體運算式爲

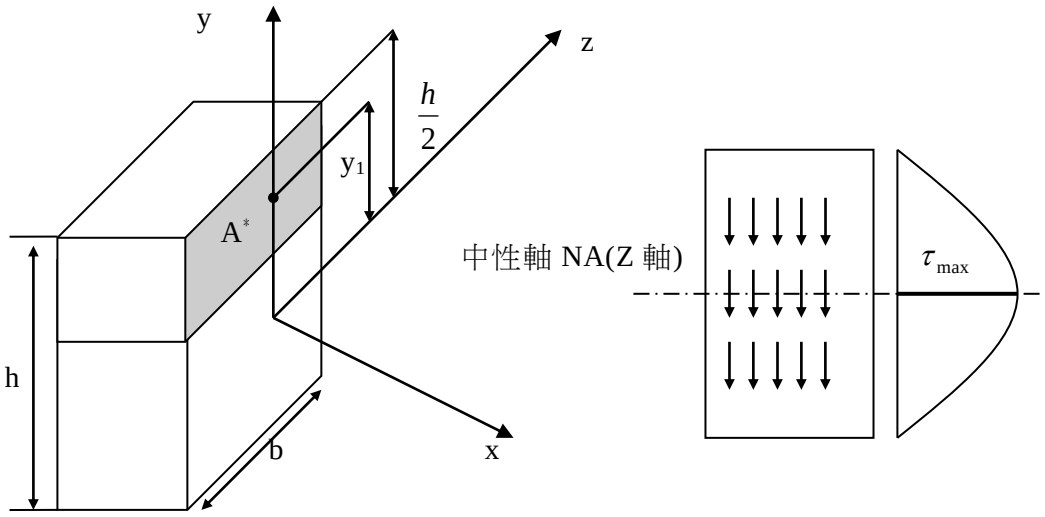
$$\tau=\frac{V}{2I_z}\left(\frac{h^2}{4}-y^2\right)$$

當 $y=\pm\frac{1}{2}h$ 時，即矩形截面的上、下邊緣處剪應力 $\tau=0$；當 y=0 時，截面中性軸上的剪應力爲最大值：

$$\tau_{\max}=\frac{Vh^2}{8I_z}\Rightarrow\boxed{\tau_{\max}=\frac{3V}{2bh}=\frac{3V}{2A}}$$

直接用彎曲剪應力公式：

$$\boxed{\tau=\frac{VQ}{I_z b}=\frac{V(h/4)(bh/2)}{[\frac{1}{12}bh^3]b}=\frac{3V}{2bh}=\frac{3V}{2A}}$$ 與上面分析相同



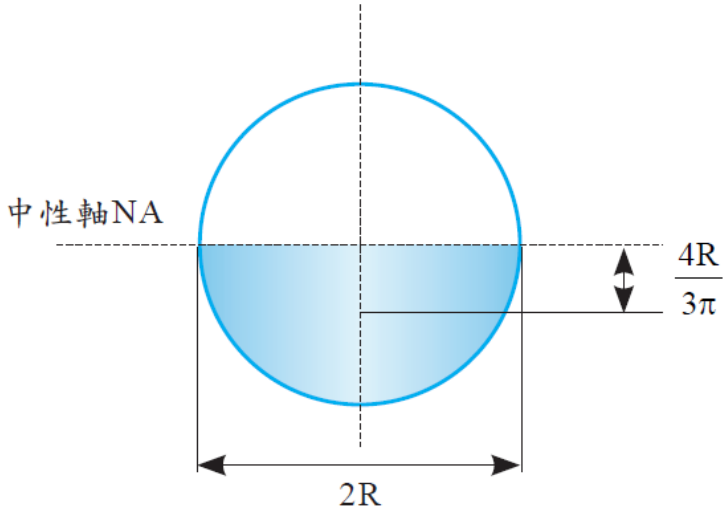
(二)圓形斷面

圓形斷面剪應力

圓形截面上，任一平行於中性軸的橫線，剪應力的方向必切於圓周，並相交於y軸上，因此在中性軸上各點剪應力的方向皆平行於剪力V，剪應力值為最大，直接用彎曲剪應力公式：

$$Q=\frac{\pi R^2}{2}\times\frac{4R}{3\pi}=\frac{2}{3}R^3 \text{ 、 } b=2R \text{ 、 } I_z=\frac{\pi R^2}{4}$$

$$\tau_{max}=\frac{VQ}{I_z b}=\frac{4}{3}\frac{V}{\pi R^2}=\frac{4V}{3A}$$



(三)I 字形斷面

工字形截面可以看作由三個矩形截面組成，因此其彎曲剪應力計算與矩形截面梁類似仍然沿用矩形截面梁彎曲剪應力計算公式 $\tau=\frac{VQ}{I_z b}$

$$Q=B\left(\frac{H}{2}-\frac{h}{2}\right)\left[\frac{h}{2}+\frac{1}{2}\left(\frac{H}{2}-\frac{h}{2}\right)\right]+b\left(\frac{h}{2}-y\right)\left[y+\frac{1}{2}\left(\frac{h}{2}-y\right)\right]$$

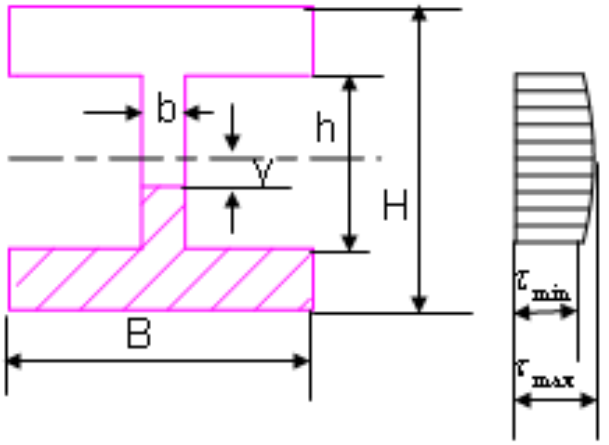
將此式代入彎曲剪應力公式，可得腹板上彎曲剪應力的計算公式

$$\tau=\frac{V}{I_z b}\left[\frac{B}{8}\left(H^2-h^2\right)+\frac{b}{2}\left(\frac{h^2}{4}-y^2\right)\right]$$

將 $y=0$ 時，在截面中性軸上時，在腹板與翼緣的交 $y=\pm\frac{h}{2}$ 帶入上式，得

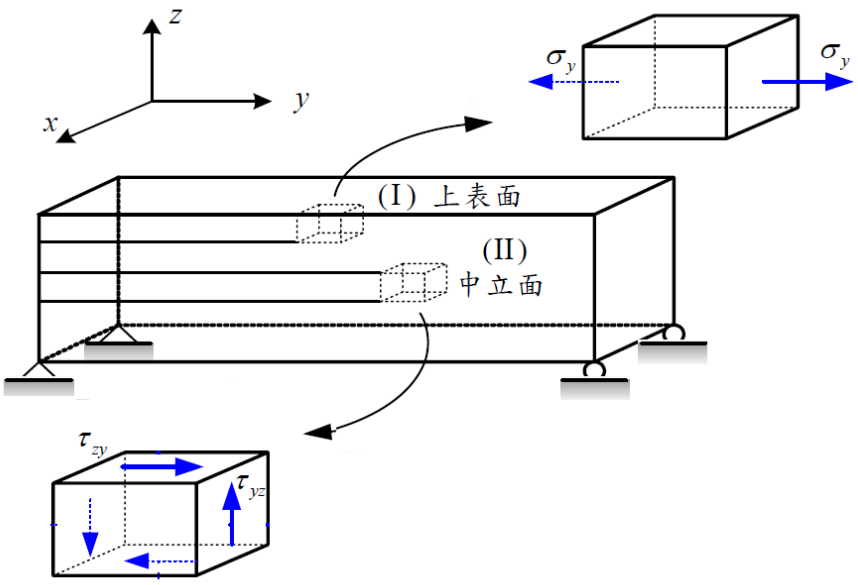
$$\tau_{max}=\frac{V}{I_z b}\left[\frac{BH^2}{8}-(B-b)\frac{h^2}{8}\right]$$

$$\tau_{min}=\frac{Q}{I_z b}\left[\frac{BH^2}{8}-\frac{Bh^2}{8}\right]$$



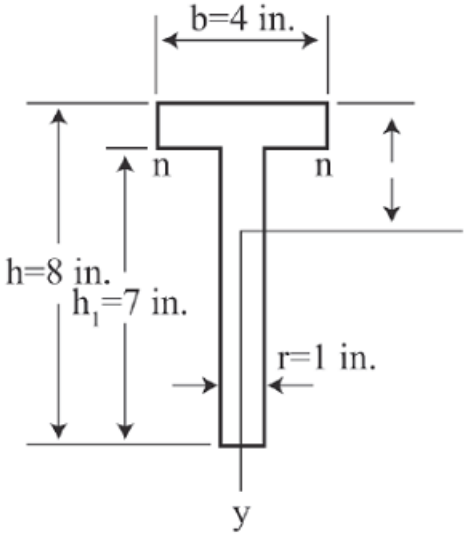
【觀念說明】

1. 如圖自由表面，就是完全沒有應力作用的面，若我們在上表面割一元素如(I)：很明顯的圖中的 z 面是自由表面，所以正交應力和剪應力皆為零 $\sigma_z = \tau_{zx} = \tau_{zy} = 0$，而 x 面也是自由表面，所以 $\sigma_x = \tau_{xz} = \tau_{xy} = 0$
2. y 面不是自由表面，所以可以存在應力(σ_y 、 τ_{yz} 、 τ_{yx})，但是因為應力的對稱性 $\tau_{xy} = \tau_{yx} = 0 = \tau_{yz} = \tau_{zy}$，所以最後只有 σ_y 可以存在。
3. 中立面上割一元素如(II) ，在中立軸上，沒有彎曲正交應力，但有水平剪應力。
4. 對於樑的問題，在上下表面只有彎曲正交應力，沒有水平剪應力，此時彎曲應力值最大。相反的，此時水平剪應力值最大，若元素落在中立面與上下表面間，則同時具有剪應力與正交應力。



範例 14

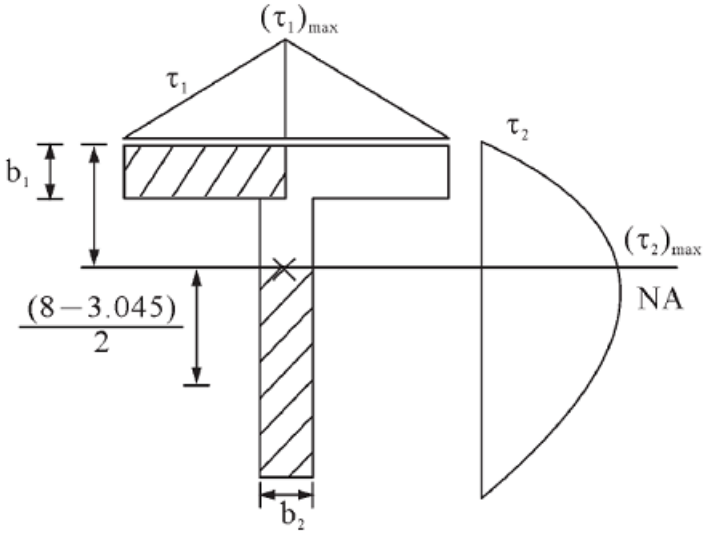
已知某一樑結構之T形截面，截面慣性矩(moment of inertia) I = 69.66 in^4，如圖所示，其形狀中心的位置 c = 3.045 inch，受到10000lb的剪力(shear force)，試問：該樑在截面的最大剪應力？



【中山機械】

【解析】

(C)。

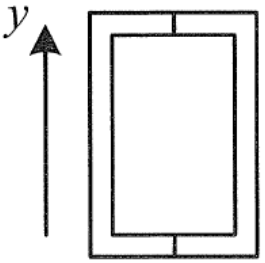


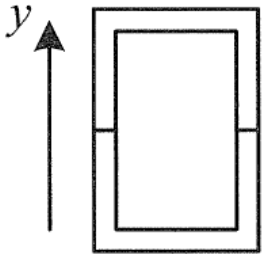
$$(\tau_1)_{max} = \frac{VQ_1}{Ib_1} = \frac{10000\times[1\times2\times(3.045-0.5)]}{69.66\times1} = 730.7\ (psi)$$

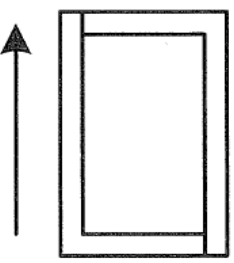
$$(\tau_2)_{max} = \frac{VQ_2}{Ib_2} = \frac{10000\times[1\times(8-3.045)\times(\frac{8-3.045}{2})]}{69.66\times1} = 1762.28(psi)$$

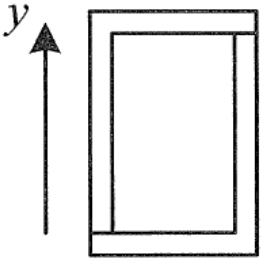
範例 15

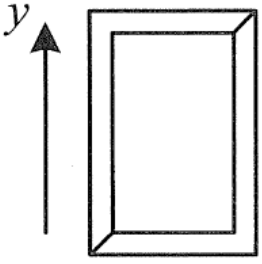
由兩根完全相同的型鋼熔接而成一根樑，當此樑承受負 y 軸方向的外力時，請指出對熔接強度要求最高的斷面。

(A) 

(B) 

(C) 

(D) 

(E) 

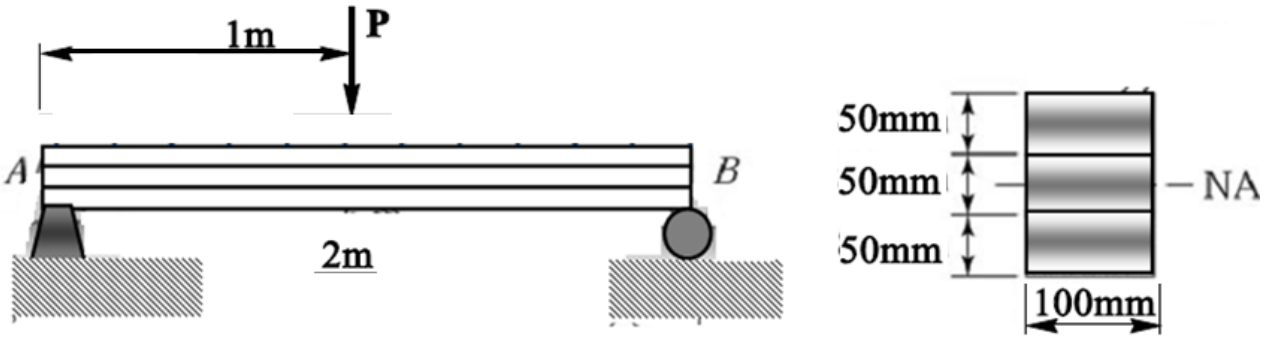
【台大機械】

【解析】

箱型斷面在中性軸之剪應力最大故選 B

範例 16

A laminated wood beam on simple supports is built up by gluing together three 50×100mm . boards (actual dimensions) to form a solid beam 100 $\times150$ mm . in cross section, as shown in the figure. The allowable shear stress in the glued joints is 0.35 MPa and the allowable bending stress in the wood is11 MPa. If the beam is 2 m long, what is the allowable load P acting at the midpoint of the beam? (Disregard the weight of the beam.)



【中央機械】

【中文翻譯】

一疊層木樑受簡支，由三片 50×100mm （ 實際尺寸 ） 板子膠合成堅固的樑，橫截面為100 $\times150$ mm ，如圖所示。膠合接面的容許剪應力為0.35 MPa，且木材的容許彎曲應力為11 MPa。如果樑2 m長，作用於樑中點的容許負載 P 為何？(忽略樑的自重)

【解析】

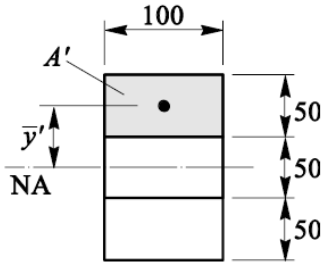
(一)內力分析

$$V_{\max}=\frac{P}{2}\ \text{N}\cdot\ M_{\max}=\frac{PL}{4}=\frac{P(2000)}{4}=500P\ \text{N-mm}$$

(二)斷面分析

$$I=\frac{1}{12}(100)(150)^3=28.125\times10^6\ \text{mm}^4$$

(三)考慮膠接面之容許剪應力



$$Q = A'\bar{y}' = (100\times 50)(50) = 0.250\times 10^6\ \text{mm}^3$$

$$\tau = \frac{VQ}{Ib}\ ,\ 0.35 = \frac{\left(\frac{P}{2}\right)(0.250\times 10^6)}{(28.125\times 10^6)(100)}\ ,\ P = 7875\ \text{N}$$

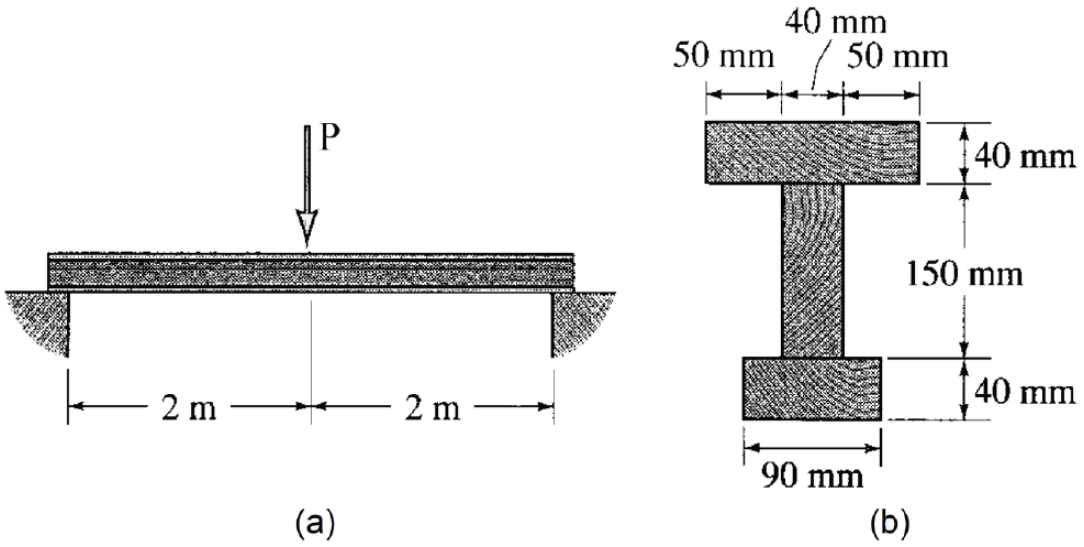
(四)考慮木材之容許彎曲應力

$$M = \sigma S\ ,\ 500P = (11)\left(\frac{1}{6}\times 100\times 150^2\right)\ ,\ P = 8250\ \text{N}$$

故容許負荷 $P = 7875$ N(取較小者)

範例 17

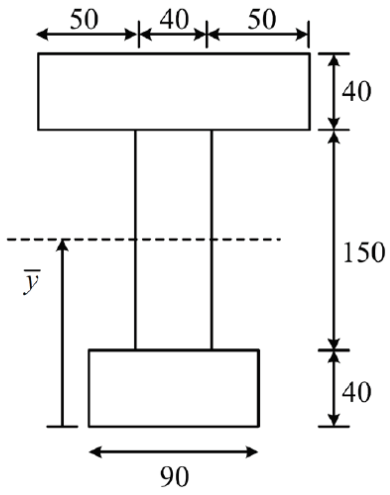
如圖(a)之簡支樑在跨距中點承受集中負載P，樑截面如圖(b)由三塊矩形於接合面膠合而成，若膠合處之容許剪應力為600kPa，在不計樑本身自重下，試求在膠合處剪強度(shear strength)考量下所能容許之最大負載P。



【103 地特四等】

【解析】

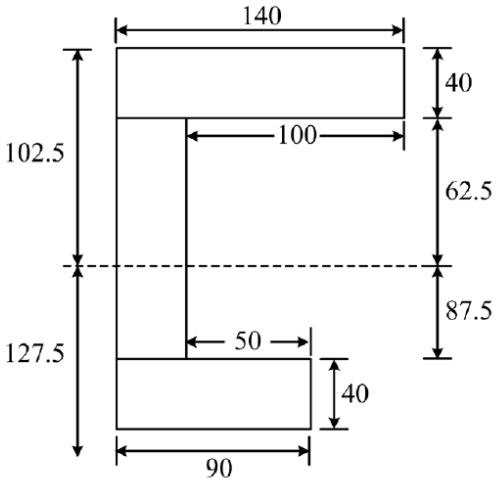
答：



$$\bar{y}=\frac{90\times40\times\frac{40}{2}+150\times40\times(\frac{150}{2}+40)+140\times40\times(190+\frac{40}{2})}{90\times40+150\times40+140\times40}$$

$$=\frac{1938000}{15200}=127.5(\text{mm})$$

$$\bar{y}=40+40+150-127.5=102.5(\text{mm})$$



$$I=\frac{1}{3}\times 140\times 102.53-\frac{1}{3}\times 100\times 62.53+\frac{1}{3}\times 90\times 127.53-\frac{1}{3}\times 50\times 87.53$$

$$=93131666(mm^4)$$

$$\text{上接合面}Q=140\times 40\times(62.5+\frac{40}{2})=462000(mm^3)$$

$$\text{下接合面}Q=90\times 40\times(127.5-\frac{40}{2})=387000(mm^3)$$

上接合面

$$\tau=\frac{VQ}{It}\le\tau_{all}\Rightarrow\frac{\frac{P}{2}\times 462000}{93131666\times 40}\le 0.6\Rightarrow P\le 9676(N)$$

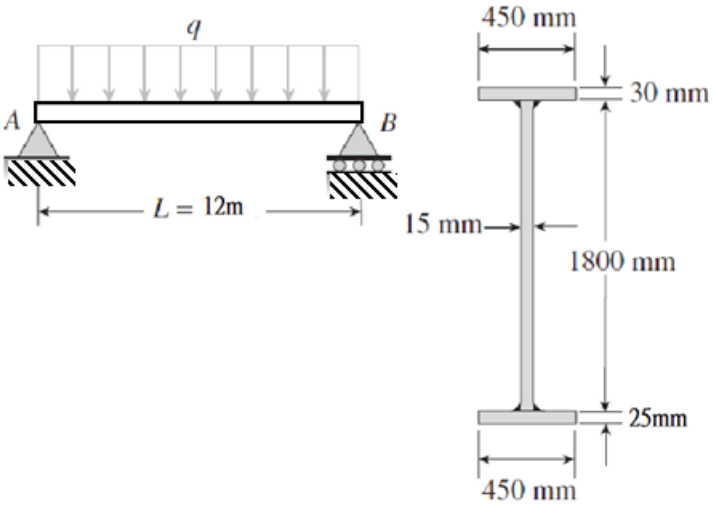
下接合面

$$\tau=\frac{VQ}{It}\le\tau_{all}\Rightarrow\frac{\frac{P}{2}\times 387000}{93131666\times 40}\le 0.6\Rightarrow P\le 11551(N)$$

取$P=9676(N)$

範例 18

A bridge girder AB on a simple span of length L=12m supports a distributed load of intensity qthat includes the weight of the girder (see fig). The girder is constructed of three plates welded to form the cross section in the fig. Determine the maximum permissible load q based on, (a) and allowable bending stress $\sigma_{allow} = 90\text{MPa}$, and b) an allowable shear stress $\tau_{allow} = 50MPa$.



【100 台大應力】

【中文翻譯】

橋的樑 AB 簡支跨距長 $L = 12$ m ，支持包括自重強度 q 的均勻負載。樑由三片焊接而成如圖橫截面；基於如下求最大許可負載：(a) 容許彎曲應力 $\sigma_{allow} = 90\text{MPa}$ 及 (b) 容許剪應力 $\tau_{allow} = 50MPa$ 。

【解析】

(一)斷面分析

$$I = \frac{1}{12}(bh^3 - bh_1^3 + th_1^3) = 26.03 \times 10^{-3}\ m^4$$

$$S = I/c = 28.14 \times 10^{-3}\ m^3$$

(二)應力分析

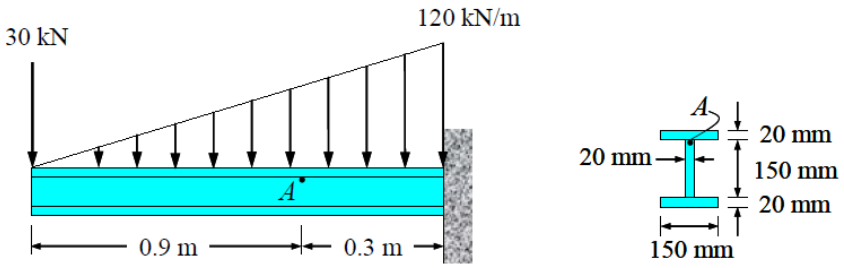
$$M_{max} = \frac{qL^2}{8} = \sigma_{allow} S$$

$$q_{allow} = \frac{8S\sigma_{allow}}{L^2} = 141\ kN/m$$

$$\tau_{allow} = \frac{VQ}{Ib} \Rightarrow 50 = \frac{(\frac{1}{2}q \times 12000)(450 \times 25 \times 912.5 + 900 \times 15 \times 450)}{I \times 15} \Rightarrow q = 199.1(N/mm)$$

範例 19

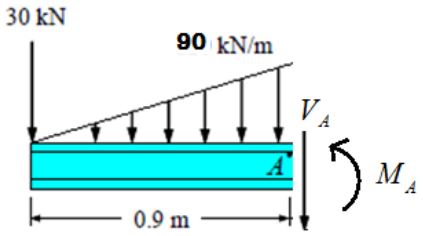
一懸臂梁右端固定，受力狀況及其截面之形狀與尺寸如下圖所示：
㈠試求圖中 A 點處之彎曲應力及剪應力，並繪製其應力元素圖。（12 分）
㈡試求圖中 A 點處的主應力，並繪製其應力元素圖。（8 分）



【105 地特三等】

【解析】

(1)在 A 處分布力值為 $90(kN/m)$



$$\sum F_Y = 0 \Rightarrow V_A + \frac{1}{2}\times 0.9\times 90 + 30 = 0 \Rightarrow V_A = -70.5(kN)$$

$$\sum M_A = 0 \Rightarrow 30\times 0.9 + \frac{1}{2}\times 90\times 0.9\times(\frac{1}{3}\times 0.9) + M_A = 0 \Rightarrow M_A = -39.15(kN-m)$$

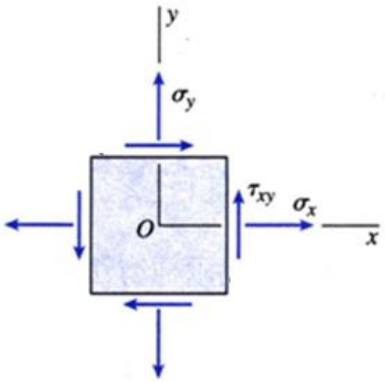
(2)

$$I = \frac{1}{12}\times 150\times 190^3 - \frac{1}{12}\times(150-20)\times 150^3 = 49175000(mm^4)$$

$$\sigma_A = \frac{My}{I} = \frac{39.15\times 10^3\times 10^3\times 75}{49175000} = 59.71(Mpa) = \sigma_x$$

$$\tau_A = \frac{VQ}{It} = \frac{70.5\times 10^3\times[(150\times 20)\times(10+75)]}{49175000} = 18.28(Mpa) = \tau_{xy}$$

$$\sigma_y = 0$$

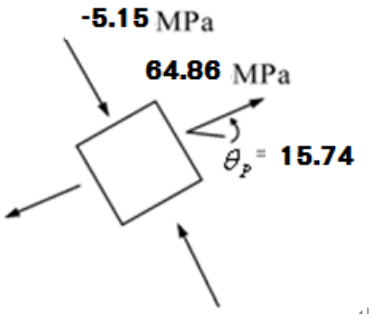


（3）

$$\sigma_{1,2}=\frac{\sigma_x+\sigma_y}{2}\pm\sqrt{(\frac{\sigma_x-\sigma_y}{2})^2+\tau_{xy}^2}=64.86/-5.15(Mpa)$$

$$\theta_P=\frac{1}{2}\tan^{-1}(\frac{2\tau_{xy}}{\sigma_x-\sigma_y})=\frac{1}{2}\tan^{-1}(\frac{2\times18.28}{59.71})=15.74^0$$

$$\sigma_{\theta=15.74}=\frac{\sigma_x+\sigma_y}{2}+\frac{\sigma_x-\sigma_y}{2}\cos2\theta+\tau_{xy}\sin2\theta=64.86(Mpa)=\sigma_1$$



10-4.樑的剪力流及剪力中心

一、剪力流

(一)剪力流

剪力流是指當梁內彎矩有變化時（即非純彎曲），斷面內將出現剪力，同時，在梁內沿軸線方向也會有相應的剪力存在。此種在梁內沿軸線方向單位長度的剪力稱為“剪力流（shear flow）***q:***剪力流為單位長度的剪力，單位 N/m。

剪力流公式：

$$q=\frac{VQ}{I}$$

(二)接合面的剪應力

剪力流為沿樑中縱軸方向之每單位長度所量測的力，其值可由剪力公式求得，且是用來獲知樑中將不同區段組合用之連接件與膠合面中，所產生的剪力大小，如下圖所示，上部橫元件接縫的剪力流為:

1.斷面慣性矩：

$$\boldsymbol{I=\frac{1}{12}\left(bh^3-(b-2t)(h-2t)^3\right)}$$

$$\boldsymbol{Q=(b-2t)t\left(\frac{1}{2}\left(\frac{h}{2}+\left(\frac{h}{2}-t\right)\right)\right)}$$

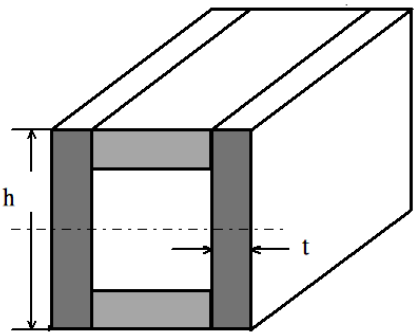
$$\boldsymbol{=\frac{1}{2}t(b-2t)(h-t)}$$

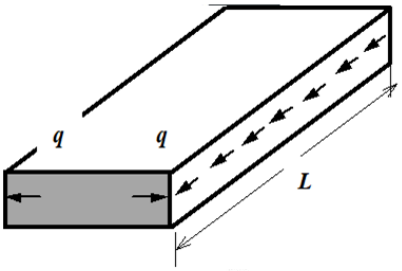
接縫有兩邊；每一邊的剪力流為

$$q=\frac{1}{2}\frac{VQ}{I}$$

若為膠合面；膠合面的剪力$\boldsymbol{V_G}$

$V_G=qL$





(三)**薄壁構件內剪力流**

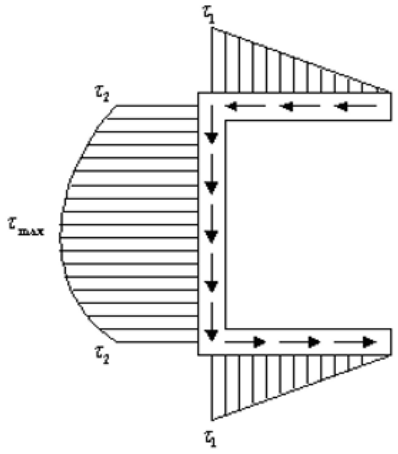
薄壁元件剪力流特性：

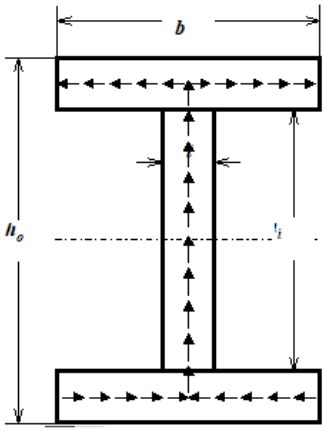
(1)剪力流起於流源(source)，集於流匯(sink)。在流源與流匯上為零。

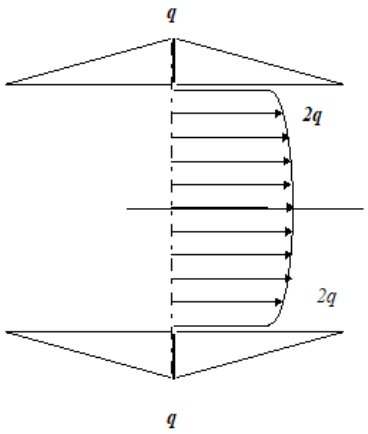
(2)合流為各合成支流的代數和。

(3)總向量和的方向為斷面剪力的方向。

(4)薄壁構件內剪力流下圖所示

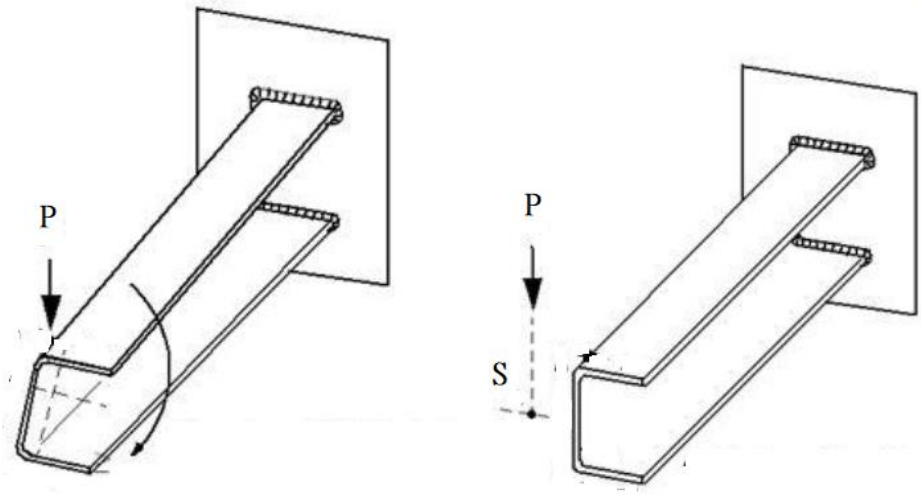






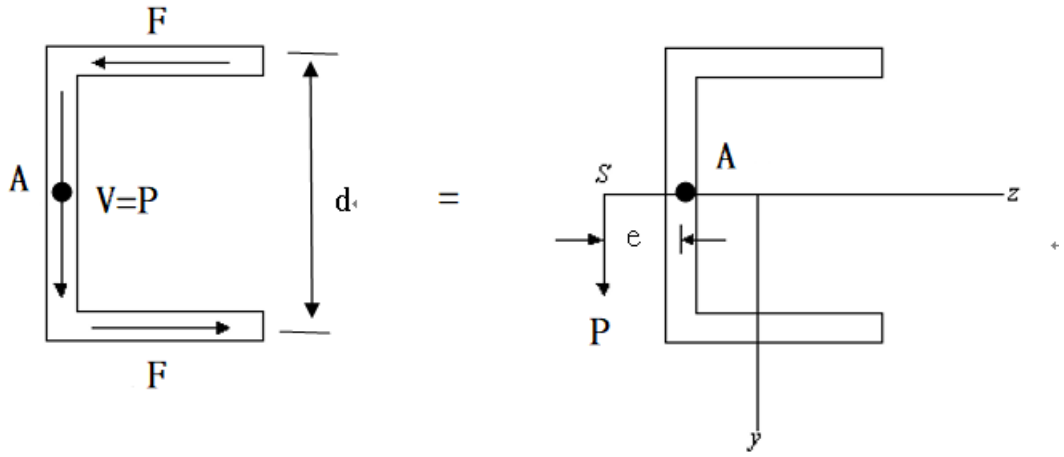
二、剪力中心 (shear center)

剪力彎曲時，梁的橫截面上不僅有正應力還有剪應力，對於有對稱截面的梁，當外力作用在形心主慣性平面內時，剪應力的合力，即剪力作用線通過形心，梁發生平面彎曲，對於非對稱截面（特別是薄壁截面）梁，橫向外力即使作用在形心主慣性平面內，剪應力的合力作用線並不一定通過截面形心，當剪應力的合力作用線不通過截面形心時，梁不僅發生彎曲變形，而且還將產生扭轉，如圖所示，只有當橫向力作用在某一特定點 S 並和形心主慣性平面平行時，該梁才只產生平面彎曲而無扭轉，如圖所示，這一特定 S 稱為剪力中心 (shear center) 或彎曲中心 (flexural center)。



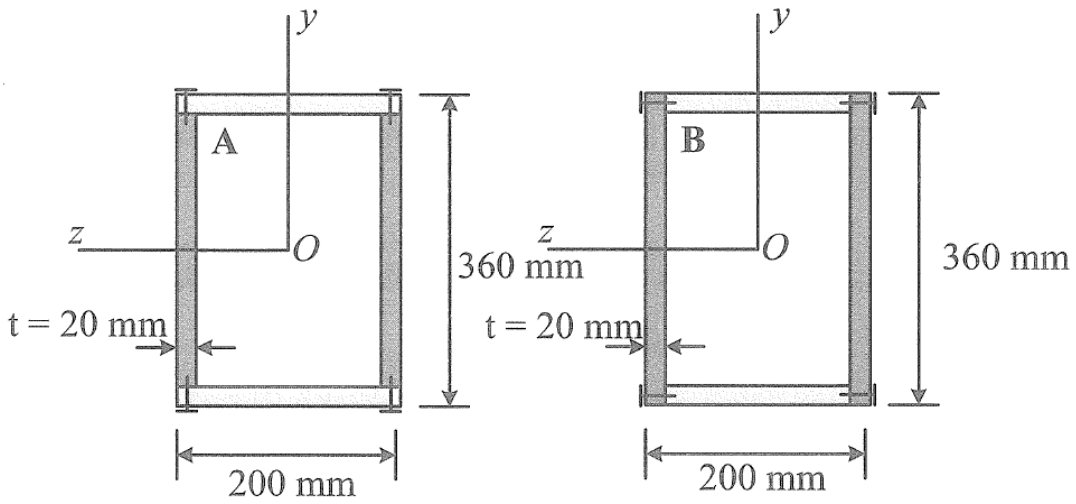
P 作用在距槽型腹板 e 之點 S 上，將此分佈對整個翼緣及腹板面積積分，則可得各翼緣內合力 F 及腹板內 V = P 若，對 A 點取力矩和，由翼緣力產生之力偶或扭矩乃造成構件之扭轉，如圖所示，因平衡反力所產生之扭轉，為阻止此扭轉而形成平衡:

$$\sum \mathrm{M}_{\mathrm{A}} = 0 \Rightarrow F \times d = Pe$$



範例 20

Two wood box beams (beams A and B) have the same outside dimensions $(200\times360\text{mm})$ and the same thickness $t=20\text{mm}$ throughout, as shown in the figure on the next page. Both beams are formed by nailing, with each nail having an allowable shear load of 250 N. The beams are designed for a shear force $V=3.2\text{kN}$. (a) What is the maximum longitudinal spacing s_A for the nails in beam A? (b) What is the maximum longitudinal spacing s_B for the nails in beam B? (c) Which beam is more efficient in resisting the shear force?



【成大航太】

【中文翻譯】

兩木盒形樑（A及B）具相同外尺寸$(200\times360\text{mm})$（厚$t=20\text{mm}$），如圖所示。均由釘子所結合，每支釘子的容許剪負載為 250 N。樑設計為了剪力$V=3.2\text{kN}$。(a) 樑A釘子的長度方向最大間距s_A為多大？(b) 樑B釘子的長度方向最大間距s_B為多大？(c) 那種（A或B）樑對抵抗剪力較具效率？

【解析】

(一)斷面分析

$$b=200 \quad b_1=200-2(20)=160$$
$$h=360 \quad h_1=360-2(20)=320$$
$$t=20$$

$$I=\frac{1}{12}(bh^3-b_1h_1^3)=340.69\times10^6\ \text{mm}^4$$

(二)計算s_A

$$Q=A_pd_p=(bt)\left(\frac{h-t}{2}\right)=(200)(20)\left(\frac{1}{2}\right)(340)$$
$$=680\times10^3\ \text{mm}^3$$

剪力流：$q=\frac{VQ}{I}=\frac{(3200)(100\times20\times170)}{I}$

$$s_A=\frac{250}{q}=78.3(\text{mm})$$

(三)計算 s_B

$$Q=A_fd_f=(b-2t)(t)\left(\frac{h-t}{2}\right)=(160)(20)\frac{1}{2}(340)$$
$$=544\times10^3\,\text{mm}^3$$

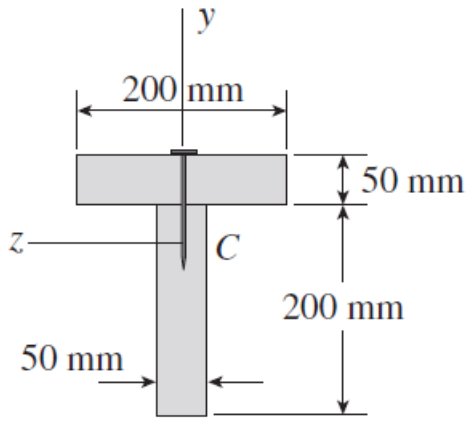
剪力流：$q=\frac{VQ}{I}=\frac{(3200)(80\times20\times170)}{I}$

$$s_B=\frac{250}{q}=97.9(\text{mm})$$

因爲 $s_B>s_A$，所以 B 梁之抗剪能力較佳。

範例 21

A beam of T cross section is formed by nailing together two boards having the dimensions shown in the figure. If the total shear force V acting on the cross section is 1800 N and each nail may carry 800 N in shear, what is the maximum allowable nail spacing s?

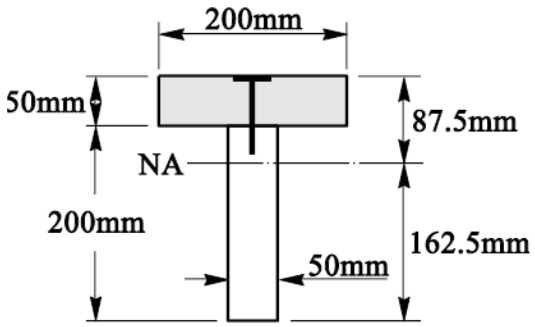


【中文翻譯】

圖中 T 型斷面之組合梁，已知斷面所承受之剪力 V = 1800 N。設每根釘子之容許剪力為 800 N，試求釘子之最大容許間距。

【解析】

(一)斷面分析



$$\bar{y} = c_2 = \frac{(200\times 50)(225) + (50\times 200)(100)}{200\times 50 + 50\times 200} = 162.5 \text{ mm}$$

$$c_1 = 250 - c_2 = 87.5 \text{ mm}$$

$$I = \frac{1}{3}(200)(50)^3 + \frac{1}{3}(50)(200)^3 - (200\times 100)(87.5-50)^2$$

$$= 113.5 \times 10^6 \text{ mm}^4$$

(二)計算 s

$$Q = (200 \times 50)(87.5 - 25) = 0.625 \times 10^6 \text{ mm}^3$$

$$f = \frac{VQ}{I} = \frac{(1800)(0.625 \times 10^6)}{113.5 \times 10^6} = 9.91 \text{ N/mm}$$

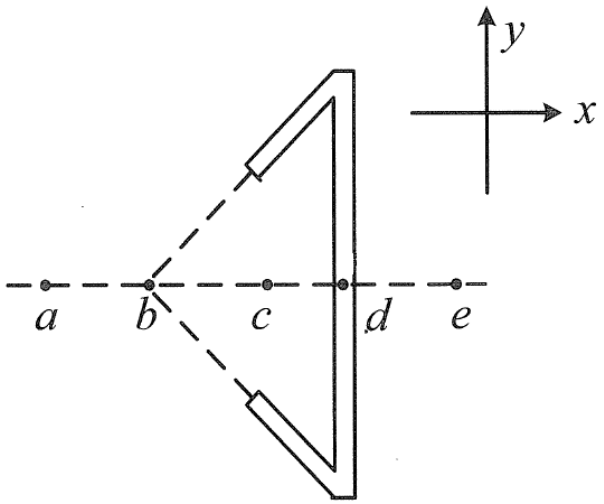
$$s = \frac{F}{f} = \frac{800}{9.91} = 80.7 \text{ mm}$$

範例 22

圖爲某懸臂梁的截面，當在正 y 軸方向施加外力且通過 d 點時，請指出此斷面的位移方式。

(A) 向上並且順時針旋轉　(B) 向上並且逆時針旋轉　(C) 向右上平移不旋轉

(D) 向左上平移不旋轉　(E) 以上皆非



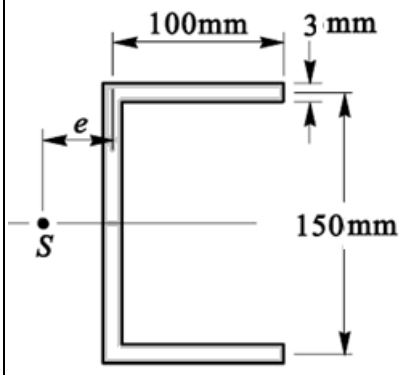
【台大機械】

【解析】

(A)

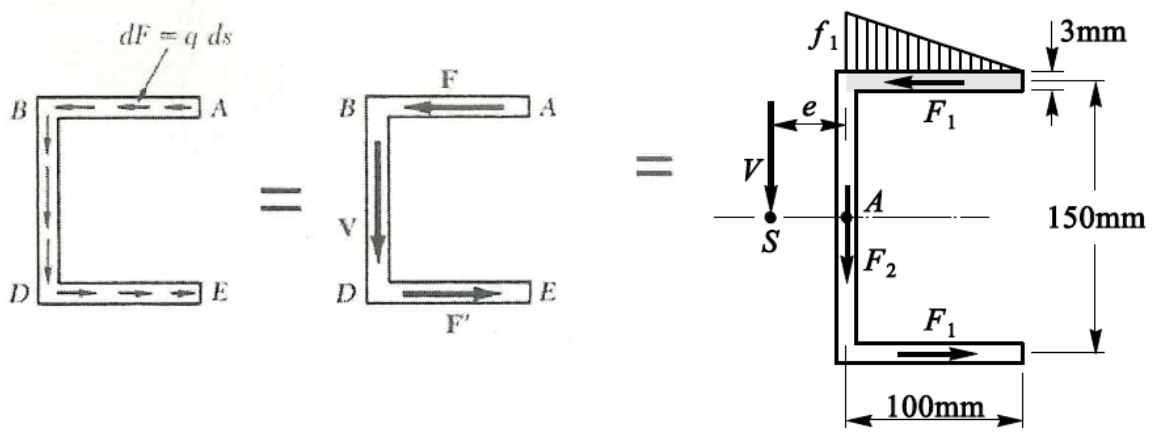
範例 23

圖中厚度均勻之槽型薄壁斷面，試求其剪力中心之位置。若剪力中心承受 800N 之垂直剪力，試求最大剪應力。



【106 成大土木】

【解析】



慣性矩：$I = \frac{1}{12}(3)(150)^3 + 2[(100 \times 3)(75)^2] = 4.219 \times 10^6 \text{ mm}^4$

$Q_1 = (100 \times 3)(75) = 22500 \text{ mm}^3$

$f_1 = \frac{VQ_1}{I} = \frac{(800)(22500)}{4.219 \times 10^6} = 4.266 \text{ N/mm}$

$F_1 = \frac{1}{2}(4.266)(100) = 213.3 \text{ N}$

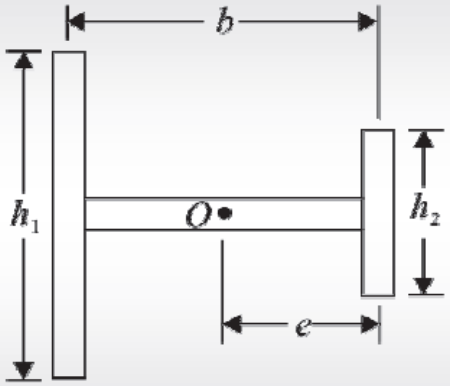
$Ve = F_1 h \quad \cdot \quad e = \frac{F_1}{V} h = \left(\frac{213.3}{800}\right)(150) = 40 \text{ mm}$

$Q_A = Q_1 + (3 \times 75)\left(\frac{75}{2}\right) = 30937.5 \text{ mm}^3$

$\tau_{\max} = \frac{VQ_A}{It} = \frac{(800)(30937.5)}{(4.219 \times 10^6)(3)} = 1.955 \text{ MPa}$

範例 24

求剪力中心 O 點之位置 e 值，
假設構件各段厚度均為 t。



【成大土木】【107 成大土木】

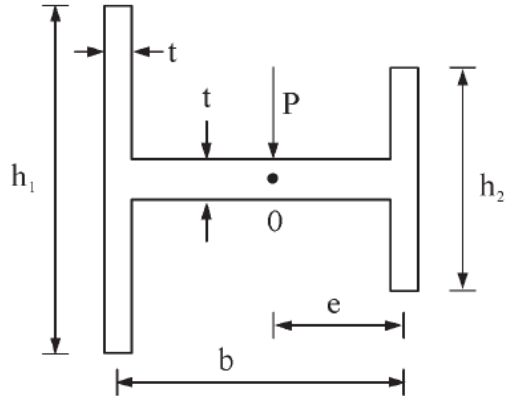
【解析】

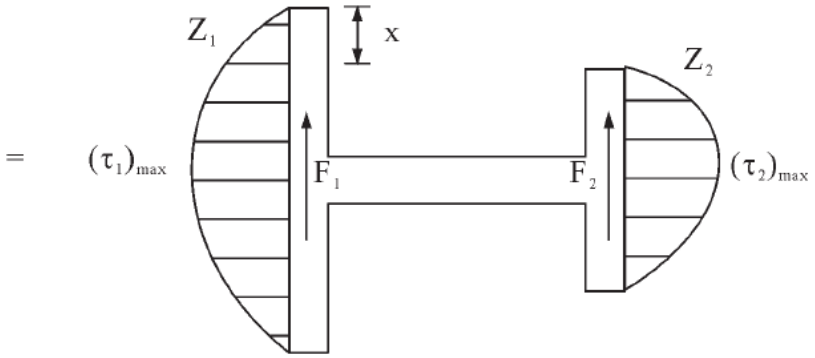
觀念說明

$$(\tau_1)_{max} = \frac{VQ}{It} = \frac{P\times(\frac{h_1}{2}\times t\times\frac{h_1}{4})}{It} = \frac{Ph_1^2}{8I}$$

同理也可求得 $(\tau_2)_{max}$

解





$$I=\frac{t}{12}(h_1^3+h_2^3)\ ,\ V=P$$

$$Q_1=t\cdot x(\frac{h_1}{2}-\frac{x}{2})\ ,\ b_1=t$$

$$\tau_1=\frac{VQ_1}{Ib_1}=\frac{P\times t\cdot x(\frac{h_1}{2}-\frac{x}{2})}{It}=\frac{6Px(h_1-x)}{t(h_1^3+h_2^3)}$$

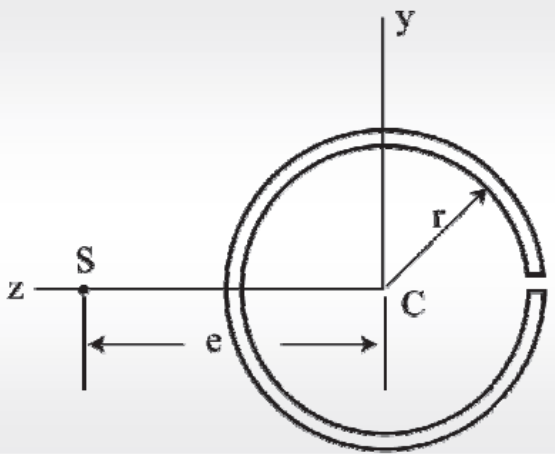
$$F_1=\int_0^{h_1}\tau_1 dA=\int_0^{h_1}\tau_1 t\cdot dx=\frac{6P}{(h_1^3+h_2^3)}\int_0^{h_1}(h_1x-x^2)dx=\frac{6P}{h_1^3+h_2^3}$$

$$\times(\frac{h_1^3}{2}-\frac{h_1^3}{3})=\frac{Ph_1^3}{h_1^3+h_2^3}$$

又 $P\cdot e=F_1\cdot b$　則求得 $e=\frac{h_1^3 b}{h_1^3+h_2^3}$

範例 25

有縫圓管具有等壁厚者之剖面示於圖中。示出從圓的中心至剪力中心 S 的距離等於 2r。



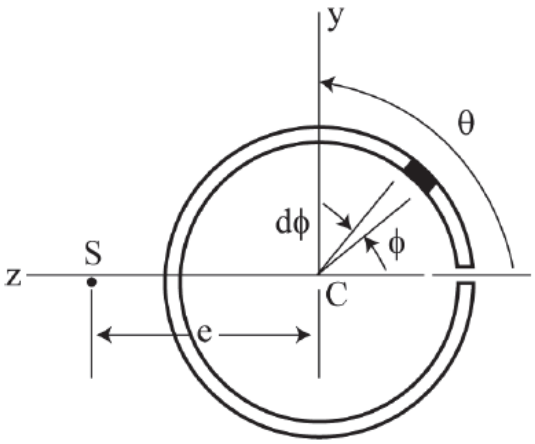
【103 北科土木】【交大機械】【成大土木】

【解析】

解 $Q_A=\int ydA=\int_0^{\varphi}(r\sin\varphi)t\varphi$

$=r^2t(1-\cos\theta)$

$\tau_A=\dfrac{VQ_A}{I_zt}=\dfrac{Vr^2(1-\cos\theta)}{I_z}$



$I_Z=\pi r^3t,\tau_A=\dfrac{V(1-\cos\theta)}{\pi rt}$

A 點：$dA=rtd\theta$

$T_c=\int\tau ArdA=\int_0^{2\pi}\dfrac{Vr}{\pi}(1-\cos\theta)d\theta=2Vr$

$\therefore\sum M_c=V_e=T_c,e=T_c/V=2r$

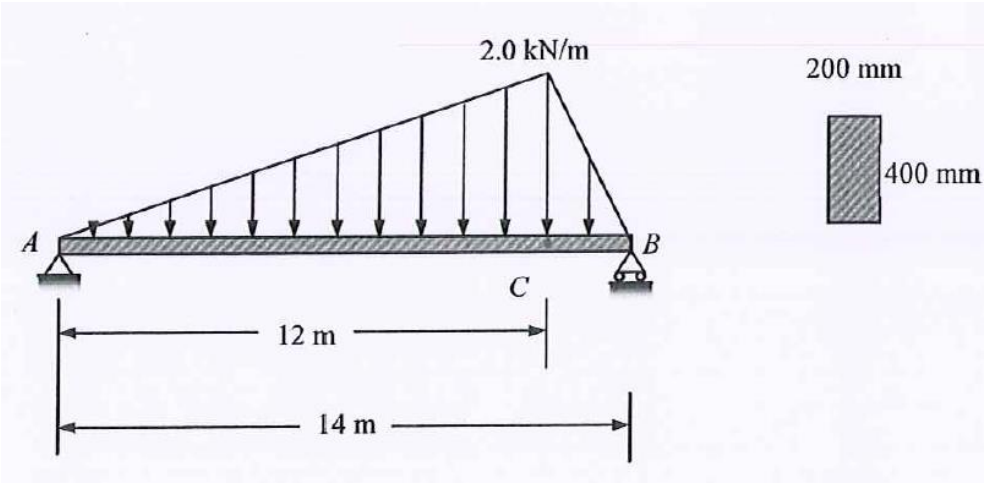
實戰演練

例 1

The simple beam *ACB* shown in the figure is subjected to a triangular load of maximum intensity 2.0 kN/m.

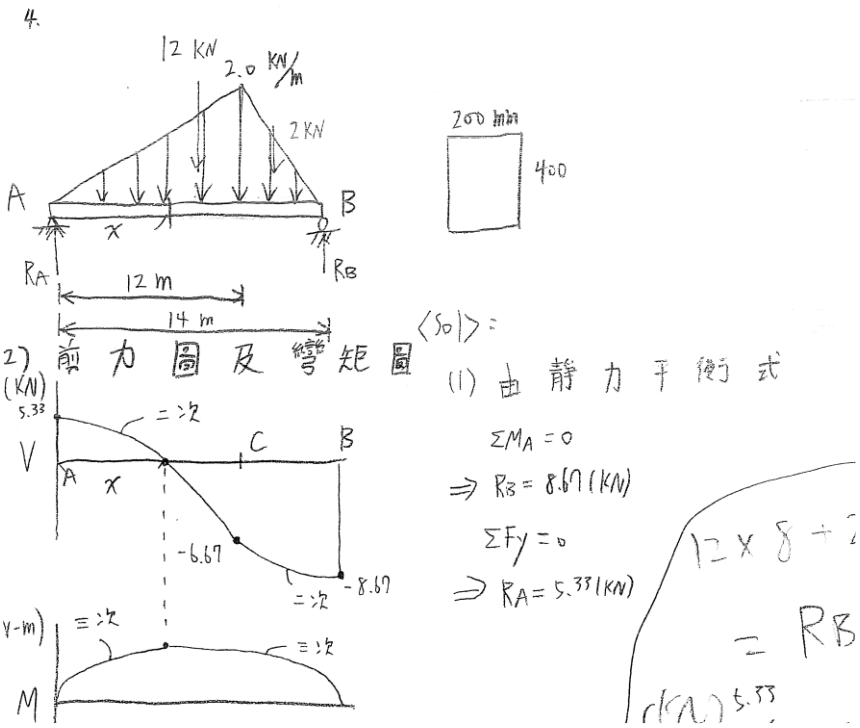
a. Draw the shear force and bending-moment diagrams for this beam. (20%)

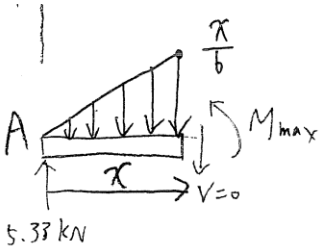
b. Calculate the maximum bending stress σ_{max} if the beam has a rectangular cross section with width $b = 200$ mm and height $h = 400$ mm. (5%)



【106 北科土木】

【解析】





$\Sigma F_y = 0 \Rightarrow 5.33 = \frac{x^2}{12} \Rightarrow x = 8\ (m)$

$\Sigma M_A = 0 \Rightarrow M_{max} = (\frac{x^2}{12}) \times \frac{2x}{3} = 28.44\ (kN\text{-}m)$

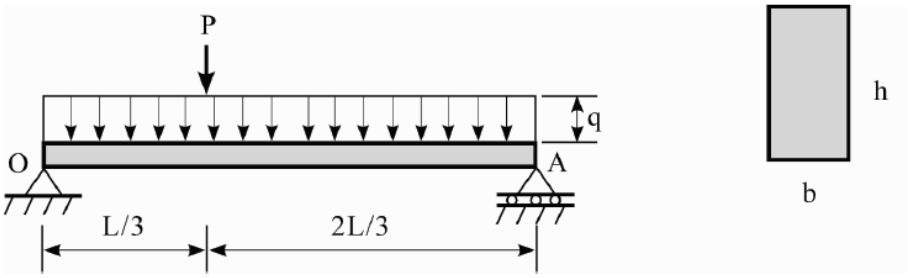
(3) 由 $\sigma = \frac{My}{I}$

$$\sigma_{max} = \frac{(28.44 \times 10^6) \times 200}{\frac{1}{12} \times 200 \times 400^3}$$

$$= 5.33\ (MPa)$$

例 2

如下左圖所示之OA簡支樑，L＝6m，受均勻分佈負載，q＝2kN／m，並於距左端L／3處受有一集中力，P＝9kN，其截面形狀如下右圖所示，b＝40mm，h＝90mm。



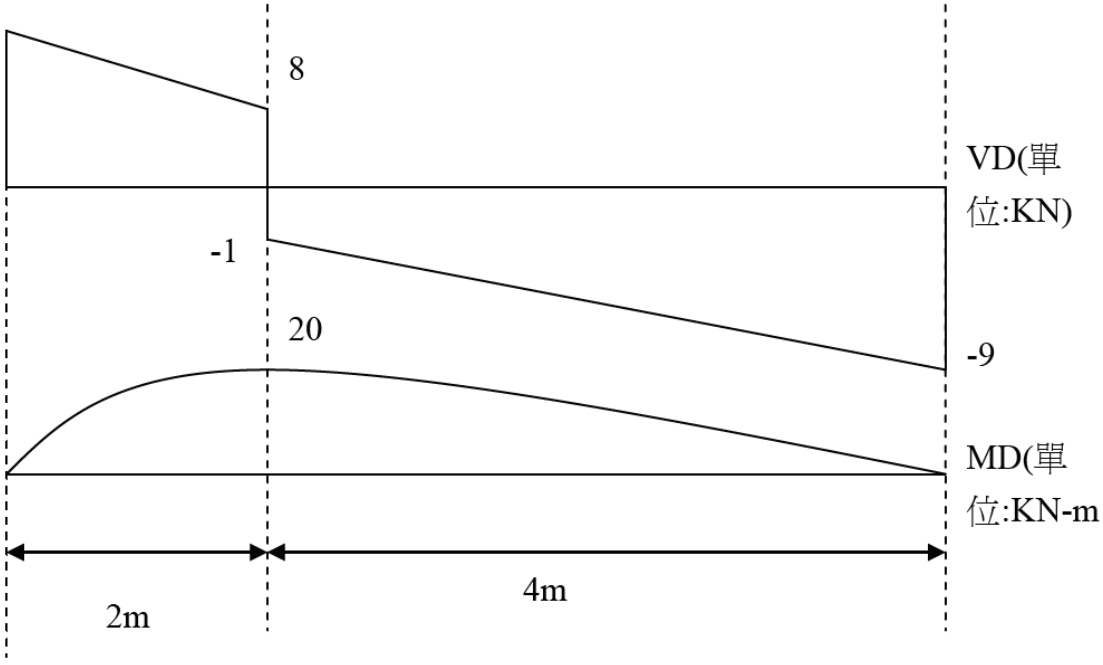
(1) **試繪出OA簡支樑之剪力圖與彎矩圖。**

(2) **最大的拉應力（**tensile stress, σ t**）為何？**(100普考)

【解析】

答：(1)計算支承反力

R_O=12(KN)　　R_A=9(KN)



(2) $\sigma_t = \dfrac{My}{I} = 370.37\ (MPa)$

例 3

如圖所示之懸臂梁的斷面呈 T 形，如圖 1 所示，則在圖 2 所示的 B 點處所產生的最大拉應力為多少？（102 關三）

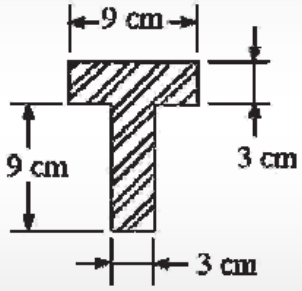


圖 1

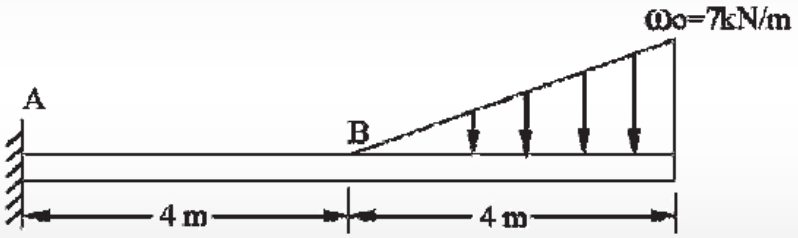
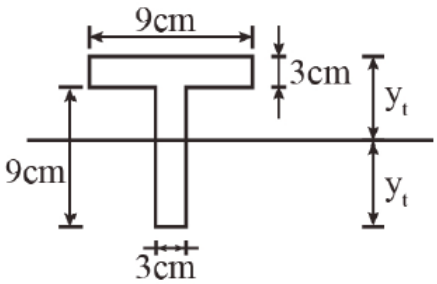
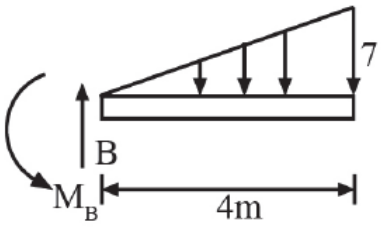


圖 2

【解析】





解 慣性矩 $I=\frac{1}{3}\times[9\times4.5^3-6\times1.5^3+3\times7.5^3]$

$=688.5(cm^4)=688.5\times10^4(mm^4)$

$(9\times3)\times2\times y_t=9\times3\times1.5+9\times3\times(3+4.5)$

$y_t=4.5(cm)$

$MB=7\times4\times\frac{1}{2}\times\frac{2}{3}\times4$

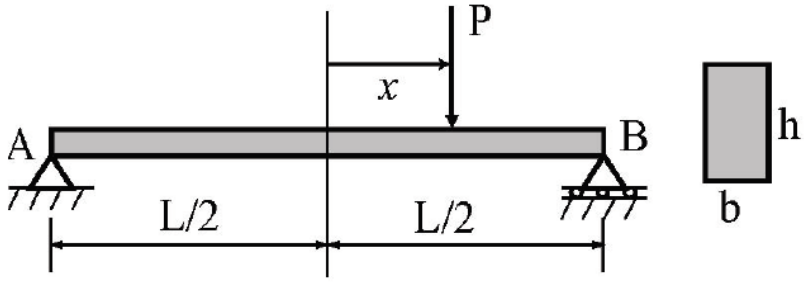
$=37.33(KN-m)$

$=37333.33(N-m)$

$\sigma=\frac{My_t}{I}=\frac{37333.33\times10^3\times4.5\times10}{688.5\times10^4}=244(MPa)$

例 4

如下圖所示之簡支樑AB，長度為L，在距離中間（L/2）算起右方x處，受一集中力P。該樑的斷面為長方形，寬度為一固定值b，高度為h，h隨x而變。假設只考慮由彎曲（bending）所造成的垂直應力（normal stresses），而能承受的最大應力值（allowable stress）為σallow。若要使簡支樑的重量達到最小，h應該為何？(100地四)



【解析】

答：$R_A = \dfrac{P \times (\frac{L}{2} - x)}{L}$

$$M = \frac{P \times (\frac{L}{2} - x) \times (\frac{L}{2} + x)}{L}$$

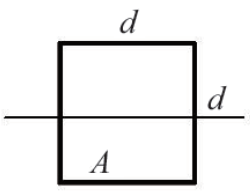
$$S = \frac{1}{6} \times b \times h^2(x)$$

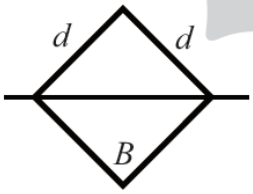
$$\sigma = \frac{M}{S} = \frac{P \times (\frac{L}{2} - x) \times (\frac{L}{2} + x)}{L \times \frac{1}{6} \times b \times h^2(x)}$$

$$h(x) = [\frac{P \times (\frac{L}{2} - x) \times (\frac{L}{2} + x)}{L \times \frac{1}{6} \times b \times \sigma}]^{\frac{1}{2}}$$

例 5

一正方形截面的橫樑承受單純彎力矩（pure bending moment）。此橫樑之截面可以右圖之A或B的方式水平橫置，以承受單純彎力矩。其材料特性以線性彈性估算之。





(1) 如果此橫樑截面以A的方式橫置，請求出截面A的慣性矩（area moment of inertia）。請將答案以截面邊長“d”表示之。

(2) 如果此橫樑截面以B的方式橫置，請求出截面B的慣性矩（area moment of inertia）。請將答案以截面邊長“d”表示之。

(3) 請求出以上A、B兩放置的截面情況所能承受的最大彎力矩（M_{max}）的比值。亦即“M_A/M_B”的值為何？

【102 地特三等】

【解析】

答：(1) $I_A = \dfrac{d^4}{12}$

(2) $I_B = \dfrac{d^4}{12}$

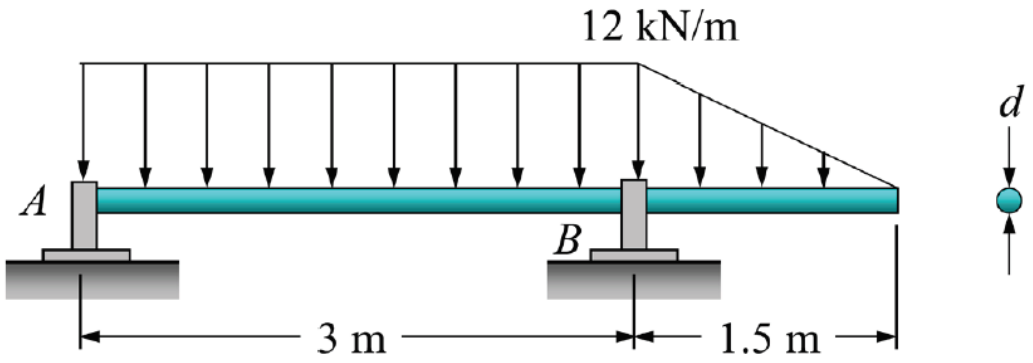
(3) $\dfrac{M_A \times \dfrac{d}{2}}{I_A} = \dfrac{M_B \times \dfrac{\sqrt{2}d}{2}}{I_B} \Rightarrow \dfrac{M_A}{M_B} = \sqrt{2}$

例 6

如下圖所示，一根受垂直方向荷載之軸件以光滑軸承支撐於A和B兩點，假設兩軸承僅能提供垂直方向之支撐力，軸件允許之最大彎曲應力為 $\sigma_{allow}=180MPa$，軸件之截面形狀為直徑d之實心圓：

(一) 沿軸之長度方向繪製軸之剪力分布圖及彎矩分布圖。

(二) 試求軸件能承受外力而不損壞時，最小之直徑d須為多少？



【105 地特四等】

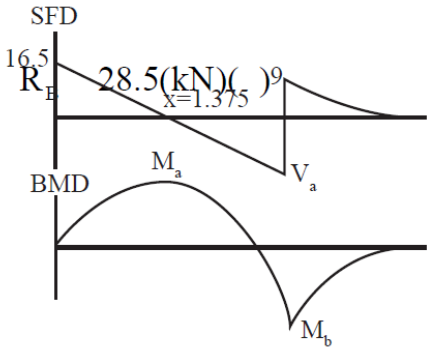
【解析】

答：(一) $\sum M_B=0\Rightarrow R_A\times 3+\frac{1}{2}\times 12\times 1.5\times(\frac{1}{3}\times 1.5)=12\times 3\times\frac{3}{2}\Rightarrow R_A=16.5(kN)(\uparrow)$

$\Rightarrow R_A=16.5(kN)(\uparrow)$

$\sum F_Y=0\Rightarrow R_A+R_B=\frac{1}{2}\times 12\times 1.5+12\times 3$

$\Rightarrow R_B=28.5(kN)(\uparrow)$



求距左端A處距離d剪力為零

$\sum F_Y=0\Rightarrow R_A=12\times d\Rightarrow d=1.375(m)$

$$V_{右}=V_{左}-\int_{左}^{右}wdx\Rightarrow V_a=16.5-12\times3=-19.5(kN)$$

$$M_{右}=M_{左}+\int_{左}^{右}Vdx\Rightarrow M_a=0+\frac{1}{2}\times16.5\times1.375=11.34375(kN-m)=M_{max}$$

$$=11.34375(kN-m)=M_{max}$$

$$M_{右}=M_{左}+\int_{左}^{右}Vdx\Rightarrow M_b=11.34375+\frac{1}{2}\times(-19.5)\times1.625=-4.5(kN-m)$$

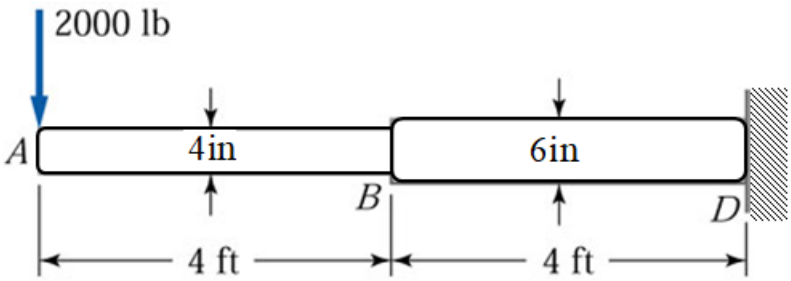
$$=-4.5(kN-m)$$

(二) $$\sigma_{max}=\frac{32M_{max}}{\pi d^3}\le\sigma_{all}\Rightarrow d\ge86.26(mm)$$

取 $d=87(mm)$

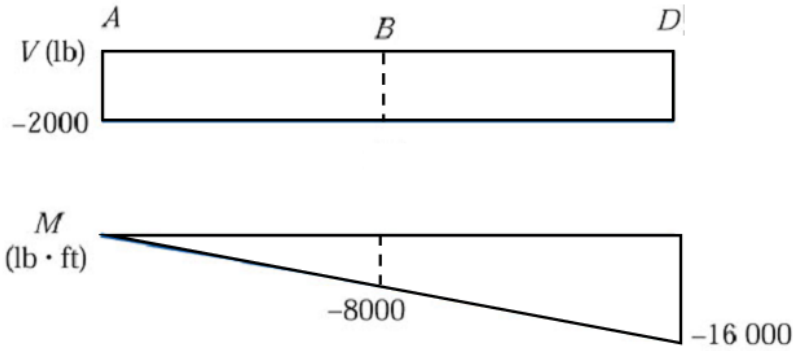
例 7

The cantilever beam in Fig.is composed of two segments with rectangular cross sections. The width of the each section is 2 in.,but the depths are different, as shown in the figure. Determine the maximum bending stress in the beam.



【解析】

(a)內力分析



(b)斷面分析

$$S_{AB} = \frac{bh^2{}_{AB}}{6} = \frac{(2)(4)^2}{6} 5.333 in.^3$$

$$S_{BD} = \frac{bh^2{}_{BD}}{6} = \frac{(2)(6)^2}{6} 12.0 in.^3$$

(c)應力分析

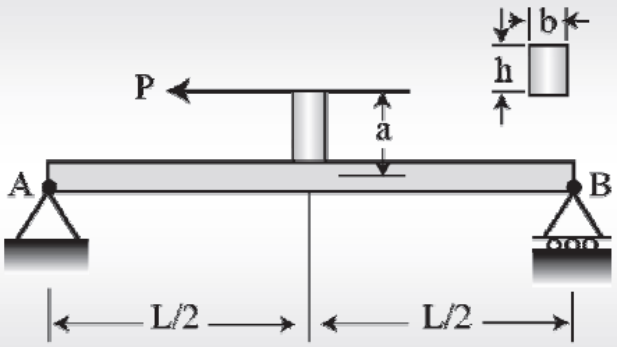
$$(\sigma_B)_{\max} = \frac{[M_B]}{S_{AB}} = \frac{8000 \times 12}{5.333} = 18000 psi$$

$$(\sigma_D)_{\max} = \frac{[M_D]}{S_{BD}} = \frac{16000 \times 12}{12.0} = 16000 psi$$

$\sigma_{\max} = 18000$ psi

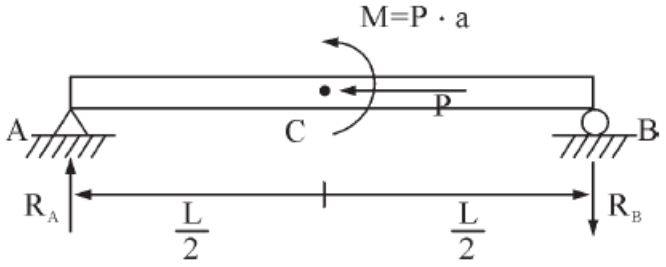
例 8

如下圖所示，一簡支樑 AB，其斷面為矩形(寬 b 高 h)，長度為 L。當負荷 P 作用在長為 a 的臂上時，試求樑的最大拉應力及最大壓應力。(96 年機械高考、101 年普考)



【解析】

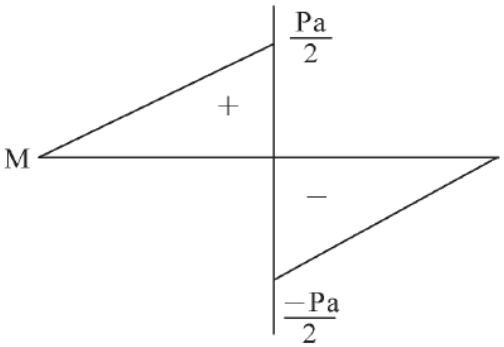
解 1. 先求支承反力：



$$\sum M_A = 0 \Rightarrow R_B = \frac{p.a}{L}(\downarrow)$$

$$\sum M_B = 0 \Rightarrow R_A = \frac{p.a}{L}(\uparrow)$$

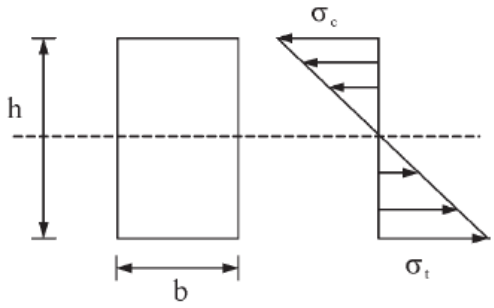
畫出彎矩圖



畫軸力圖



2. **梁於 C 點有最大之彎矩，取 AC 段來看：**



$$\sigma_c = \frac{-My}{I} = \frac{\frac{-Pa}{2}\times\frac{h}{2}}{\frac{1}{12}\times b\times h^3} = \frac{-3Pa}{bh^2}\text{(壓應力)}$$

$$\Rightarrow \text{最大壓應力} = \sigma_c + \frac{-P}{bh} = -(\frac{3Pa}{bh^2}+\frac{P}{bh})$$

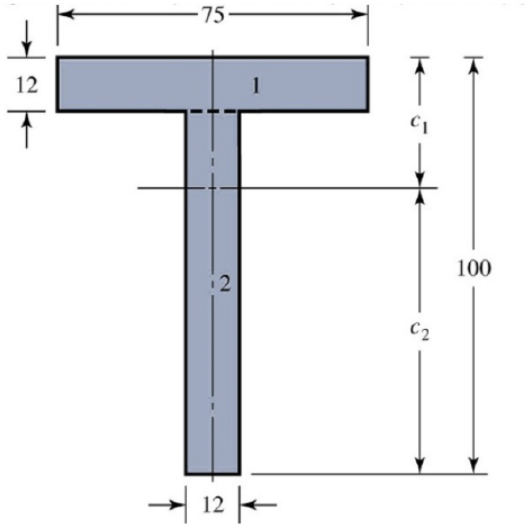
$$\sigma_t = \frac{My}{I} = \frac{Pa\times\frac{n}{2}}{\frac{1}{12}\times b\times h^3} = \frac{3Pa}{bh^2}\text{(拉壓力)}$$

$$\Rightarrow \text{拉應力} = \sigma_t + \frac{-P}{bh} = \frac{3Pa}{bh^2} - \frac{P}{bh}$$

3. **取 BC 段來看，則樑底部拉應力** $\sigma_t = \frac{3Pa}{bh^2} \Rightarrow$ **故最大拉應力** $= \frac{3Pa}{bh^2}$

例 9

如下圖所示之T形剖面的樑及其尺寸，若承受一彎矩1500N-m使得在頂面處產生拉應力。試求發生於頂面之最大拉應力為何？（圖中所示長度單位為mm）



【105 普考】

【解析】

解 $$c_2 = \frac{12\times 88\times \frac{88}{2} + 12\times 75\times (100-\frac{12}{2})}{12\times 88 + 12\times 75} = 67(\text{mm})$$

$$c_1 = 100 - 67 = 33(\text{mm})$$

$$I = \frac{1}{3}(12\times 67^3 + 75\times 33^3 - 63\times (33-12)^3) = 1906996(\text{mm}^4)$$

$$\sigma_{\max,\text{top}} = \frac{Mc_2}{I} = \frac{1500000\times 33}{1906996} = 25.957(\text{Mpa})$$

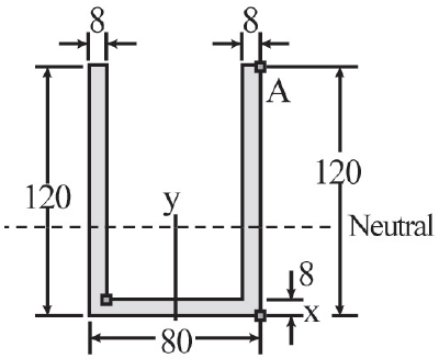
$$\sigma_{\max,\text{bottom}} = \frac{Mc_2}{I} = \frac{1500000\times 67}{1906996} = 52.7(\text{Mpa})$$

例 10

某鋁合金製成的橫樑（beam）承受正向彎曲力矩（positive bending moment），其截面形狀如圖三中所示，長度單位為 mm。若允許彎曲應力為 150 MPa，試求：

(一) 該結構之慣性截面矩（area moment of inertia）。

(二) 該樑可承受的最大彎曲力矩。（提示：於 A 點處）



圖三

【102 普考】

【解析】

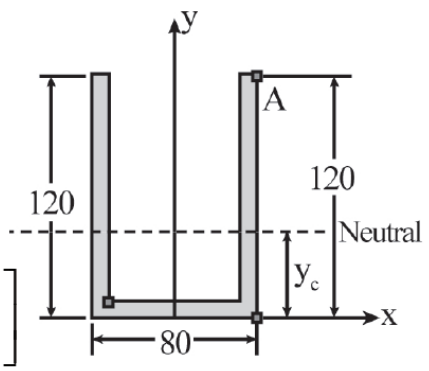
解 (一) $120\times8\times60\times2 + 64\times4\times8$

$= (120\times8\times2 + 64\times8)\times y_c$

$y_c = 48.2(\text{mm})$

$$I_c = \frac{64}{12}\times8^3 + 64\times8\times(48.2-4)^2 + 2\left[\frac{8\times(120)^3}{12} + 8\times(120)\times(60-48.2)^2\right]$$

$= 3574335\ \text{mm}^4$

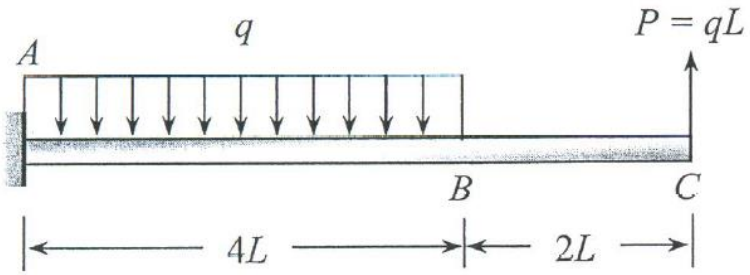


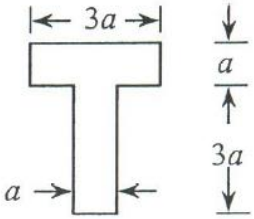
$$(二)\ \sigma = \frac{My}{I} \Rightarrow 150\times10^6 = \frac{M\times(120-48.2)\times10^{-3}}{3.574335\times10^{-6}}$$

$$\Rightarrow M = 7468\ (\text{N-m})$$

例 11

3. (20%) A channel section is used as a cantilever beam to support a uniformly distributed load of intensity q and a concentrated load of $P - qL$, as shown in the figure (a) below. The relevant dimensions of the cross section are shown in the figure (b). (a) Sketch the shear and moment diagrams for the beam and (b) determine the maximum tensile and compressive flexural stresses (please specify both the values and locations).





【100 交大機械】

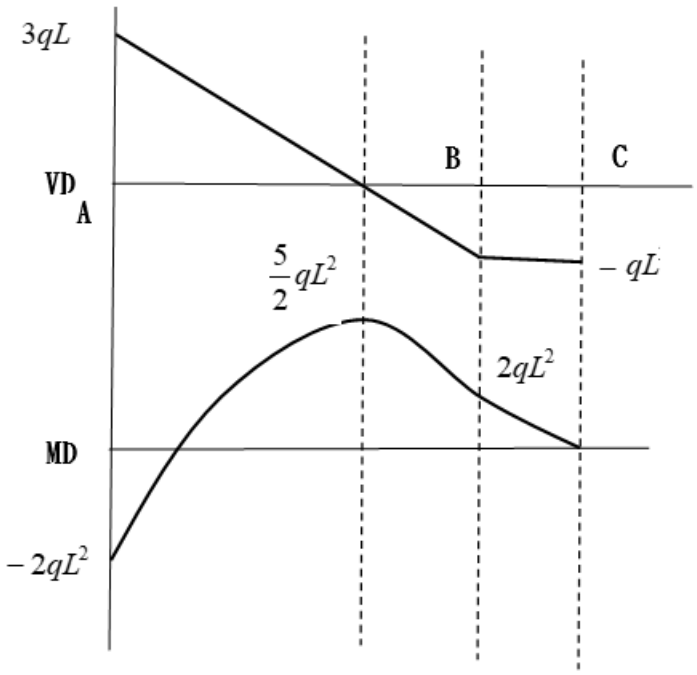
【解析】

(一)內力分析

$$\sum F_Y = 0 \Rightarrow V_A - 4qL + qL = 0 \Rightarrow V_A = 3qL$$

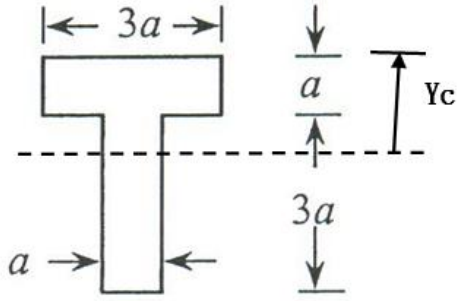
$$\sum M_A = 0 \Rightarrow -4qL \times 2L + qL \times 6L + M_A = 0 \Rightarrow M_A = 2qL^2$$

(二) 剪力圖與彎矩圖



(三)斷面分析

$$(3a\times a\times 2)\times Y_C = 3a\times a\times 2.5a + 3a\times a\times 0.5a \Rightarrow Y_C = 1.5a$$



$$I = \frac{1}{3}\times a\times (2.5a)^3 + \frac{1}{3}\times 3a\times (1.5a)^3 - \frac{1}{3}\times 2a\times (0.5a)^3 = 8.5a^4$$

比較應力後

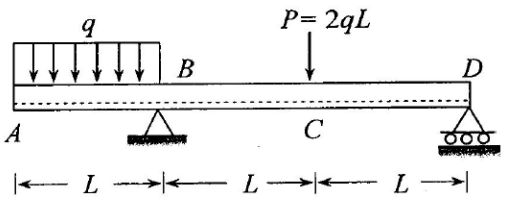
$$(\sigma_t)_{\max} = \frac{My}{I} = \frac{2.5qL^2\times 2.5a}{8.5a^4} = \frac{25qL^2}{34a^3}$$

$$(\sigma_C)_{\max} = \frac{My}{I} = \frac{2qL^2\times 2.5a}{8.5a^4} = \frac{10qL^2}{17a^3}$$

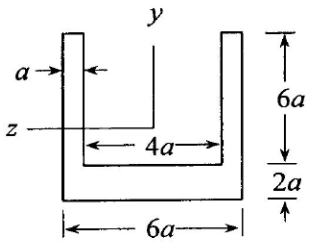
例 12

1. A steel beam *ABCD* is simply supported at points *B* and *D* and has an overhang *AB* as shown in figure (a). The beam is subjected to a uniformly distributed load of intensity *q* over region *AB* and a concentrated force *P* at point *C* (see the same figure). The beam has U-shaped cross section with associated dimensions given in figure (b). Finish the following:

(a) Draw the shear force and bending moment diagrams of the beam. (10%)
(b) Find the values and locations of maximum tensile and compressive normal (bending) stresses of the beam. (10%)



(a)



(b)

【101 交大機械】

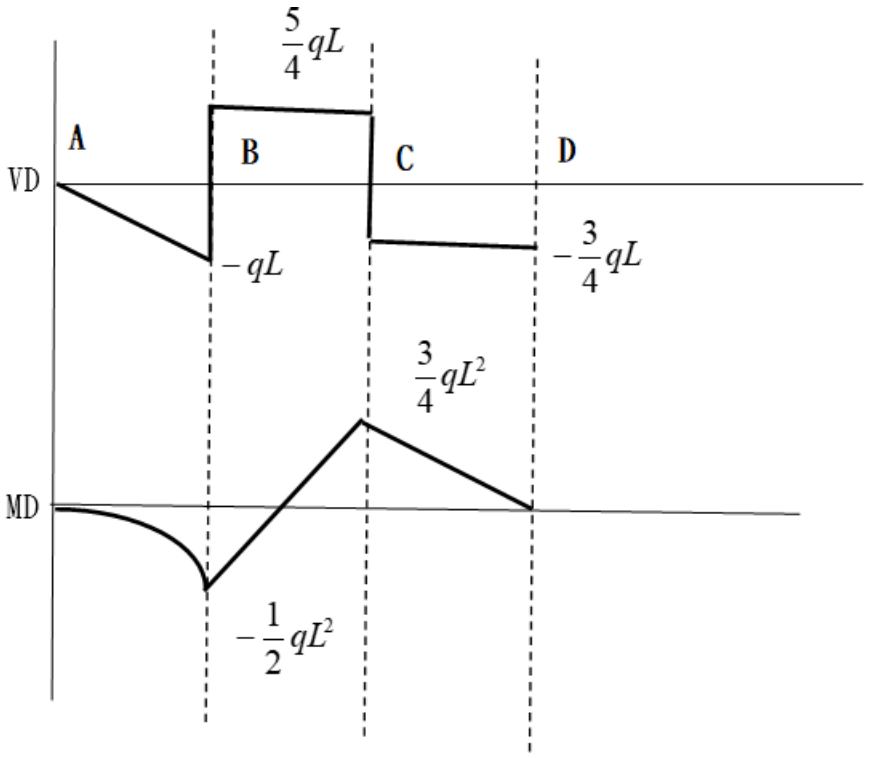
【解析】

(一)內力分析

$$\sum M_B = 0 \Rightarrow -qL \times \frac{L}{2} + 2qL \times L - R_D \times 2L = 0 \Rightarrow R_D = \frac{3}{4}qL$$

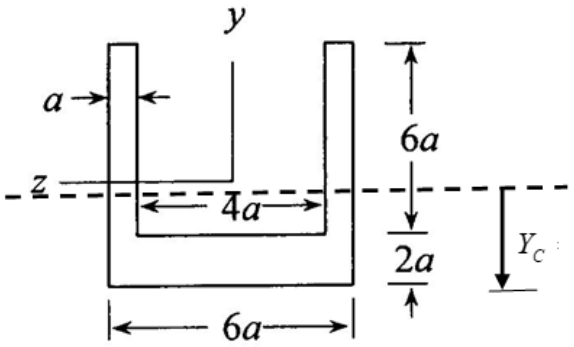
$$\sum F_Y = 0 \Rightarrow R_B = \frac{9}{4}qL$$

(二) 剪力圖與彎矩圖



(三)斷面分析

$$(6a\times a\times 2+6a\times 2a)\times Y_C=6a\times 2a\times a+6a\times a\times 5a\times 2\Rightarrow Y_C=3a$$



$$I=\frac{1}{3}\times 6a\times(3a)^3+\frac{1}{3}\times 2a\times(5a)^3-\frac{1}{3}\times 4a\times(a)^3=136a^4$$

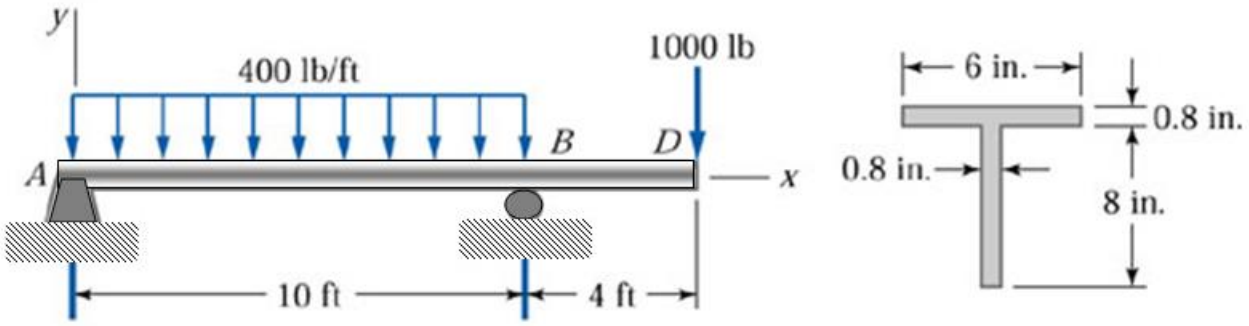
比較應力後

$$(\sigma_t)_{\max}=\frac{My}{I}=\frac{2.5qL^2\times 5a}{136a^4}=\frac{5qL^2}{272a^3}$$

$$(\sigma_C)_{\max}=\frac{My}{I}=\frac{\frac{3}{4}qL^2\times 3a}{136a^4}=\frac{15qL^2}{544a^3}$$

例 13

The simply supported beam in Fig. has the T-shaped cross section shown. Determine the values and locations of the maximum tensile and compressive bending stresses.

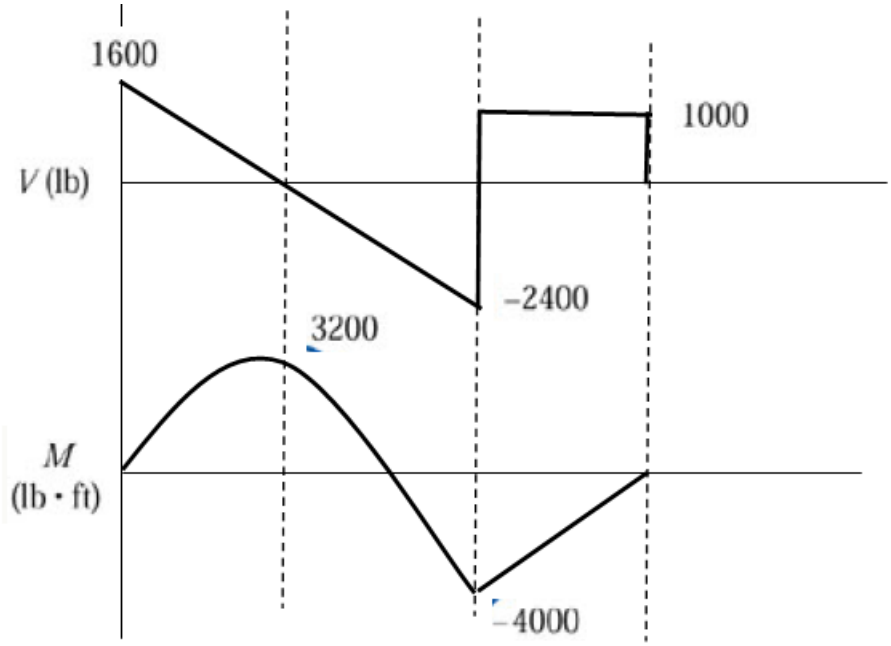


【交大機械類題】

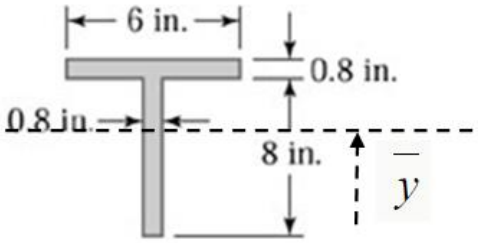
【解析】

(d)內力分析

$R_A = 1600\,(lb)$　$R_B = 3400\,(lb)$



(e)斷面分析



$$\bar{y} = \frac{A_1\overline{y_1} + A_2\overline{y_2}}{A_1 + A_2} = \frac{6.4(4) + 4.8(8.4)}{6.4 + 4.8} = 5.886in.$$

$$I = \left[\frac{0.8(8)^3}{12} + 6.4(4 - 5.886)^2\right] + \left[\frac{6(0.8)^3}{12} + 4.8(8.4 - 5.886)^2\right] = 87.49in.^4$$

(f)應力分析

$$\sigma_{top} = -\frac{Mc_{top}}{I} = -\frac{(3200 \times 12)(2.914)}{87.49} = -1279\,psi$$

$$\sigma_{bot} = -\frac{Mc_{bot}}{I} = \frac{(3200 \times 12)(-5.886)}{87.49} = 2580\,psi$$

$$\sigma_{top} = -\frac{Mc_{top}}{I} = \frac{(-4000 \times 12)(2.914)}{87.49} = 1599\,psi$$

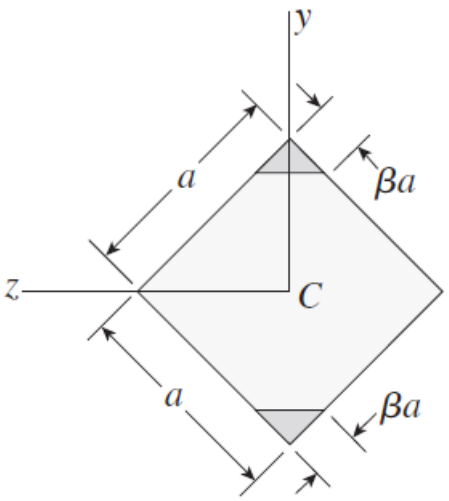
$$\sigma_{bot} = -\frac{Mc_{bot}}{I} = \frac{(-4000 \times 12) - (5.886)}{87.49} = -3230\,psi$$

$(\sigma_T)_{max} =$ 2580 psi

$(\sigma_c)_{max} =$ 3230 psi

例 14

A beam of square cross section (a=length of each side) is bent in the plane of a diagonal (see figure). By removing a small amount of material at the top and bottom corners, as shown by the shaded triangles in the figure, we can increase the section modulus and obtain a stronger beam, even though the area of the cross section is reduced. (a) Determine the ratio β defining the areas that should be removed in order to obtain the strongest cross section in bending. (b) By what percent is the section modulus increased when the areas are removed?

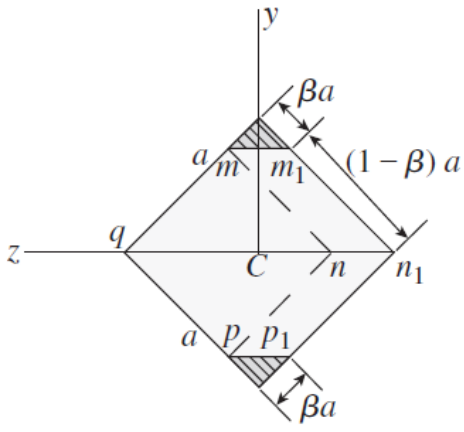


【台大應力所】

【中文翻譯】

方形樑（a 為每邊長 ），以對角受平面上的彎曲（ 見圖 ）。如果如圖示上下角隅之有陰影的三角形部份被移除，而增加剖面模數並獲得較強的樑。(a) 求比率 β，使得能夠獲彎曲之較強的橫截面。(b) 面積移除後，剖面模數增加的百分率為何？

【解析】



$$I_0 = \frac{a^4}{12} \quad c_0 = \frac{a}{\sqrt{2}} \quad S_0 = \frac{I_0}{c_0} = \frac{a^3\sqrt{2}}{12}$$

矩形mnpq之慣性矩

$$I_1 = \frac{(1-\beta)^4 a^4}{12}$$

平行四邊形mm_1nn_1之慣性矩

$$I_2 = \frac{1}{3}(\beta a\sqrt{2})\left[\frac{(1-\beta)a}{\sqrt{2}}\right]^3 = \frac{\beta a^4}{6}(1-\beta)^3$$

$$I = I_1 + 2I_2 = \frac{a^4}{12}(1+3\beta)(1-\beta)^3$$

$$c = \frac{(1-\beta)a}{\sqrt{2}} \quad S = \frac{I}{c} = \frac{\sqrt{2}\,a^3}{12}(1+3\beta)(1-\beta)^2$$

(a)

$$\frac{dS}{d\beta} = \frac{\sqrt{2}}{12}a^3(1-\beta)(1-9\beta) = 0 \qquad \text{故}\beta = \frac{1}{9}$$

(b)

$$S_1 = S\big|_{\beta=0} = \frac{\sqrt{2}}{12}a^3 \qquad S_2 = S\big|_{\beta=\frac{1}{9}} = \frac{64\sqrt{2}}{729}a^3$$

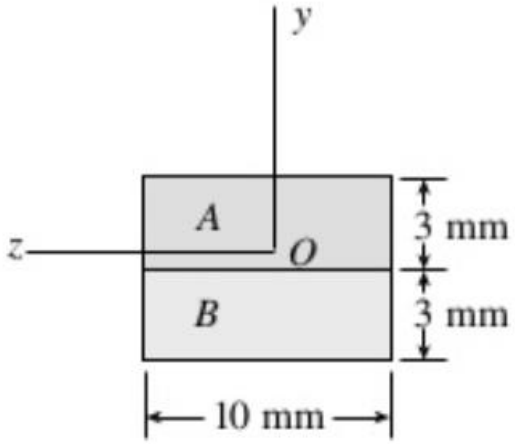
$$\frac{S_2 - S_1}{S_1} \times 100\% = 5.35\%$$

例 15

. The cross section of a bimetallic strip is shown in Fig. 2. Assume that the moduli of elasticity for metals A and B are E_A=196 GPa and E_B=98 GPa, respectively.

(a) Determine the section moduli for metals A and B of the beam, respectively. (15%) (Note that the section moduli is defined as the bending moment divided by the maximum bending stress)

(b) In which material does the maximum stress occur? (10%)



【104 中正機械】

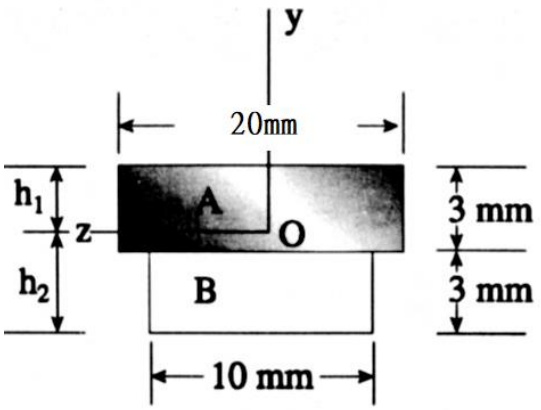
【中文翻譯】

如圖所示雙金屬之剖面。假定金屬 A 與 B 分別有彈性模數 $E_A = 196\text{GPa}$ 及 $E_B = 98\text{GPa}$ ；求樑之剖面模數中之較小者。(回想，剖面模數等於彎矩除以最大彎曲應力)。在那一種材料發生最大應力呢？

【解析】

(一)斷面幾何分析

$$n = \frac{196}{98} = 2$$



$$h_2 = \frac{10\times 3\times 1.5 + 20\times 3\times 4.5}{10\times 3 + 20\times 3} = 3.5(mm)$$

$$h_1 = 6 - h_2 = 2.5(mm)$$

$$I = \frac{1}{3}\times 20\times 2.5^3 + \frac{1}{3}\times 10\times 3.5^3 + \frac{1}{3}\times 10\times 0.5^3 = 247.5(mm^4)$$

(二)應力分析

$$\sigma_{\max,A} = n\times\frac{Mh_1}{I} \Rightarrow S_A = \frac{I}{nh_1} = 49.5(mm^3)$$

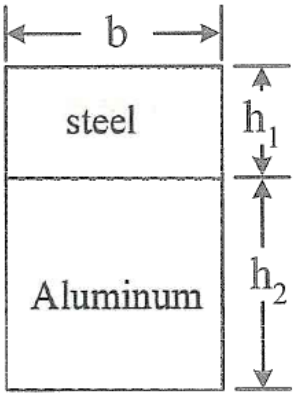
$$\sigma_{\max,B} = \frac{Mh_2}{I} \Rightarrow S_B = \frac{I}{h_2} = 70.7(mm^3)$$

(三)最大發生位置

$S_A < S_B \Rightarrow$ 故最大應力發生在 A材料

例 16

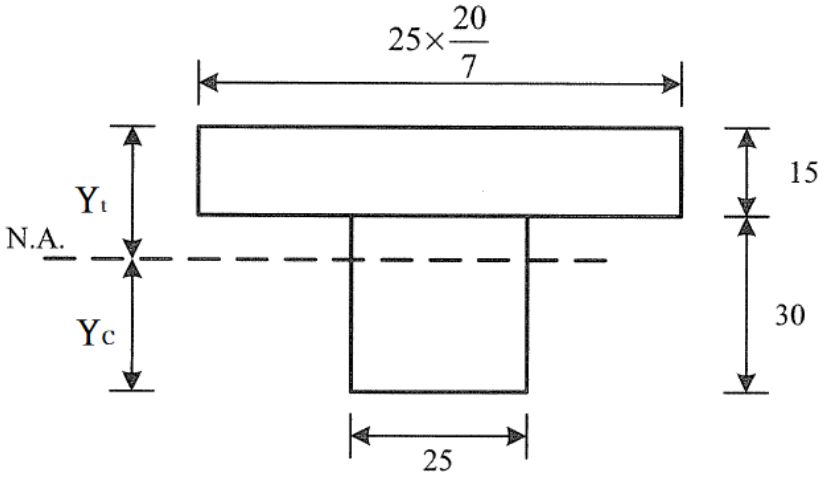
A composite beam is formed by securely bonding the steel and aluminum bars shown. If the allowable stresses in the aluminum and steel are 100 MPa and 170 MPa, respectively, determine the maximum allowable moment about a horizontal axis. Take $E_{st} = 200\,\text{GPa}$, $E_{al} = 70\,\text{GPa}$, $b = 25\,\text{mm}$, $h_1 = 15\,\text{mm}$, and $h_2 = 30\,\text{mm}$.



【中央機械】

【解析】

(一)轉換斷面求中性軸位置



$$Y_c = \frac{(25\times\frac{20}{7}\times 15)\times 37.5 + (30\times 25)\times 15}{30\times 25 + 25\times\frac{20}{7}\times 15} = 28.2(\text{mm})$$

$$Y_t = 45 - Y_c = 16.8(\text{mm})$$

(二)面積慣性矩

$$I_t = \frac{1}{12}(25\times\frac{20}{7})(15)^3 + (25\times\frac{20}{7}\times 15)(9.3)^2$$
$$+\frac{1}{12}(25)(30)^3 + (25\times 30)(13.2)^2 = 299687(\text{mm}^4)$$

(三)應力分析

以 steel 計算

$$\sigma_{\max} = \frac{M Y_t}{I_t}\times\frac{20}{7} \le 170(\text{MPa}) \Rightarrow M \le 1061.4(\text{N-m})$$

以 aluminum 計算

$$\sigma_{\max} = \frac{M Y_c}{I_t} \le 100(\text{MPa}) \Rightarrow M \le 1062.7(\text{N-m})$$

$$\therefore M_{\max} = 1061.4(\text{N-m})$$

【另解】

假設中性軸距離下表面距離為 y_C

$$E_S\int_S ydA + E_A\int_A ydA = 0$$

$$200(25\times 15) + (37.5 - y_C) + 70(25\times 30)(15 - y_C) = 0$$

$$y_C = 28.235\,\text{mm}$$

$$I_S = \frac{1}{12}(25)(15^3) + 375\times(37.5 - 28.235)^2 = 3.922\times 10^4\ \text{mm}^4$$

$$I_A = \frac{1}{12}(25)(30^3) + 750\times(15 - 28.235)^2 = 1.876\times 10^5\ \text{mm}^4$$

以 steel 計算

$$(\sigma_S)_{\max} = \frac{M(0.016765)E_S}{E_S I_S + E_A I_A} \qquad 170 = \frac{M(0.016765)(200)}{200\times 3.922\times 10^{-2} + 70\times 1.876\times 10^{-1}}$$

$M = 1063.5\ \mathrm{N\cdot m}$

以 aluminum 計算

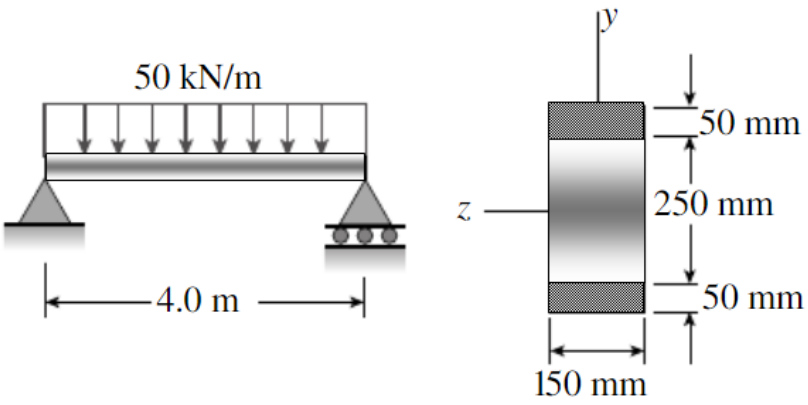
$$(\sigma_A)_{\max} = \frac{M(0.028235)E_A}{E_S I_S + E_A I_A} \qquad 100 = \frac{M(0.028235)(70)}{200\times 3.922\times 10^{-2} + 70\times 1.876\times 10^{-1}}$$

$M = 1061.3\ \mathrm{N\cdot m}$

縱合上面的結果： $M_{\text{allow}} = 1061.3\ \mathrm{N\text{-}m}$

例 17

The composite beam shown in the figure is simply supported and carries a total uniform load of 50 kN/m on a span lengthof 4.0 m. The beam is built of a wood member having cross-sectionaldimensions $150\times250mm$ and two steel plates of cross-sectionaldimensions $50\times150\text{mm}$.Determine the maximum stress in the steel and wood,respectively, if the moduli of elasticity are $E_s=209\text{GPa}$ and $E_w=11GPa$. (Disregard the weight of the beam.)



【中文翻譯】

複合樑示於圖者受簡支並支持均勻負載50 kN/m，其跨距為4m。樑由木，截面為$150\times250mm$，及兩鋼板，尺寸為$50\times150\text{mm}$所構成。如果鋼與木的彈性模數分別為$E_s=209\text{GPa}$及$E_w=11GPa$，求鋼與木之最大應力σ_s及σ_w。(忽略樑自重)

【解析】

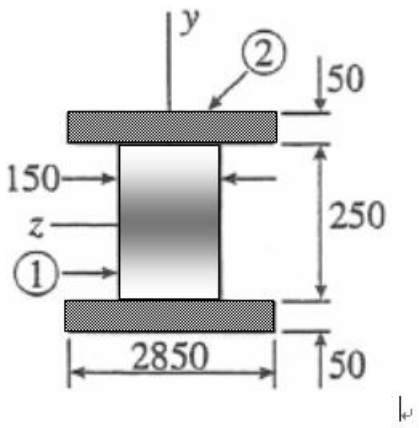
(一)轉換斷面

$$n=\frac{E_s}{E_w}=\frac{209}{11}=19$$

$$nb=(19)\,(150\text{ mm})=2850\text{ mm}$$

$$I_T=\frac{1}{12}(2850)(350)^3-\frac{1}{12}(2850-150)(250)^3$$

$$=6.667\times10^9\,\text{mm}^4$$



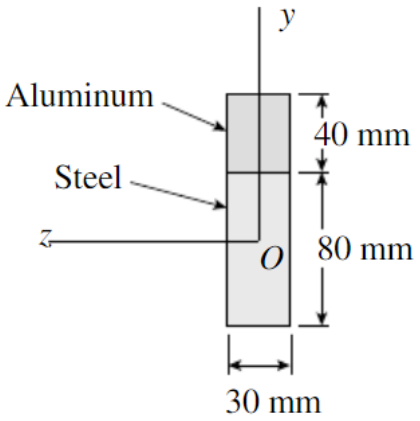
(二)應力分析

$$\sigma_w = \sigma_1 = \frac{M_{\max}(h_1/2)}{I_T} = 1.9 \text{ MPa}$$

$$\sigma_s = \sigma_2 = \frac{M_{\max}(h_2/2)n}{I_T} = 49.9 \text{ MPa}$$

例 18

The cross section of a composite beam made of aluminum and steel is shown in the figure. The moduli of elasticity are $E_a = 75\text{GPa}$ and $E_s = 200\text{GPa}$ Under the action of a bending moment that produces a maximum stressof 50 MPa in the aluminum, what is the maximum stress ssin the steel?



【中文翻譯】

複合樑由如圖示以鋁及鋼所製成。彈性模數 $E_a = 75\text{GPa}$ 及 $E_s = 200\text{GPa}$ 。若在彎矩作用下，鋁的最大彎矩為 50 MPa，鋼之最大應力 σ_s 為何？

【解析】

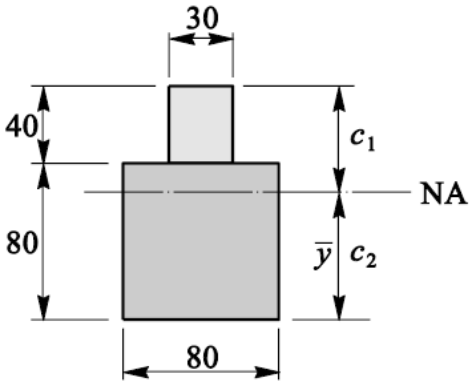
(一)轉換斷面

$$n = \frac{E_s}{E_a} = \frac{200}{75} = \frac{8}{3}$$

鋼之寬度轉為 $30 \times \frac{8}{3} = 80 \text{ mm}$

$$\bar{y} = \frac{(30 \times 40)(100) + (80 \times 80)(40)}{30 \times 40 + 80 \times 80} = 49.47 \text{ mm}$$

$c_2 = \bar{y} = 49.47 \text{ mm}$ ， $c_1 = 120 - 49.47 = 70.53 \text{ mm}$



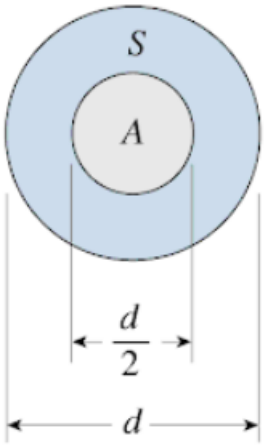
(二)應力分析

$$\sigma_a = \frac{Mc_1}{I} \quad , \quad \sigma_s = n\frac{Mc_2}{I}$$

$$\frac{\sigma_a}{\sigma_s} = \frac{c_1}{nc_2} \quad , \quad \frac{50}{\sigma_s} = \frac{70.53}{(8/3)(49.47)} \quad , \quad \sigma_s = 93.5 \text{ MPa}$$

例 19

A round steel tube of outside diameter d and an aluminum core of diameter d/2 are bonded to form a composite beam as shown in the figure. Derive a formula for the allowable bending moment M that can be carried by the beam based upon an allowable stress in the steel of σ_s. (Assume that the moduli of elasticity for the steel and aluminum are Es and Ea, respectively.)



【104 中興】

【中文翻譯】

圓鋼管外直徑 d，有一鋁核直徑 $d/2$，接合而成複合樑，如圖所示。求基於鋼的容許應力 σ_s 之樑的容許彎矩 M 的公式。(假設鋼與鋁之彈性模數分別為 E_s 及 E_a)。

【解析】

$$I_S=\frac{\pi}{64}\left[d^2-\left(\frac{d}{2}\right)^4\right]=\frac{15\pi}{1024}d^4$$

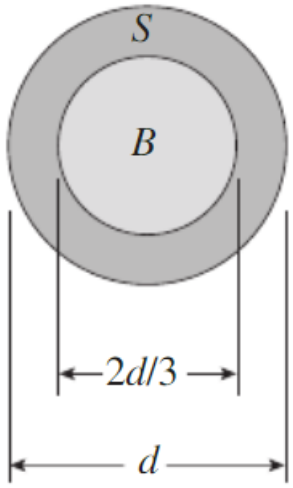
$$I_A=\frac{\pi}{64}\left(\frac{d}{2}\right)^4=\frac{\pi}{1024}d^4$$

$$(\sigma_S)_{\max}=\frac{M(d/2)E_S}{E_SI_S+E_AI_A}$$

$$M=\frac{\sigma_S(I_S+I_A(E_A/E_S))}{d/2}=\frac{\pi d^3}{512}\sigma_S\left(15+\frac{E_A}{E_S}\right)$$

例 20

圓鋼管外直徑 d ，有一鋁核直徑 $2d/3$ ，接合而成複合樑，如圖所示。求基於鋼的容許應力 σ_s 之樑的容許彎矩 M 的公式。(假設鋼與鋁之彈性模數分別為 E_s 及 E_a)。



【解析】

$$I_1 = \frac{\pi}{64}\left[d^4 - \left(\frac{2d}{3}\right)^4\right] = \frac{65}{5184}\pi d^4$$

$$I_2 = \frac{\pi}{64}\left(\frac{2d}{3}\right)^4 = \frac{\pi d^4}{324}$$

$$E_1 I_1 + E_2 I_2 = E_s I_1 + E_b I_2$$

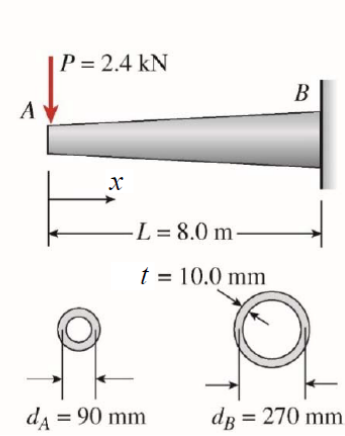
$$= \frac{\pi d^4}{5184}(65E_s + 16E_b)$$

$$M_{\text{allow}} = \sigma_s\left[\frac{E_1 I_1 + E_2 I_2}{\left(\frac{d}{2}\right)E_s}\right]$$

$$M_{\text{allow}} = \frac{\sigma_s \pi d^3}{2592}\left(65 + 16\frac{E_b}{E_s}\right)$$

例 21

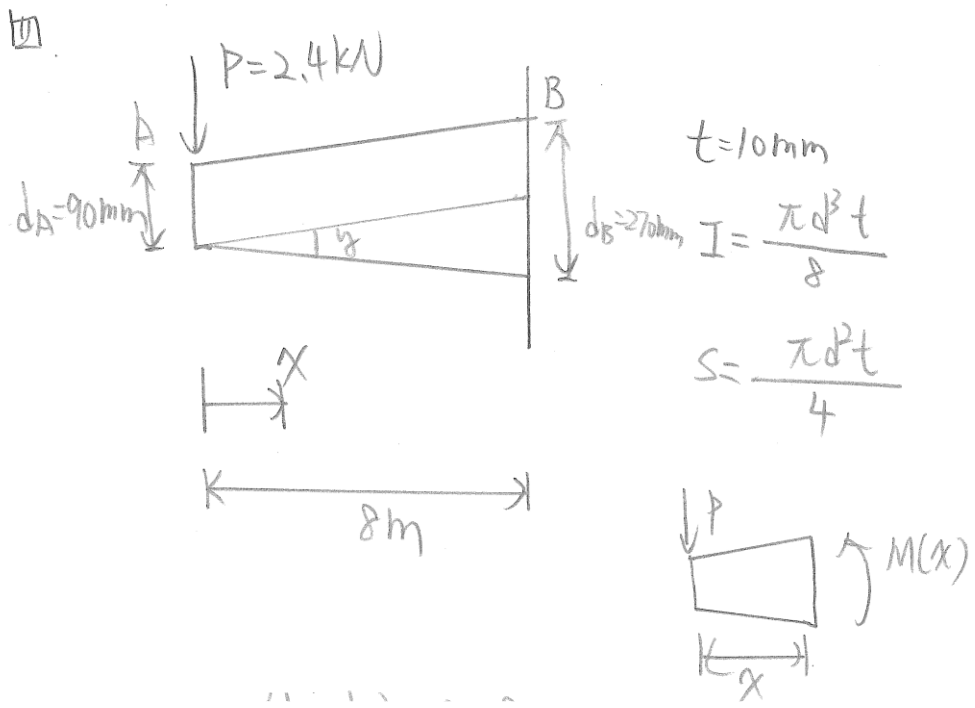
如圖所示之懸臂樑承受自由端之集中負載 P =2.4 kN，樑由一錐形管所構成，長度 8 m，A、B 處之斷面如圖中下方所示，壁厚 t =10 mm。假設斷面可視為薄壁圓管，適用面積慣性矩 $I=\dfrac{\pi d^3 t}{8}$、斷面模數（section modulus）$S=\dfrac{\pi d^2 t}{4}$，試求此懸臂樑中之最大彎曲應力。（25 分）



【106 鐵路員級】

【解析】

(一)斷面分析



$$y:(d_B-d_A)=x:\delta$$

$$\Rightarrow y=\frac{x(d_B-d_A)}{\delta} \quad \Rightarrow d=d_A+\frac{(d_B-d_A)x}{\delta}$$

$$M(x)=P\cdot x$$

$$\sigma=\frac{My}{I}=\frac{P\cdot x\cdot(\frac{d}{2})}{\frac{\pi d^3 t}{\delta}}=\frac{P\cdot x\cdot\frac{1}{2}\times\left[d_A+\frac{(d_B-d_A)}{8\times10^3}x\right]}{\frac{\pi\times\left[d_A+\frac{(d_B-d_A)}{8\times10^3}x\right]^3 t}{\delta}}$$

(二)應力分析

$$\sigma(x)=\frac{4Px\left[90+\frac{180}{8\times10^3}x\right]}{\pi\times\left[90+\frac{180}{8\times10^3}x\right]^3\cdot t}=\frac{4Px}{\pi\times\left[90+\frac{180}{8\times10^3}x\right]^2 t}$$

$$=\frac{960x}{\pi\left[90+0.0225x\right]^2}$$

最大應力發生在 $\frac{d\sigma(x)}{dx}=0$ 之處

$$\Rightarrow 960\times\pi\times\left[90+0.0225x\right]^2-960x\cdot2\pi\left[90+0.0225x\right]\times0.0225$$

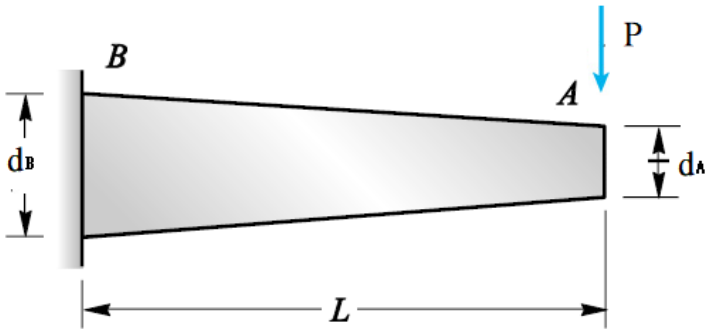
$$=0$$

$$\Rightarrow x=4000(mm)$$

$$\sigma_{max}=\sigma(4000)=37.7(MPa)$$

例 22

A tapered cantilever beam AB of solid circular cross section supports a load P at the free end with $\frac{d_B}{d_A}=2$ determine σ_B and σ_{max}



【清大動機】【108 台科機械】

【解析】

(一)斷面分析

$$d_x = d_A + (d_B - d_A)\frac{x}{L}$$

$$S_x = \frac{\pi d_x^3}{32} = \frac{\pi}{32}\left[d_A + (d_B - d_A)\frac{x}{L}\right]^3$$

(二)應力分析

$$\sigma_1 = \frac{M_x}{S_x} = \frac{32 P x}{\pi \left[d_A + (d_B - d_A)(x/L)\right]^3}$$

$$d_B = 2 d_A, \quad x = L$$

$$\sigma_B = \frac{4 P L}{\pi d_A^3}$$

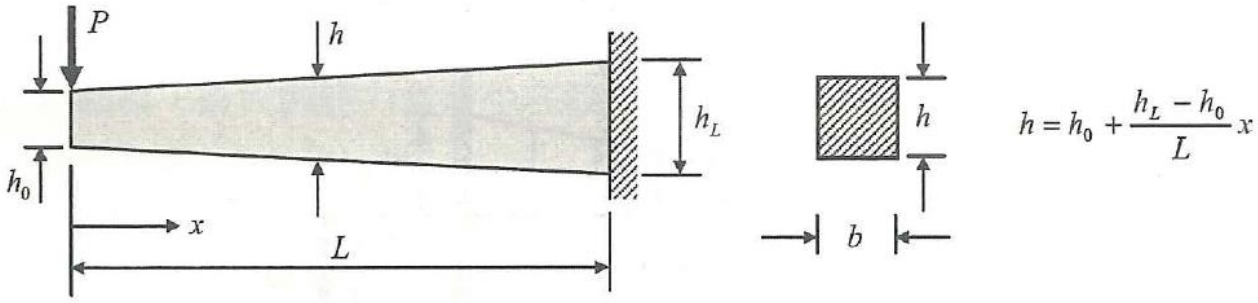
$$d\sigma_1 / dx = 0$$

$$x = L / 2$$

$$\sigma_{max} = \frac{128 P L}{27 \pi d_A^3} = 4.741 \frac{P L}{\pi d_A^3}$$

例 23

The beam shown in the figure has a slight taper. For what value of h_L/h_0 does the largest flexural stress appear at $x = L/2$? What then is the ratio of flexural stress at $x = L/2$ to flexural stress at $x = L$? State clearly your all assumptions.



【成大機械】

【解析】

(一)內力分析

彎矩： $M(x) = Px$

(二)斷面分析

高度： $h(x) = h_0 + \frac{x}{L}(h_L - h_0)$

斷面模數： $S(x) = \frac{bh^2(x)}{6} = \frac{b}{6}[h_0 + \frac{x}{L}(h_L - h_0)]^2$

(三)應力分析

$$\sigma(x) = \frac{M(x)}{S(x)} = \frac{6Px}{b[h_0 + \frac{x}{L}(h_L - h_0)]^2}$$

最大應力產生於 $x = \frac{L}{2}$

$$h_0 + \frac{h_L - h_0}{L} \times \frac{L}{2} = 2 \times \frac{L}{2} \times \frac{h_L - h_0}{L} \qquad h_0 + \frac{h_L - h_0}{2} = h_L - h_0 \qquad \frac{h_L}{h_0} = 3$$

當 $x = \frac{L}{2}$ $\qquad h = h_0 + \frac{3h_0 - h_0}{L}\frac{L}{2} = 2h_0 \qquad S = \frac{1}{6}bh^2 = \frac{2}{3}bh_0^2$

$$M = \frac{1}{2}PL \qquad \sigma_1 = \frac{M}{S} = \frac{3PL}{4bh_0^2}$$

當 $x = L$ $h = 3h_0$ $S = \frac{1}{6}bh^2 = \frac{3}{2}bh_0^2$ $M = PL$

$\sigma_2 = \frac{M}{S} = \frac{2PL}{3bh_0^2}$ $\frac{\sigma_1}{\sigma_2} = \frac{3/4}{2/3} = \frac{9}{8}$

例 24

有一矩形均勻厚度斷面懸臂梁受線性變化之均佈載重如圖 4 所示，此梁斷面厚度為 b，但梁深h_x可變化。如梁於各斷面之強度均一致且梁深在固定端為h_O，請將梁深h_x表為x、L及h_O之函數。如L＝80 cm，b＝3.5 cm，h_O＝20 cm且梁之容許應力為σ_{allow}＝200 MPa，則此梁可承受載重之最大q_O值為何？（20 分）

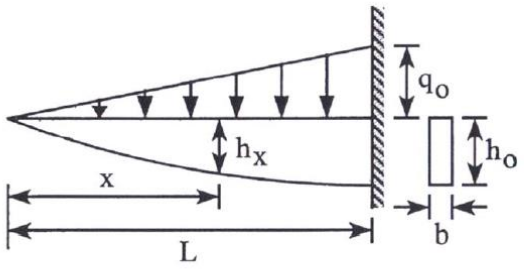
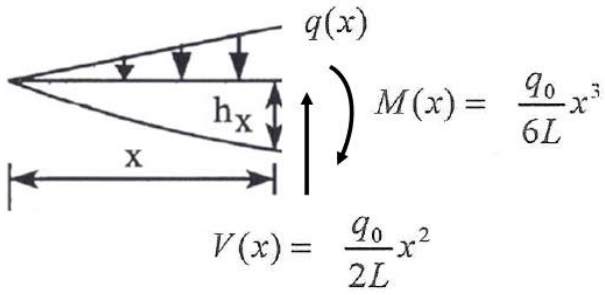


圖 4

【土木高考】

【解析】

(二)內力分析



(二)斷面分析

$$S(x)=\frac{1}{6}bh_x^2$$

(三)應力分析

應力均相同

$$\sigma=\frac{M(x)}{S(x)}=\frac{\frac{q_0}{6L}x^3}{\frac{1}{6}bh_x^2}=\frac{\frac{q_0}{6L}L^3}{\frac{1}{6}bh_0^2}$$

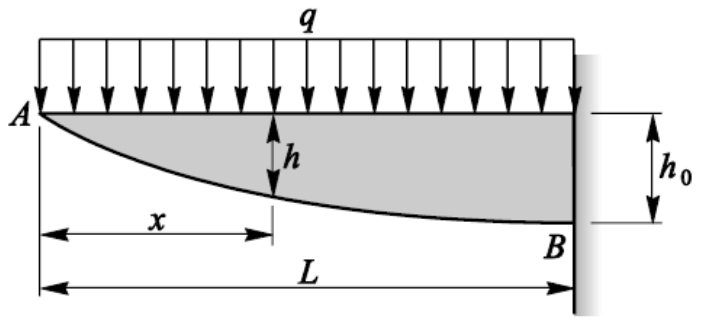
$$\left(\frac{h_x}{h_0}\right)^2 = \left(\frac{x}{L}\right)^3 \qquad h_x = h_0\left(\frac{x}{L}\right)^{2/3}$$

$$M_0 = \frac{1}{6}q_0L^2 \qquad S_0 = \frac{1}{6}bh_0^2 \qquad \sigma = \frac{M_0}{S_0} = \frac{q_0L^2}{bh_0^2}$$

$$q_{\max} = \frac{\sigma_{allow}bh_0^2}{L^2} = \frac{\left(200\times10^6\right)\left(0.035\right)\left(0.2^2\right)}{0.8^2}\,\text{N/m} = 437.5\,\text{kN/m}$$

例 25

圖中承受均佈負荷 q 之懸臂梁長度為 L，斷面寬度 b 保持不變，若欲設計為等強度梁，則斷面所需之高度 h 與 x 之關係為何？(以 x、L 及 h_0 表示)。



【解析】

(一)內力分析

在 x 位置之斷面：$M_x = \frac{1}{2}qx^2$ ，

(二)斷面分析

$$S_x = \frac{1}{6}bh_x^2$$

(三)應力分析

$$S_x = \frac{M_x}{\sigma} \ ， \ \frac{1}{6}bh_x^2 = \frac{qx^2}{2\sigma}$$

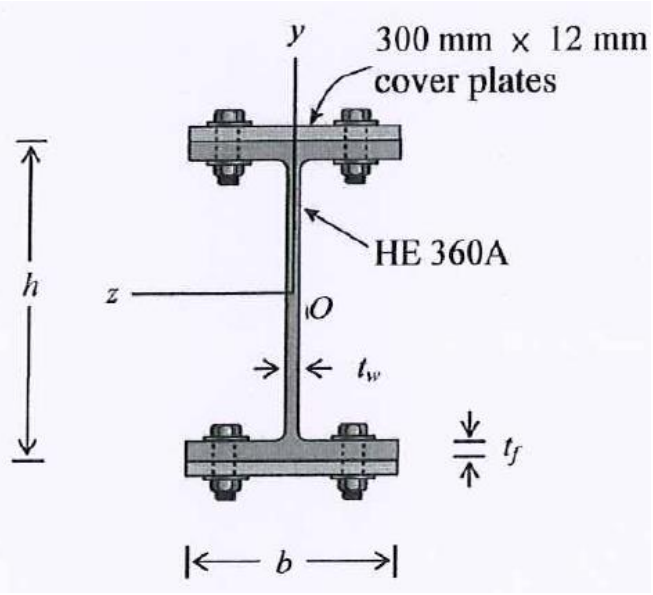
在固定端斷面：$M_B = \frac{1}{2}qL^2$ ， $S_B = \frac{1}{6}bh_0^2$

$$S_B = \frac{M_B}{\sigma} \ ， \ \frac{1}{6}bh_0^2 = \frac{qL^2}{2\sigma}$$

$$\frac{h_x^2}{h_0^2} = \frac{x^2}{L^2} \ ， \ h_x = \frac{x}{L}h_0$$

例 26

A steel beam is built up from a HE 360A wide-flange beam (b = 300 mm, h = 350 mm, t_w = 10 mm, t_f = 17.5 mm, and I = 330,900,000 mm^4 for HE 360A) and two 300 mm × 12 mm cover plates. The allowable load in shear on each bolt is 12.0 kN. What is the required bolt spacing s in the longitudinal direction if the shear force V = 200 kN? (25%)



【106 北科土木】

【解析】

(1) 剪力流

$$q = \frac{VQ}{I} = \frac{200000 \times (300 \times 12 \times 181)}{330900000 + \left(\frac{1}{12} \times 300 \times 12^3 + 300 \times 12 \times 181^2\right) \times 2}$$

$$= 230 \left(N/mm\right)$$

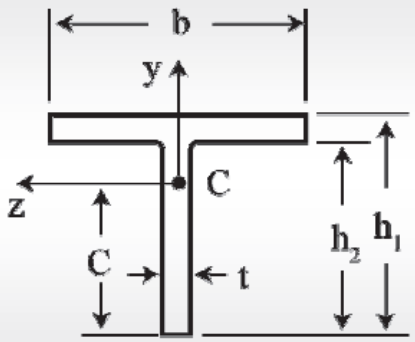
(2) $\frac{q}{②} = \frac{V_{釘}}{S}$

$$\Rightarrow \frac{230}{2} = \frac{12000}{S}$$

$$\Rightarrow S = 104.3 \ (mm)$$

例 27

T 型樑的結構如圖所示，並承受剪力 V。試求此樑的最大剪應力。已知：b=200mm，t=15mm，h_1=175mm，h_2=150mm，V=22kN



【解析】

解 $c=\dfrac{200(25)(162.5)+150(15)(75)}{200(25)+150(15)}=135.35\text{mm}$

$$I=\frac{1}{12}(200)(25)^3+200\times25(27.15)^2+135.35\times15\left(\frac{135.35}{2}\right)^2=13.24\times10^6\text{mm}^4$$

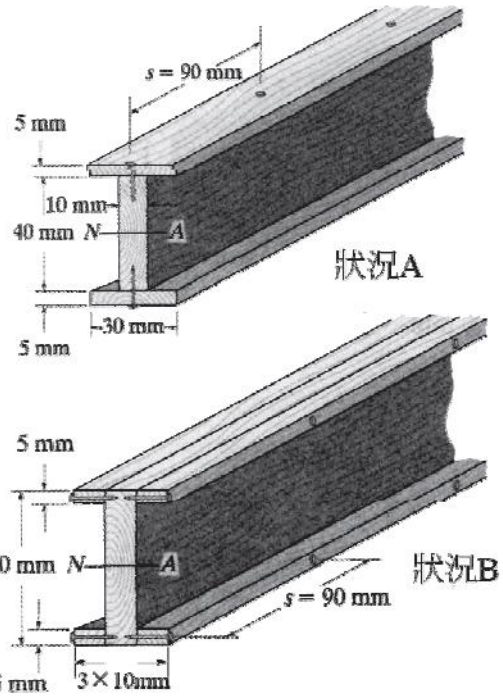
$$Q_{max}=135.35(15)\left(\frac{135.35}{2}\right)=137.4\times10^3\text{mm}^3$$

$$\tau_{max}=\frac{VQ}{Ib}=\frac{22(10^3)(137.4\times10^{-6})}{13.24(10^{-6})(0.015)}=15.22\text{MPa}$$

例 28

55. 一組合梁係由釘子組成，組成方式如右圖之狀況 A 與狀況 B 所示，狀況 A 之組合梁由 2 塊橫斷面為 30mm×5mm 與 1 塊 10mm×40mm 之木條所組成，釘子之剪力強度為 80N，若各釘間距均為 90mm，狀況 A 時組合梁所能承受之最大垂直剪力為何？

(A)27.1N (B)28.2N (C)54.2N (D)81.3N。



56. 承第 55 題，狀況 B 之組合梁由 4 塊橫斷面為 10mm×5mm 與 1 塊 10mm×50mm 之木條所組成，此梁所能承受之最大垂直剪力為何？

(A)162.6N (B)110.2N (C)81.3N (D)66.2N。

【102 台電】

【解析】

55. (C)。 $I=\frac{1}{12}\times 30\times 50^3-2\times[\frac{1}{12}\times(10)\times(40)^3]=20.58\times 10^4 mm^4$

$Q=(22.5)\times(30\times 5)=3375mm^3$

$q=\frac{VQ}{I}\Rightarrow\frac{80}{90}=\frac{V(3375)}{20.58\times 10^4}$

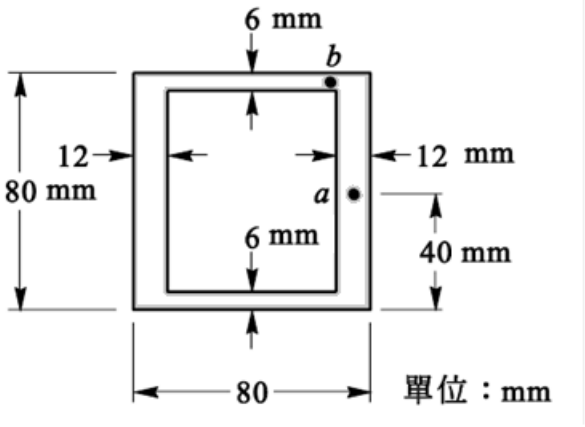
$V=54.2(N)$

56. (A)。 $Q=22.5\times 10\times 5=1125mm^3$

$\frac{80}{90}=\frac{V(1125)}{20.58\times 10^4}\Rightarrow V=162.6(N)$

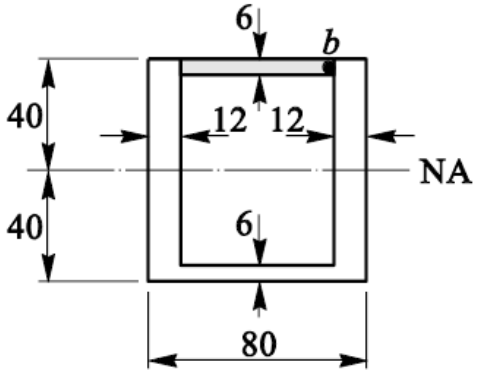
例 29

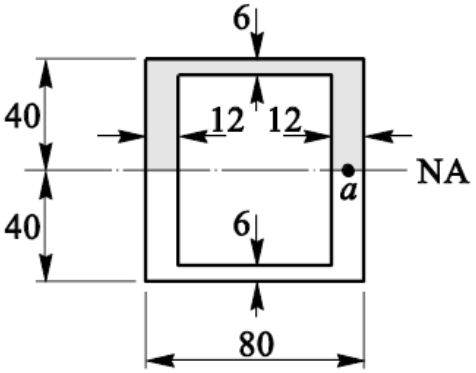
圖中箱型薄壁斷面承受垂直剪力 V = 75 kN，試求 a、b 兩點之剪應力



【成大土木類題】

【解析】





慣性矩：$I = \frac{1}{12}(80)(80)^3 - \frac{1}{12}(56)(68)^3 = 1.946\times10^6\ \text{mm}^4$

$Q_b = (56 \times 6)(37) = 12432\ \text{mm}^3$，$t_b = 6$ mm

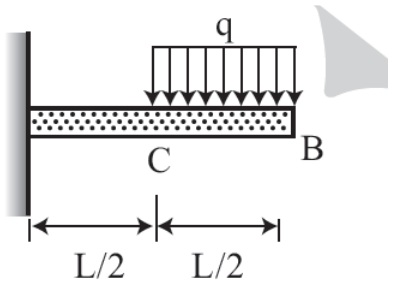
$$\tau_b = \frac{VQ_b}{I(2t_b)} = \frac{(75\times10^3)(12.432\times10^3)}{(1.946\times10^6)(2\times6)} = 39.9\ \text{MPa}$$

$Q_a = 12432 + 2(12 \times 40)(20) = 31632\ \text{mm}^3$，$t_a = 12$ mm

$$\tau_a = \frac{VQ_a}{I(2t_a)} = \frac{(75\times10^3)(31.632\times10^3)}{(1.946\times10^6)(2\times12)} = 50.8\ \text{MPa}$$

例 30

如圖所示之樑長L，斷面積為b×h之矩形，材料之楊氏模數為E，在CB段上受一均勻分布載重q。樑重不計，試求樑內之最大正應力及剪應力及其所在之位置。(96年機械高考)



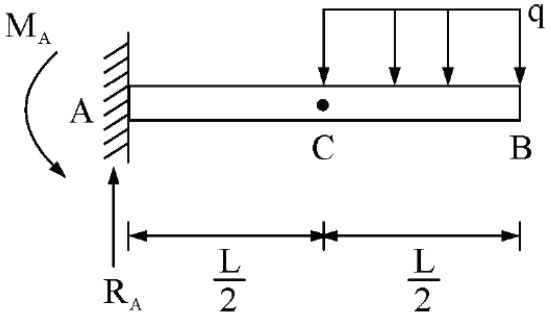
【解析】

答：(1)計算支承反力：

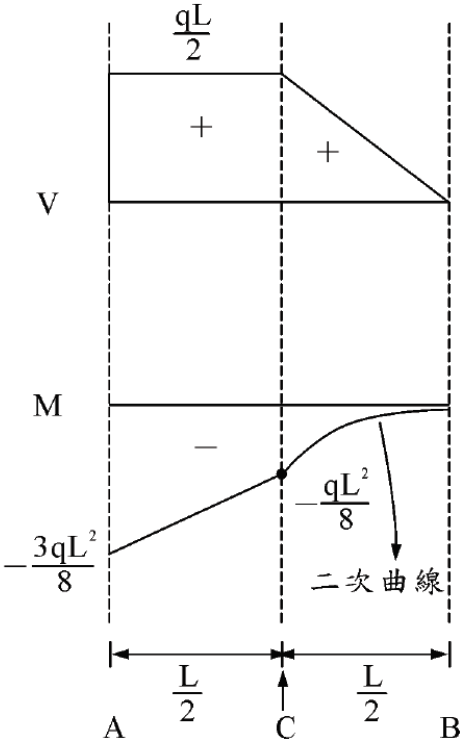
$$\Sigma F_y = 0 \Rightarrow R_A = \frac{qL}{2}$$

$$\Sigma M_A = 0$$

$$\Rightarrow M_A = \frac{qL}{2} \times \left(\frac{L}{2} + \frac{1}{2} \times \frac{L}{2}\right) = \frac{3qL^2}{8} (\curvearrowleft)$$



(2)畫剪力彎矩圖：



(3)如圖所示樑內受最大剪應力於AC段內，受最大正應力於A端：

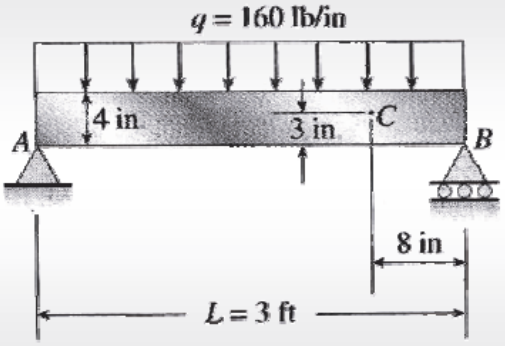
$$\tau_{max} = \frac{V_{max} Q}{Ib} = \frac{3}{2} \frac{V_{max}}{A}$$

$$= \frac{3 \times \frac{qL}{2}}{2 \times b \times h} = \frac{3qL}{4b \times h}$$

$$\sigma_{max} = \frac{M_{max} y}{I} = \frac{\frac{3qL^2}{8} \times \frac{h}{2}}{\frac{1}{12} bh^3} = \frac{9qL^2}{4bh^2}$$

例 31

一簡支樑與其斷面如下圖所示，樑長 L=3ft，樑斷面寬 b=1in，高 h=4in 之矩形，承受一均佈外力 q = 160 lb/in。假設圖中 C 點距離頂部 1in，距離 B 點 8in，請計算 C 點之正向應力 σ_c 與剪切應力 τ_c，並繪出其元素圖來表示。(100 年高考)

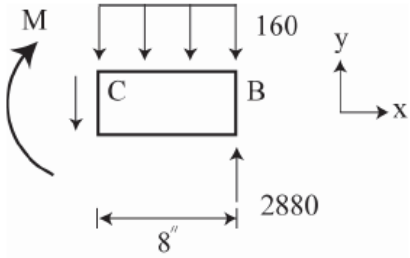


【100 高考】

【解析】

解 支承反力 R_A=2880(lb)，R_B=2880(lb)

取 BC 自由體圖



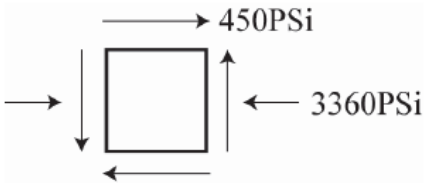
$$\Sigma F_y = 0 \Rightarrow V + 160 \times 8 = 2880 \Rightarrow V = 1600(\text{lb})$$

$$\Sigma M_B = 0 \Rightarrow M = 1600 \times 8 + 160 \times 8 \times \frac{8}{2} = 17920(\text{lb-in})$$

$$\sigma_c = \frac{My}{I} = \frac{17920 \times 1}{\frac{1}{12} \times 1 \times 4^3} = 3360(\text{psi})\text{壓}$$

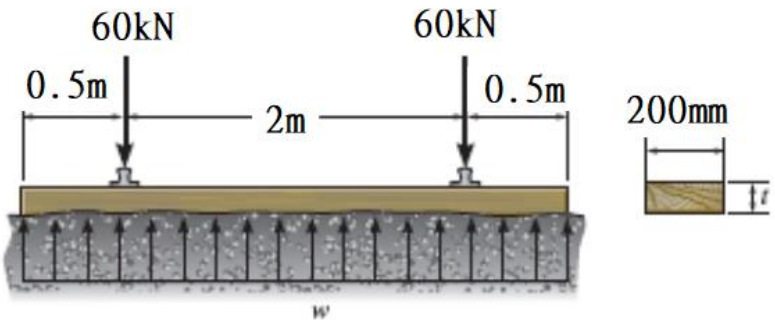
$$\tau_c = \frac{VQ}{Ib} = \frac{16000 \times (1 \times 1 \times 1.5)}{\frac{1}{12} \times 1 \times 4^3 \times 1} = 450(\text{psi})$$

壓力元素圖



例 32

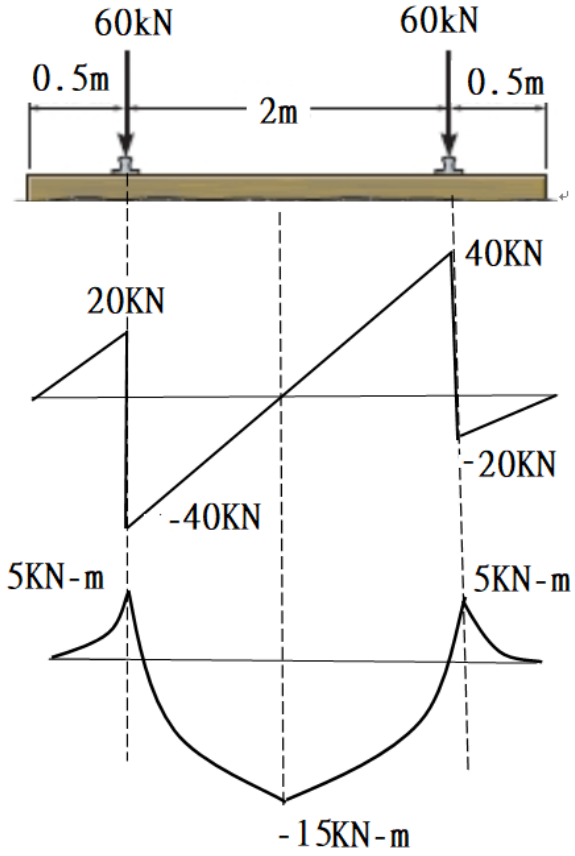
As shown in **Fig. 2**, assume the reaction on the railway tie is uniformly distributed over its length. (a) Draw the shear and moment diagram for the railway tie; and (b) If the maximum allowable bending stress of the wood is 6 MPa, determine the minimum thickness *h* of the rectangular cross sectional area of the tie. **(25%)**



【104 中央機械】

【解析】

(一)剪力圖與彎矩圖

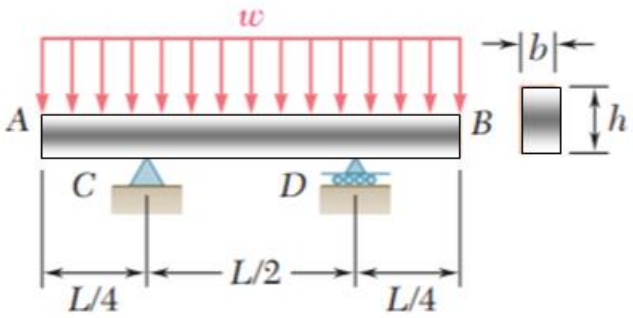


(二)求最小厚度h

$$\sigma = \frac{My}{I} \le 6Mpa \Rightarrow \sigma = \frac{15\times10^3 \times h/2}{\frac{1}{12}\times 200\times h^2} \le 6 \Rightarrow h_{\min} = 274mm$$

例 33

A timber beam AB of length L and rectangular cross section carries a uniformly distributed load w and is supported as shown. (a) Show that the ratio τ_m / σ_m of the maximum values of the shearing and normal stresses in the beam is equal to $2h/L$, where h and L are, respectively, the depth and the length of the beam,respectively



【104 成大工科】

【解析】

(一)靜力平衡

$$\mathrm{R}_c = R_D = \frac{1}{2} wL$$

(二)由剪力圖及彎矩圖如下所示得

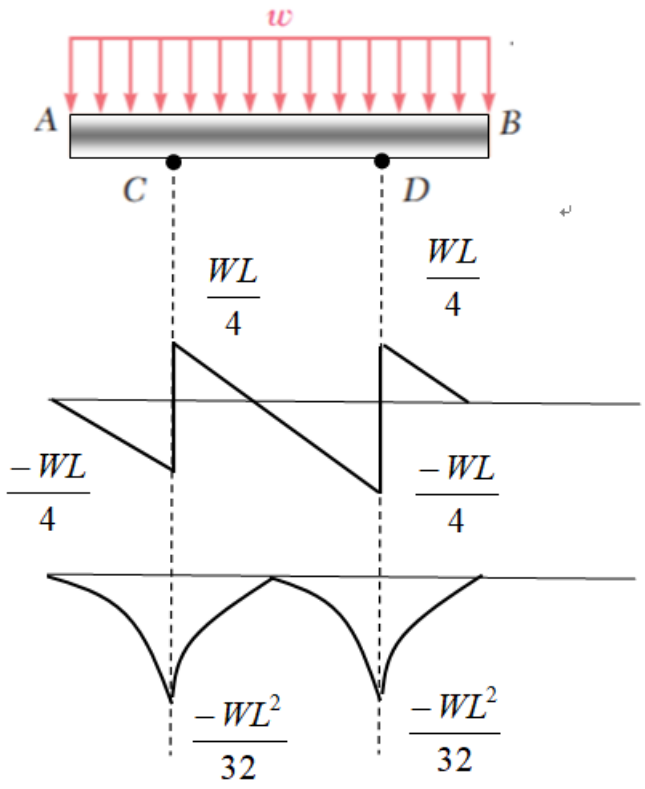
$$\mathrm{V}_{\max} = \frac{1}{4} wL \text{ 、 } \mathrm{M}_{\max} = -\frac{1}{32} wL^2$$

(三)應力分析

$$\tau_m = \frac{3\mathrm{V}_{\max}}{2A} = \frac{3}{2} \times \frac{\frac{1}{4} wL}{bh}$$

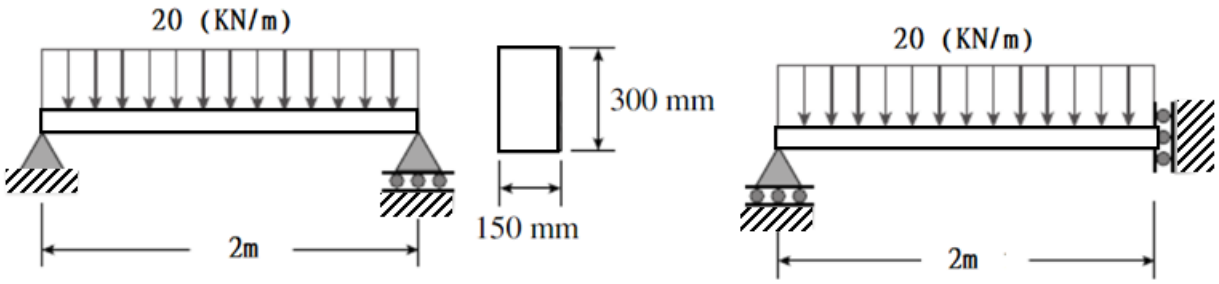
$$\sigma_m = \frac{\mathrm{M}_{\max} y}{I} = \frac{\frac{1}{32} wL^2 \times \frac{h}{2}}{\frac{1}{12} bh^3}$$

故 $\tau_m / \sigma_m = 2h/L$


w
A
B
C
D
WL/4
WL/4
−WL/4
−WL/4
−WL²/32
−WL²/32

例 34

Calculate the maximum shear stress tmaxand the maximum bending stress smaxin a wood beam (see figure) carrying a uniform load of 20 kN/m (which includes the weight of the beam) if the length is2m and the cross section is rectangular with width 150 mm and height 300 mm, and the beam is (a) simply supported as in the figure part (a) and (b) has a sliding support at right as in the figure part (b)

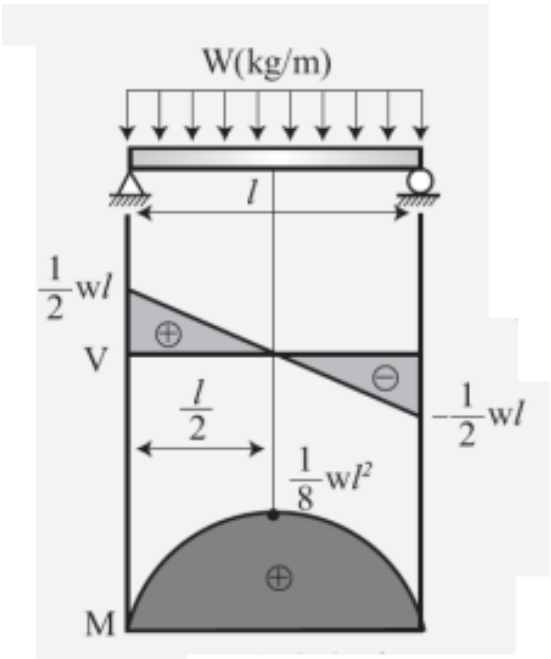


【105 成大醫工】

【解析】

(a)

(一)內力分析



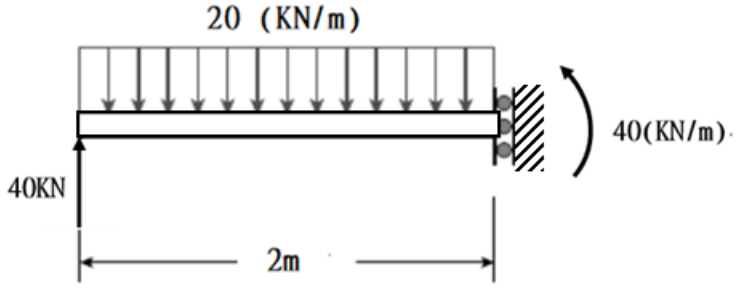
(二)應力分析

$$\sigma_{\max}=\frac{My}{I}=\frac{(\frac{1}{8}\times 20\times 2^2)\times 10^6\times 150}{\frac{1}{12}\times 150\times 300^3}=4.44(MPa)$$

$$\tau_{\max}=\frac{VQ}{Ib}=\frac{3V}{2A}=\frac{3\times(\frac{1}{2}\times 20\times 2)\times 10^3}{2\times 150\times 300}=0.667(MPa)$$

(b)

(一)內力分析



$M_{\max}=40(KN-m)$ 、 $V_{\max}=40(KN)$

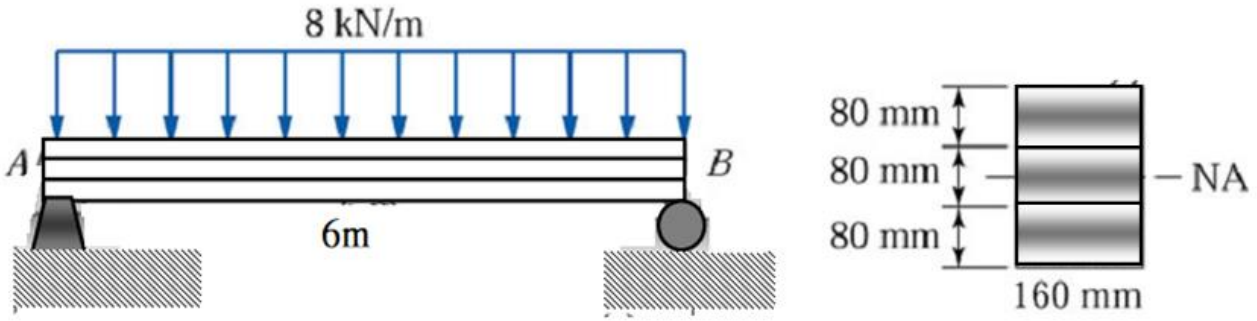
(二)應力分析

$$\sigma_{\max}=\frac{My}{I}=\frac{(40)\times 10^6\times 150}{\frac{1}{12}\times 150\times 300^3}=17.78(MPa)$$

$$\tau_{\max}=\frac{VQ}{Ib}=\frac{3V}{2A}=\frac{3\times(40)\times 10^3}{2\times 150\times 300}=1.333(MPa)$$

例 35

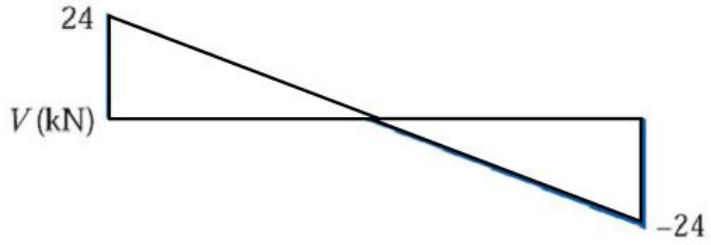
The simply supported wood beam in Fig.is fabricated by gluing together three 160-mm by 80-mm plans as shown. Calculate the maximum shear stress in (1) the glue; and (2) the wood.



【105 北科類題】【清大動機】

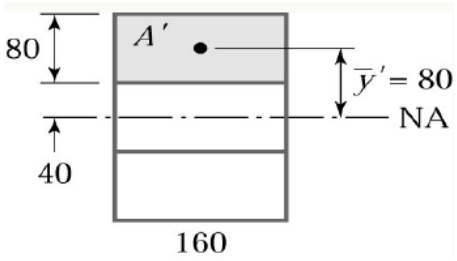
【解析】

(一)內力分析剪力圖與彎矩圖



(二)斷面分析

$$I=\frac{bh^3}{12}=\frac{160(240)^3}{12}=184.32\times10^6\,mm^4=184.32\times10^{-6}\,m^4$$



$$Q=A'\bar{y}'=(160\times80)(80)=1.024\times10^6=1.024\times10^{-3}\,m^3$$

(三)應力分析

(1) In the glue

$$\tau_{\max} = \frac{V_{\max} Q}{Ib} = \frac{(24\times10^{3})(1.024\times10^{-3})}{(184.32\times10^{-6})(0.160)}$$

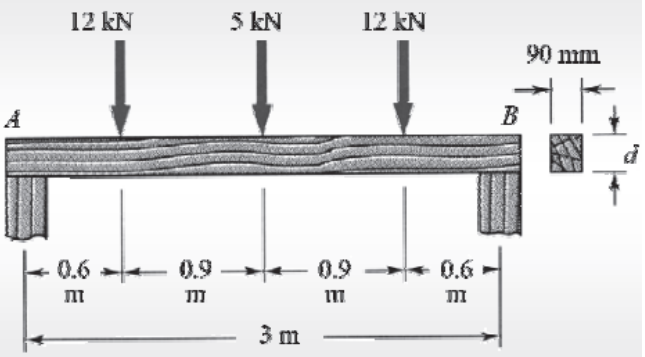
$$= 8.33\times10^{3}\,Pa = 8.33kPa$$

（2） In the wood

$$\tau_{\max} = \frac{3}{2}\frac{V_{\max}}{A} = \frac{3}{2}\frac{(24\times10^{3})}{(0.160)(0.240)} = 938\times10^{3}\,Pa = 938kPa$$

例 36

右圖顯示一跨距 3m 之木頭梁，梁斷面的寬度為 90mm、深度為 d。假設材料容許的正向應力為 σ_{all}=12MPa、容許的剪應力為 τ_{all}=0.75MPa，請問該梁斷面深度 d 至少應為何？
(99 年關三)



【99 關務三等】

【解析】

解 對於 M_{max}=10.95kN・m 及 σ_{all}=12MPa 可得

$$S=\frac{M_{max}}{\sigma_{all}}\Rightarrow 15\cdot d^2=\frac{10.95\times10^6}{12}\Rightarrow d^2=60833.333\therefore$$

$$d=246.644mm$$

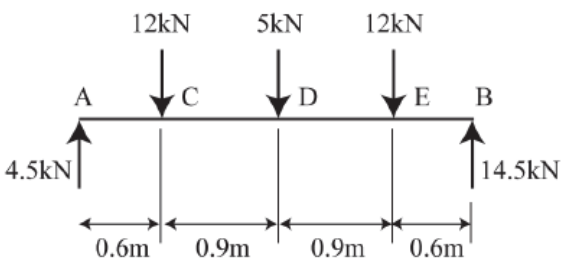
$$\tau_{max}=\frac{3}{2}\cdot\frac{V_{max}}{A}=\frac{3}{2}\cdot\frac{14.5\times10^3}{(90mm)\cdot(246.644mm)}=0.9798MPa$$

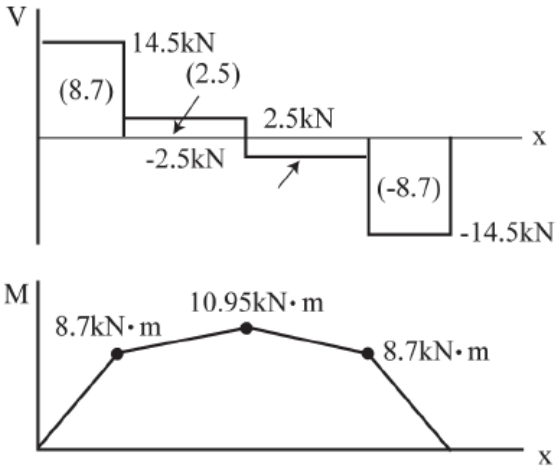
因 $\tau_{all}=0.75MPa$ ，深度 d=246.644mm 不夠，因此我們必須基於 $\tau_m\le0.75MPa$ 的要求重新設計此樑。

基於允許剪應力的設計:

$$\tau_{max}=\tau_{all}=\frac{3}{2}\cdot\frac{V_{max}}{A}\Rightarrow0.75MPa=\frac{3}{2}\cdot\frac{14.5\times10^3}{(90mm)\cdot d}\ \therefore d=322.2222mm$$

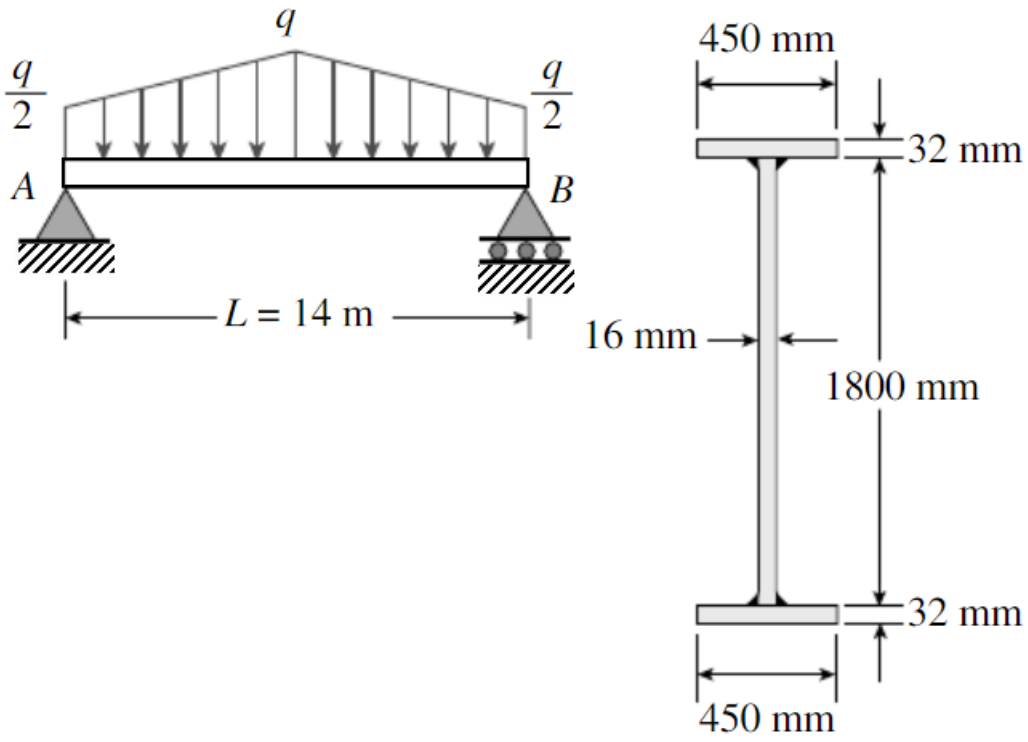
當然，正應力小於 $\sigma_{all}=12MPa$ ，因此 322.2222mm 的深度完全適用。





例 37

A bridge supports a girder AB on a simple span of length L = 14m supports a distributed load of maximum intensity q at mid-span and minimum intensity q/2 at supports A and B that includes the weight of the girder (see figure) The girder is constructed of three plates welded to form the cross section shown. Determine the maximum permissible load q based upon (a) an allowable bending stress 110MPa, and (b) an allowable shear stress 50MPa.



【105 台大工科】【108 台大土木】

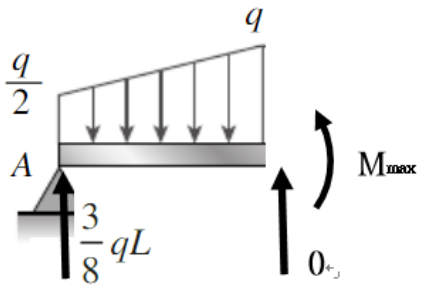
【解析】

(一)內力分析

$$R_A = R_B = \frac{q}{2}\frac{L}{2} + \frac{q}{4}\frac{L}{2} = \frac{3}{8}qL$$

中心點有最大的彎矩

切開中心點取左半邊自由體圖



$$M_{\max} = \frac{3}{8}qL\frac{L}{2} - \frac{q}{2}\frac{L}{2}\frac{L}{4} - \frac{q}{2}\frac{L}{4}\frac{L}{6}$$

$$= \frac{5}{48}qL^2$$

(二)斷面分析

$$L = 14\ \text{m}$$

$$h = 1864\ \text{mm} \quad h_1 = 1800\ \text{mm}$$

$$b = 450\ \text{mm} \quad t_f = 32\ \text{mm} \quad t_w = 16\ \text{mm}$$

$$I = \frac{1}{12}(bh^3 - bh_1^3 + t_w h_1^3)$$

$$I = 3.194 \times 10^{10}\ \text{mm}^4$$

$$S = \frac{2I}{h} \qquad S = 3.427 \times 10^7\ \text{mm}^3$$

(三)應力分析

(a)正向應力

$$\sigma = \frac{M_{\max}}{S} = \frac{\frac{5}{48}qL^2}{S}$$

$$q_{\max} = \frac{\sigma_{\text{allow}} S}{\frac{5}{48}L^2}$$

$$q_{\max} = 184.7\ \frac{\text{kN}}{\text{m}}$$

(b)剪應力

$$\tau = \frac{\text{VQ}}{\text{I}b} \Rightarrow \tau = \frac{(\frac{3}{8}q \times 14)[900 \times 16 \times 450 + 450 \times 32 \times 916]}{3.194 \times 10^{10} \times 16} = 50(MPa)$$

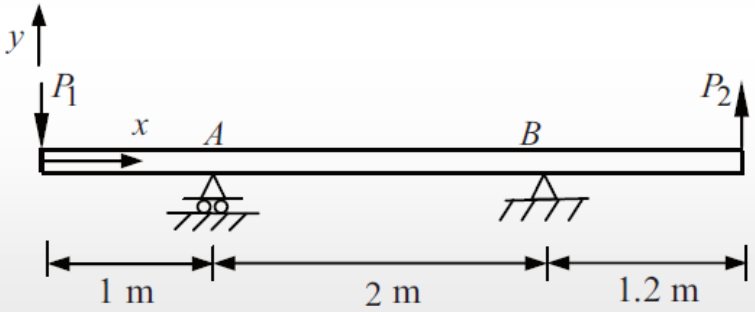
$$q = 247453\,N = 247.53(KN)$$

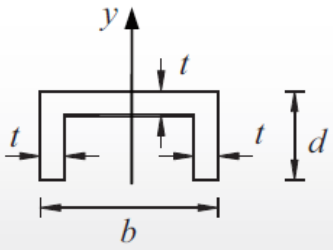
例 38

如圖所示之槽型斷面（channel section）梁的兩端受垂直力 P_1=15kN 及力 P_2=10kN 作用，已知斷面尺寸 d=0.1m、b=0.15m、t=0.01m，試求：

(1) 該梁的剪力及彎矩圖。

(2) 該梁斷面的最大拉應力、壓應力、剪應力之值及其發生的位置。



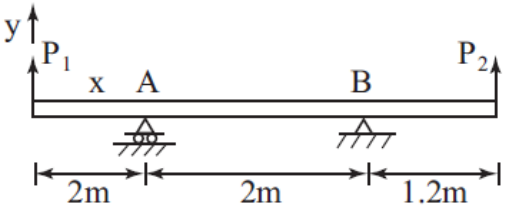


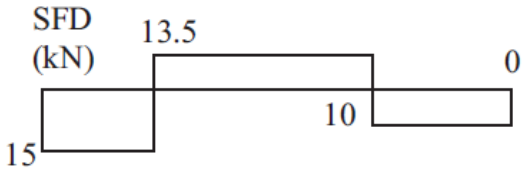
【解析】

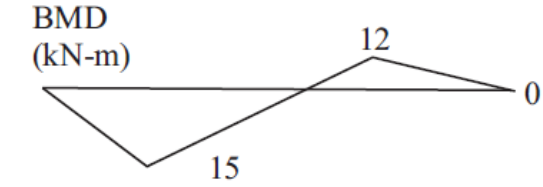
解 (1) 由計算知

$$R_B = 23.5(\text{kN})(\downarrow)$$

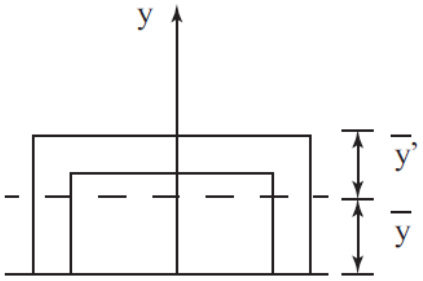
$$R_A = 28.5(\text{kN})(\uparrow)$$



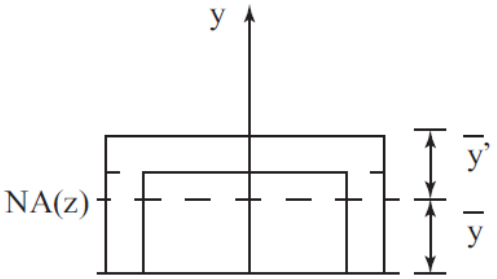




(2)



A. 求中性軸(NA)位置和 z 軸慣性矩



$$\bar{y}=\frac{150\times10\times(100-\frac{100}{2})+10\times(100-10)\times\frac{100-10}{2}\times2}{150\times10+10\times(100-10)\times2}=67.73(\text{mm})$$

$$\bar{y}'=100-\bar{y}=32.27(\text{mm})$$

$$I_z=\frac{1}{3}\times10\times67.73^3\times2+\frac{1}{3}\times150\times32.27^3-\frac{1}{3}(150-2\times10)(32.27-10)^3$$

$$=3272954.566(\text{mm}^4)$$

B. 由 SFD 中可知 $V_{max}=15(\text{kN})$

最大剪應力，取中性軸下方來作

$$Q=10\times67.73\times\frac{67.73}{2}\times2=45873.529(\text{mm}^3)$$

$$\tau_{max}=\frac{VQ}{Ib}=\frac{15\times10^3\times45873.529}{3272954.566\times(10\times2)}=10.51(\text{Mpa})$$

C. 由 BMD 中可知 $M_{max}=-15\,(\text{kN-m})$與 $M_{max}=+12\,(\text{kN-m})$，

求最大應力必需分別討論：

(a) 拉應力

$$\sigma_{max,T}=\frac{M\bar{y}'}{I_Z}=\frac{15\times10^3\times10^3\times32.27}{I_Z}=147.89(Mpa)$$

$$\sigma_{max,T}=\frac{M\bar{y}}{I_Z}=\frac{12\times10^3\times10^3\times67.73}{I_Z}=248.33(Mpa)$$

(b) 壓應力

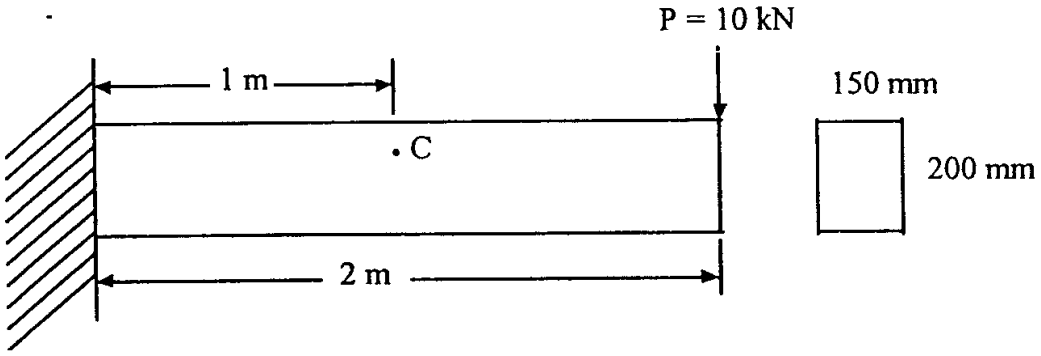
$$\sigma_{max,C}=\frac{M\bar{y}}{I_Z}=\frac{15\times10^3\times10^3\times67.73}{I_Z}=310(Mpa)$$

$$\sigma_{max,T}=\frac{M\bar{y}'}{I_Z}=\frac{12\times10^3\times10^3\times32.27}{I_Z}=118.32(Mpa)$$

得最大拉應力$\sigma_{max,T}=248.33(Mpa)$，最大壓應力$\sigma_{max,C}=310(Mpa)$

例 39

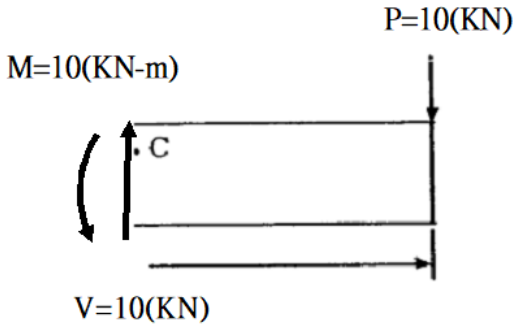
A cantilever beam of length L = 2 m supports a load of P = 10 kN (see figure). The cross-sectional dimensions of the beam are 150 mm × 200 mm. Determine the principal stresses at point C located 50 mm from the top surface of the beam. (15%)



【100 清大動機】

【解析】

(一)內力分析



(二)斷面分析

$$I = \frac{1}{12} \times 150 \times 200^3 = 10^8$$

(三)應力分析

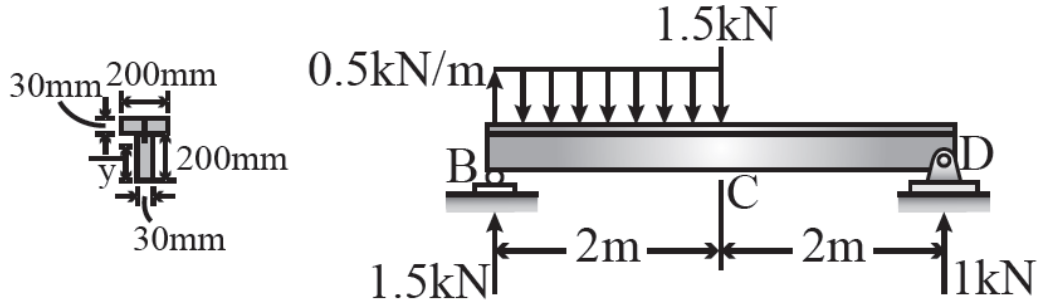
$$\sigma = \frac{My}{I} = \frac{10 \times 10^6 \times 50}{10^8} = 5(MPa)$$

$$\tau = \frac{VQ}{Ib} = \frac{10000 \times 150 \times 50 \times 75}{10^8 \times 50} = 1.125(MPa)$$

$$\sigma_{1,2} = \frac{\sigma}{2} \pm \sqrt{(\frac{\sigma}{2})^2 + \tau^2} = 5.24(MPa), -0.24(MPa)$$

例 40

如圖所示，一截面為T形的樑由兩片200mm×30mm的板材所組成。已知樑的最大容許彎曲應力σ_{allow}＝12MPa以及最大容許剪應力τ_{allow}＝0.8MPa，試問該樑能否安全地承受如圖所示的負荷？（備註：1.樑的重量可忽略不計。2.請繪出剪力圖及彎矩圖。）



【105 普考】

【解析】

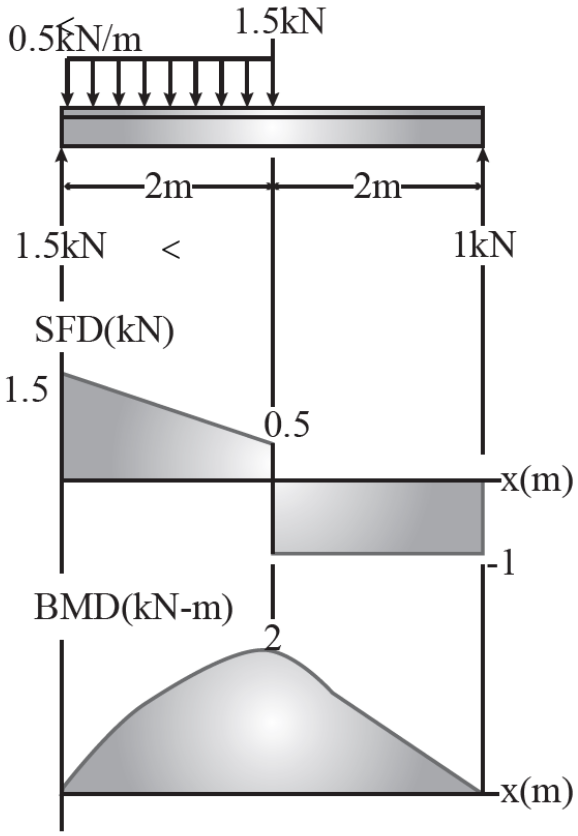
答：$\sigma = \frac{My}{I} = \frac{2(kN-m)(0.1575)}{60.125\times10^{-6}} =$

$= 5.24(Mpa) < 12(Mpa)$

$\tau = \frac{VQ}{It} = \frac{1.5(kN-m)(0.372\times10^{-3})}{60.125\times10^{-6}\times0.03} =$

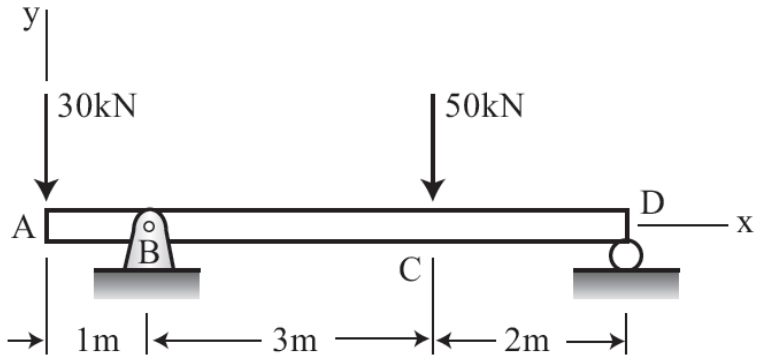
$= 0.309(Mpa) < 0.8(Mpa)$

負載可以承載圖式負載



例 41

圖顯示一樑承受兩集中荷重，請問該樑沿著長軸(x-軸)所受到之最大剪力及最大彎矩各為何？假設前題所示梁之斷面為200mm邊長之正方形，請問該樑所承受之最大軸向應力與剪應力各為何？ (94年機械第二次地特三等)



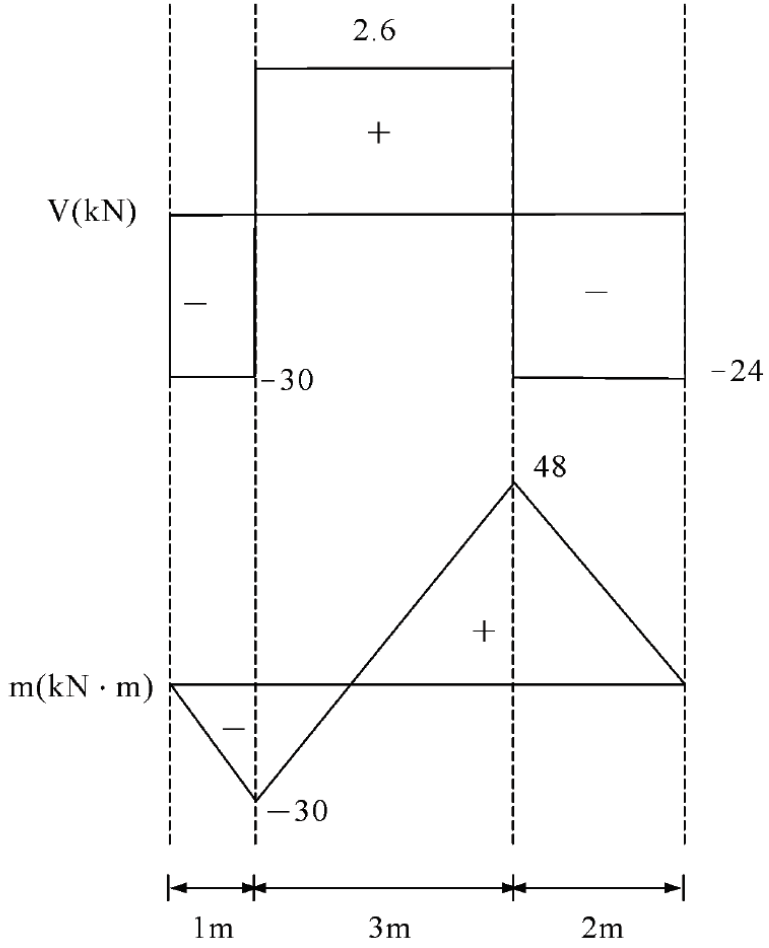
【解析】

答：(1) 先求支承反力

$$\sum M_B = 0 \Rightarrow R_D = \frac{1}{5} \times [50 \times 3 - 30 \times 1] = 24\text{kN}(\uparrow)$$

$$\sum M_D = 0 \Rightarrow R_B = 56\text{kN}(\uparrow)$$

(2) 畫剪力圖及彎矩圖：



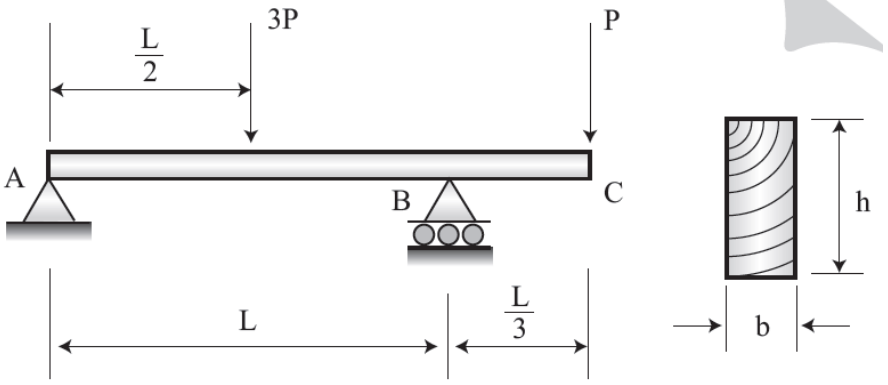
(3) $V_{max} = 30kN(\uparrow)$： $M_{max} = 48kN \cdot m(\circlearrowright)$

$$\tau_{max} = \frac{3}{2}\frac{V}{A} = \frac{3}{2} \times \frac{30 \times 10^3}{0.2 \times 0.2} = 1125000Pa = 1.125MPa$$

$$\sigma_{max} = \frac{M_{max} y}{I} = \frac{48 \times 10^3 \times 0.1}{\frac{1}{12} \times 0.2 \times 0.2^3} = 36000000Pa = 36MPa$$

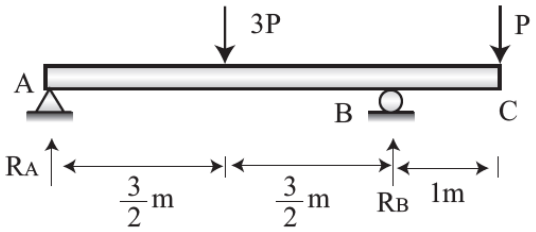
例 42

有一簡支懸臂樑如右圖所示，其斷面高度h＝300mm，樑之簡支段L＝3m，懸臂段長度為1m，分別受3P＝12kN之集中荷重於簡支段，及P＝4kN於懸臂段之自由端。如不計樑之自重，材料之容許彎曲應力（bending stress）為8MPa，容許應力為0.65MPa，則樑之斷面寬b應採用多少？(92年土木高考)



【解析】

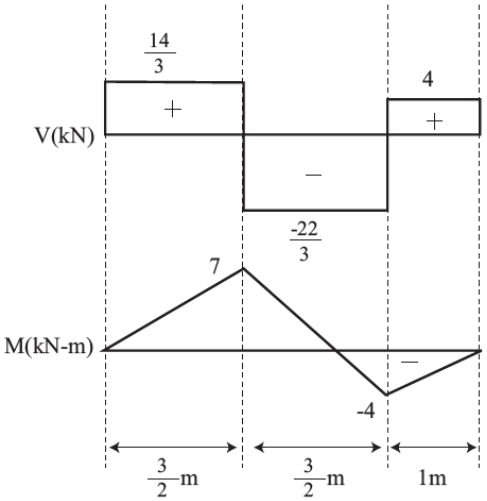
答：(1)先求支承反力



$$\Sigma M_B = 0 \Rightarrow R_A = \frac{14}{3}\text{kN}(\uparrow)$$

$$\Sigma M_A = 0 \Rightarrow R_B = \frac{34}{3}\text{kN}(\uparrow)$$

(2)畫剪力及彎矩圖：



(3)如剪力圖所示，簡支樑所受之力

最大剪力 $V_{max} = \frac{22}{3}\text{kN}(\downarrow)$　最大彎矩 $M_{max} = 7\text{kN}\cdot\text{m}(\circlearrowleft)$

$$\sigma_y = \frac{M_{max}\,y}{I} = \frac{M_{max}\times\frac{h}{2}}{\frac{1}{12}\times b\times h^3} \Rightarrow 8\times 10^6 = \frac{7\times 10^3\times\frac{(0.3)}{2}}{\frac{1}{12}\times b\times(0.3)^3} \Rightarrow b = 0.0583\text{m}$$

$$\tau_y = \frac{V_{max}Q}{I\,b} = 1.5\frac{V_{max}}{A} = 1.5\frac{V_{max}}{b\times h} \Rightarrow 0.65\times 10^3 = 1.5\times\frac{\frac{22}{3}}{0.3b}$$

$$\Rightarrow b = 0.0546\text{m}$$

(4)選擇較大之b值，可使彎曲應力及剪應力不超過容許應力

$b = 0.0583\text{m}$

例 43

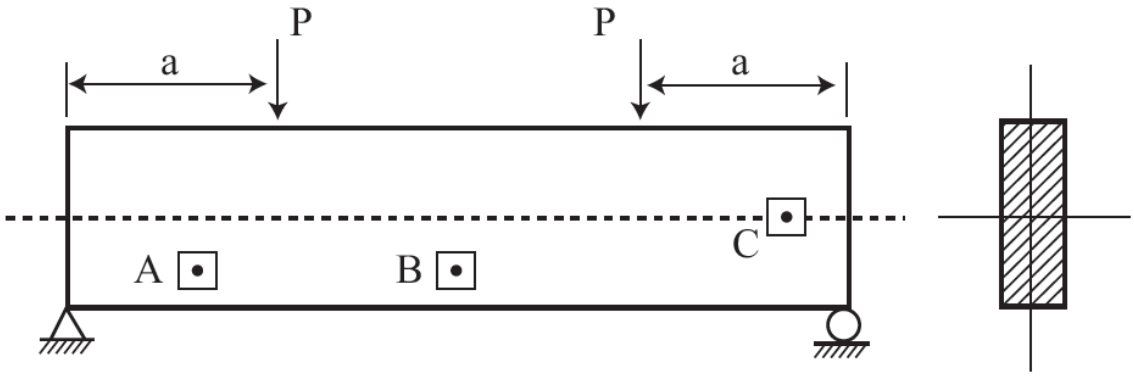
圖中是一矩形截面樑受兩個大小相同之橫向力P作用。請回答下列問題：

(1)請繪出剪力與力矩圖(shear force and moment diagrams)。

(2)請各在一正方形元素上畫出A, B, C三點上所有的應力(定性即可)。

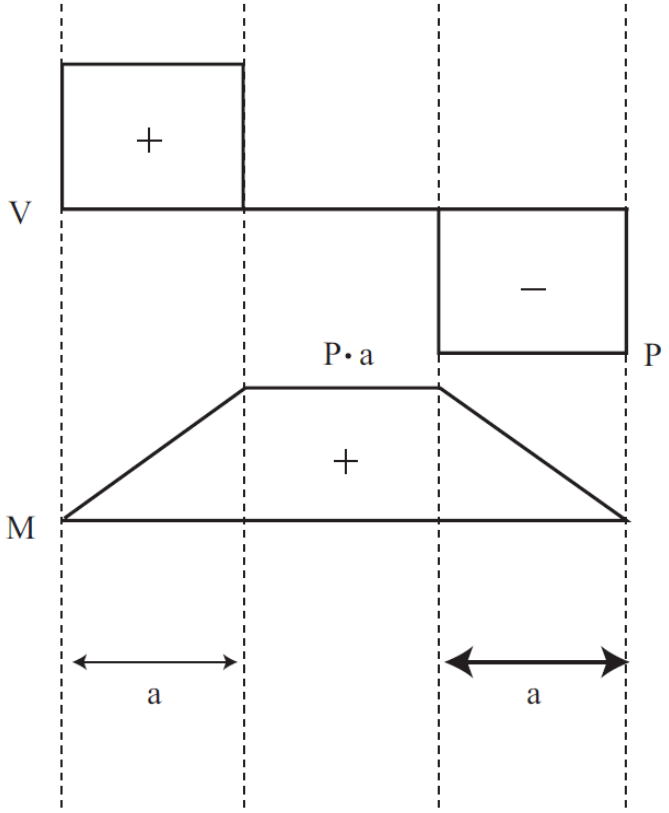
(3)請各以另一正方形元素表示主應力的方向(正確畫出與前小題元素之夾角以及應力的正負方向)。

(4)根據最大正向應力損壞判準(maximum normal-stress failure criterion)，請說明(畫圖表示)樑破壞時裂紋在B, C 兩點處的走向。(94年機械地方特考三等)

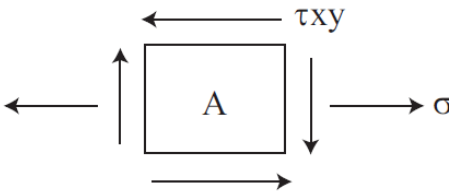


【解析】

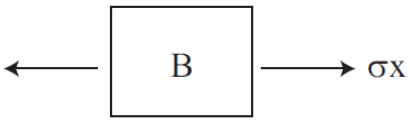
答：(1)計算兩端之支承反力均為P：



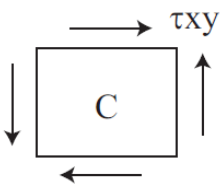
(2) A之應力元素



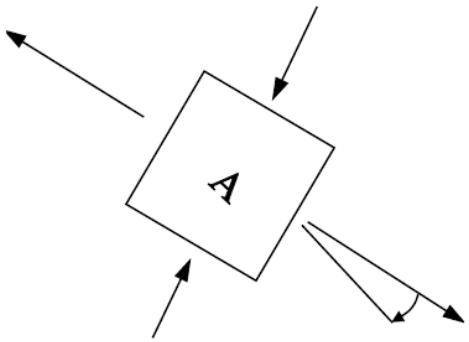
B之應力元素



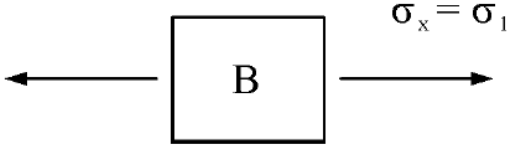
C之應力元素



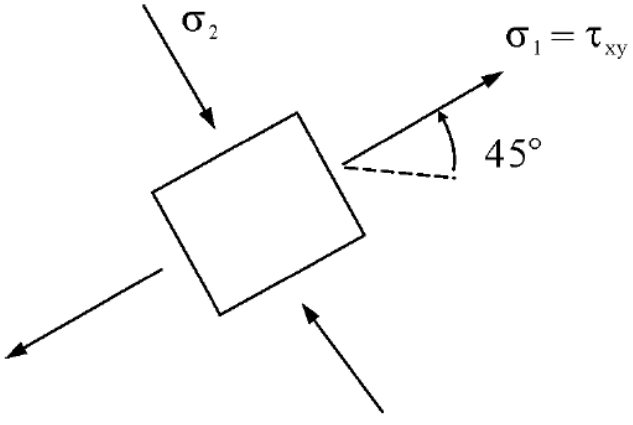
(3) A之主應力方向



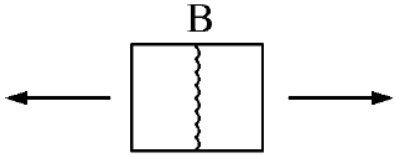
B之主應力方向



C之主應力方向



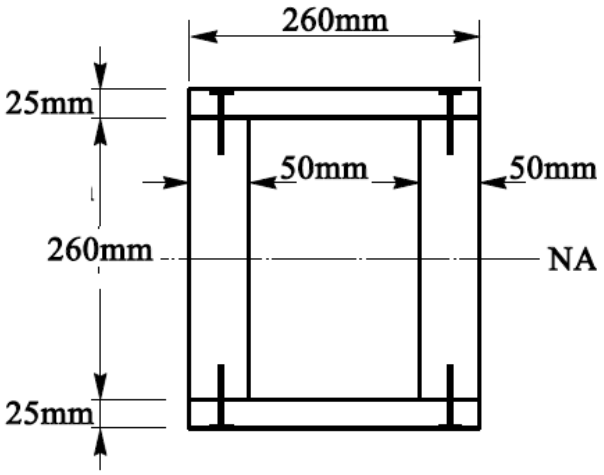
(4) B應力元素破壞之裂紋



與應力方向垂直

例 44

A box beam of wood is constructed of two $260\times50\text{mm}$ boards and two $260\times25\text{mm}$ boards (see figure). The boards are nailed at a longitudinal spacing $s=100\text{mm}$. If each nail has an allowable shear force $F=120\text{N}$, what is the maximum allowable shear force V?



【中文翻譯】

盒形木樑由兩片 $260\times50\text{mm}$ 及兩片 $260\times25\text{mm}$ 的板子釘成（ 見圖 ）。釘子間距 $s=100\text{mm}$ 。如果一支釘的容許剪力 $F=120\text{N}$ ，則最大容許剪力 V 為何？

【解析】

(一)斷面分析

$$I=\frac{1}{12}(260)(310)^3-\frac{1}{12}(160)(260)^3=411\times10^6\,\text{mm}^4$$

(二)計算 V

$$Q=(260\times25)(142.5)=0.926\times10^6\,\text{mm}^3$$

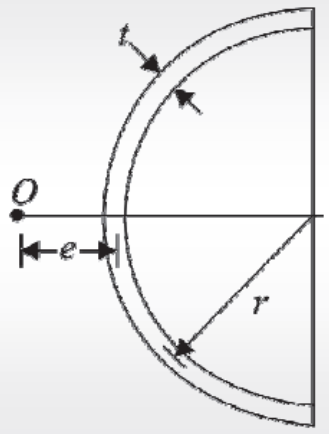
$$f=\frac{2F}{s}=\frac{2(1200)}{100}=24\ \text{N/mm}$$

$$f=\frac{VQ}{I}\ ,\ 24=\frac{V(0.926\times10^6)}{411\times10^6}\ ,$$

$$V=10.7\times10^3\,\text{N}=10.7\,\text{kN}$$

例 45

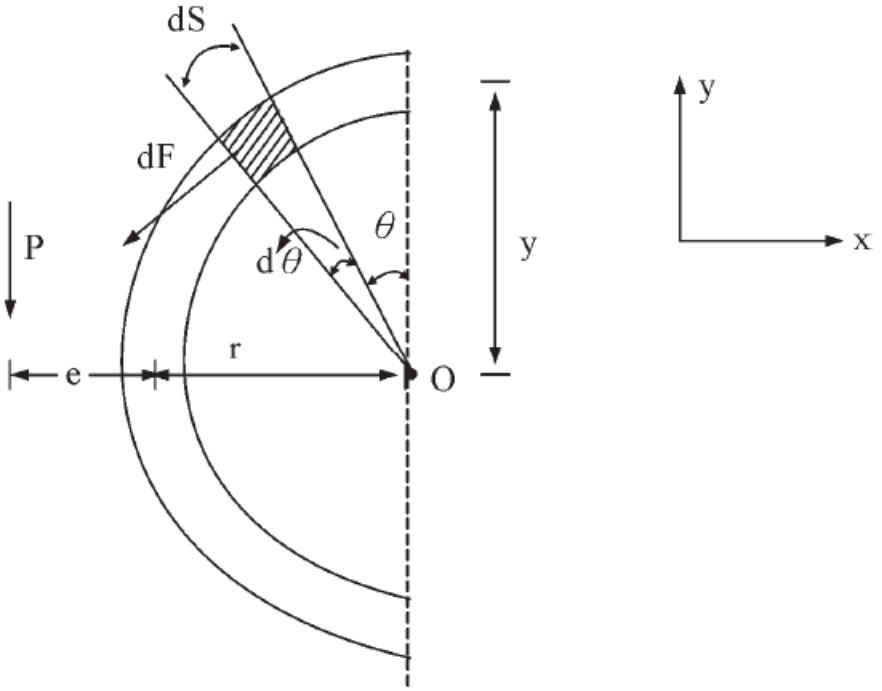
求剪力中心 O 點之位置 e 值，假設構件各段厚度均為 t。



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【解析】

解



如上圖所示

$$I = \int y^2 dA = r^3 t \int_0^{\pi} \cos^2\theta d\theta = r^3 t \int_0^{\pi} (\frac{\cos 2\theta + 1}{2}) d\theta = \frac{\pi r^3 t}{2}$$

$$Q = \int_0^{\theta} r^2 t \cos\theta d\theta = r^2 t \sin\theta \qquad \tau = \frac{VQ}{Ib} = \frac{P(r^2 t \sin\theta)}{\frac{\pi r^3 t}{2} \times t} = \frac{2P\sin\theta}{\pi r t}$$

$$F = \int \tau \cdot t \cdot r d\theta = \int_0^{\pi} \frac{2P\sin\theta}{\pi r} d\theta = \frac{4P}{\pi}$$

又 $P(e+r) = r \cdot F = \frac{4Pr}{\pi}$ ，$e = \frac{4r}{\pi} - r = 0.273r$